Investigation of Geometric and Dosimetric Effects of Bladder and Rectal Filling on the Quality of Pelvic Radiotherapy Treatment using 3DCRT, IMRT & VMAT Techniques

A project report submitted to Department of Medical Physics and Biomedical Engineering (MPBME), Faculty of Science and Engineering, Gono Bishwabidyalay in partial fulfillment of the requirement for the degree of Master of Science in Medical Physics.

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It is hereby declared that this research project has been conducted ethically. The results presented are genuine and obtained by me during the working period. It is being submitted for the Degree of Masters of Science in the University of Gono Bishwabidyalay, Dhaka, Bangladesh. It has not been submitted before for any degree, award, or examination in any other University.

Signature of the candidate



Dedicated

to

MY BELOVED PARENTS.

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LIST OF SYMBOLS

D_{max}	Maximum Dose
D_{min}	Minimum Dose
Cm	Centimeter
Mm	Millimeter
kV	Kilovolt
MV	Megavolt
D	Dose
S	Second
ML	Milliliter
Min	Minutes

LIST OF ABBREVIATIONS

Gy	Gray
QA	Quality Assurance
ICRU	International Commission of Radiation Units and Measurements
AAA	Anisotropic Analytical Algorithm
FF	Flattening Filter
TPS	Treatment Planning System
MLC	Multi Leaf Collimator
CT	Computed Tomography
PTV	Planning Target Volume
3DCRT	Three-Dimensional Conformal Radiation Therapy
DVH	Dose Volume Histogram
EBRT	External Beam Radiation Therapy
LINAC	Linear Accelerator
IMRT	Intensity-Modulated Radiation Therapy
VMAT	Volumetric-Modulated Arc Therapy
CI	Conformity Index
HI	Homogeneity Index
WHO	World Health Organization
SA-VMAT	Single Arc- Volumetric-Modulated Arc Therapy
DA-VMAT	Double Arc- Volumetric-Modulated Arc Therapy
V	Volume
MU	Monitor Unit

PDIP	Portal Dose Image Prediction
EPID	Electronic Portal Device
CBCT	Cone Beam Computer Tomography
Abs DD	Absolute Dose Difference
Avg DD	Average Dose Difference
UB	Urinary Bladder
R	Rectum
DD	Dose Difference
RTOG	Radiation Therapy Oncology Group
OARs	Organs at Risk
MLC	Multi Leaf Collimator
S D	Standard Deviation
ADR	Actual Dose Received

ABSTRACT

Background: Approximately 5%–10% of men and 5%-30% women who receive prostate and cervical cancer radiotherapy will suffer from radiation cystitis. Bladder and rectum filling before the administration of radiotherapy results in lower radiation exposure to the bladder and rectum. This study evaluated the impact of bladder and rectum filling variations on geometric and dosimetric differences using CT and CBCT data aiming to quantify discrepancies and their influence on treatment accuracy.

Methods and Materials: This study analyzed pelvic data from 20 patients: first group (14) and control group (6), following consistent protocols. An expert oncologist contoured the bladder and rectum on planning CT images. Six registered CBCT image sets per patient were randomly selected for recontouring. Geometric comparisons between CT and recontoured CBCT structures were made using Absolute Volume Difference (abs VD), Hausdorff Distance (HD), and Dice Similarity Coefficient (DSC), calculated with in-house made Python program. Replans based on recontoured structures were created using identical optimization parameters in Eclipse (ve13.6). The resulting Monitor Unit (MU) differences, Absolute Dose Difference (abs DD), and Percentage Dose Difference (% DD) were recorded.

Results and Discussion: Significant geometric and dosimetric differences were observed between the First and Control groups for bladder and rectum measurements. For the bladder, abs VD, HD, and DSC were 68.48 ± 77.11, 40.54 mm ± 35.08, 0.21 mm ± 0.34 and (First) versus 90.63 cc ± 118.03, 61.88 mm ± 14.91, and 0.27 ± 0.27 (Control). For the rectum, abs VD, HD, and DSC were 28.98 cc ± 15.31, 35.55 mm ± 79.46, and 0.58 ± 0.39 (First) versus 32.68 mm ± 24.39, 22.38 mm ± 5.51, and 0.86 ± 0.09 (Control). In dosimetry, the average MU difference was 10.82 ± 17.67 (First) and 7.96 ± 6.40 (Control). The abs DD was 639 cGy ± 1064 (First) and 601.23 cGy ± 511.94 (Control), while the % DD was 10.82% ± 17.67 (First) and 7.96% ± 6.40 (Control).

Conclusion: Significant geometric and dosimetric differences were observed, highlighting the need for accurate filling protocol to improve treatment precision.

CHAPTER-1: INTRODUCTION

1.1 General Introduction

Pelvic cancers, encompassing malignancies such as rectal, bladder, and gynecological cancers, are a significant cause of morbidity. Globally, cervical cancer is a significant concern. In 2022, there were approximately 660,000 new cases and around 350,000 deaths attributed to cervical cancer worldwide. [1]. Curative treatment for localized pelvic cancers typically involves surgery, radiation therapy (RT), and sometimes chemotherapy, with treatment regimens tailored to the specific cancer type and stage [2]. Pelvic tumors are often treated using external beam radiation therapy (EBRT), which can be challenging due to the proximity of critical structures such as the bladder, small intestine, and rectum [3].

In Bangladesh, cervical cancer remains a pressing health issue. According to the GLOBOCAN 2020 database, the age-standardized incidence rate for cervical cancer in Bangladesh was estimated at 10.6 per 100,000 women, with a mortality rate of 6.7 per 100,000 women. Furthermore, data from the National Center for Cervical & Breast Cancer Surveillance System indicates that, as of January 2025, over 5 million women have been enrolled in screening programs, with approximately 2.9% testing positive through Visual Inspection with Acetic Acid (VIA) screening. Without intervention, projections indicate that by 2070, a total of 505,703 women in Bangladesh will die from cervical cancer, with the number rising to 1,042,859 by 2120 [1].

In the case of bladder cancer, the presence of the bladder itself in the radiation field requires special attention to minimize toxicity to the surrounding organs [4]. Pelvic radiation for rectal cancer or gynecological cancers, such as cervical or endometrial cancers, also presents challenges in sparing the small bowel and femoral heads from excessive radiation exposure, which could lead to long-term complications such as bowel obstruction, urinary incontinence, or secondary malignancies [5,6]. Techniques like intensity-modulated radiation therapy (IMRT) and image-guided radiation therapy (IGRT) are commonly employed to achieve highly precise dose delivery to the tumor while minimizing dose to surrounding healthy tissue [7].

Radiation therapy for pelvic cancers is often delivered using a standard fractionation schedule of 1.8–2 Gy per fraction, five days a week, for a total dose of 45–50.4 Gy in 5–6 weeks for localized disease [8]. However, studies have indicated that hypo fractionated radiation, with fewer fractions and higher doses per fraction, may offer comparable efficacy with reduced treatment times for certain pelvic cancers [9]. The FAST-Forward trial demonstrated that hypo fractionated RT for early breast cancer was non-inferior to conventional fractionation, suggesting that similar approaches could be applied to pelvic malignancies, especially for those with poor prognoses [10].

Accurate radiation delivery in pelvic cancer treatment is crucial to avoid complications resulting from variations in tumor location, surrounding organ movement, and changes in bladder or rectal filling. The bladder's role in maintaining consistent bladder volume during treatment is particularly important for ensuring precise radiation delivery to pelvic cancers involving the bladder or other nearby structures [11]. Variability in bladder volume has been shown to affect target coverage in several studies, potentially reducing the effectiveness of RT while increasing the risk of toxicity to critical organs [12,13].

To mitigate these risks, new technologies are being developed to maintain bladder consistency throughout the course of radiation therapy. One such approach involves using bladder-filling devices or urinary catheters that maintain a consistent bladder volume, thereby reducing the chances of treatment errors and ensuring that radiation is delivered to the intended target [14]. These devices, such as catheter-based check valves and balloon-based bladder volume regulation, are still in the development phase but offer promising solutions for improving the precision of pelvic cancer radiation therapy [15].

Intensity-modulated radiation therapy (IMRT) and volumetric modulated arc therapy (VMAT) are commonly used in external beam radiation therapy (EBRT) techniques to treat sincerity is a type of conformal radiotherapy that uses a linear accelerator to deliver an advanced type of high-precision radiotherapy that shapes the radiation beam to closely fit the area of the temerity uses multiple small photon or proton beams of varying intensities to precisely irradiate a tumor. The radiation intensity of each beam is controlled, and the beam shape changes throughout each treatment. The goal of IMRT is to conform the radiation dose to the target and to avoid or reduce exposure of healthy tissue to limit the side effects of treatment. IMRT can also create a U shaped (concave) area at the edge of the radiotherapy field. This avoids high radiation doses to structures that would otherwise be damaged by the radiotherapy. IMRT allows for the delivery of high dose of radiation to targets while minimizing doses to organ at risk (OARS).

Volumetric modulated arc therapy (VMAT) is a novel radiation therapy technique that delivers the radiation dose continuously as the treatment machine rotates. This technique accurately shapes the radiation dose to the tumor while minimizing the dose to the organs surrounding the tumor and VMAT is a new type of IMRT. In recent years, significant improvement in the radiation dose delivery technique has been noticed. VMAT is particularly useful for delivering radiation therapy to tumors near sensitive body organs, and can be an effective treatment for many types of cancer. VMAT planning and treatment is technically demanding and requires a high level of precision. The VMAT procedure typically takes around 2minutes. Volumetric Modulated Arc Therapy (VMAT) uses photons generated by a medical linear accelerator. Very small beams with varying intensities are aimed at a tumor and then rotated 360 degrees around the patient. This results in attacking the target in a complete three-dimensional manner. The most modern and complex of these modes is VMAT which rotates the gantry of the linear accelerator around the patient for a partial or full arc at a constant or variable rate. The MLC are in constant motion with the radiation beam based on the incident angle. VMAT become a popular delivery option, taking advantage of shorter delivery time and smaller monitor unit (MU) Compared to IMRT. Treatment planning has now moved from the three-dimensional conformal radiotherapy to volumetric modulated arc therapy (VMAT) [5].

The aim of the study is to evaluate the impact of bladder and rectum filling variations on geometric and dosimetric differences using CT and CBCT data aiming to quantify discrepancies and their influence on treatment accuracy

1.2 Aims of the study

- To investigate the geometric and dosimetric variations caused by bladder and rectal filling during pelvic radiotherapy
- To compare the dosimetric impact on target volume coverage and organ-at-risk (OAR) doses among 3DCRT, IMRT, and VMAT techniques under varying bladder and rectal filling conditions.
- To determine the effect of bladder and rectal filling on monitor unit (MU) requirements for 3DCRT, IMRT, and VMAT techniques.
- To quantify the geometric changes in the bladder and rectum using metrics such as absolute volume difference (abs VD), Hausdorff distance (HD), and Dice similarity coefficient (DSC) for varying filling protocols.
- To assess the impact of bladder and rectal volume changes on dose distribution to critical structures
- To establish optimal bladder and rectal filling protocols for improved consistency and accuracy in pelvic radiotherapy.

1.3 Scope of the study

The idea of the project and theoretical part of this thesis were done in Medical Physics and Biomedical Engineering department Gono, Bishwabidyalay, Savar, Dhaka. This department is well equipped with various radiotherapy instruments for example phantoms, dosimeters, different ionization chambers.

The practical part of the thesis has been done in the "Oncology and Radiotherapy Centre" of the " Square Hospital Ltd", Panthapath, Dhaka, Bangladesh.

1.4 Literature Reviews

For the successful completion of this project work different literature has been collected from different books, journal articles, internet and other relevant sources. Those reviews have been discussed about the different similarity, dissimilarity and relation of the project work.

Kiran Chauhan, Daniel K. Ebner, Katherine Tzou et all [16] assessment of bladder filling during prostate cancer radiation therapy with ultrasound and cone-beam CT. Prostate cancer patients undergoing external beam radiation therapy (EBRT) benefit from a full bladder to decrease bowel and bladder toxicity. Ultrasound may offer a proxy metric for evaluation, sparing CBCT dosing. Patients were prospectively enrolled pre-simulation from January 2017 to February 2018. Bladder volume was evaluated prior to RT using US daily and CBCT for three daily treatments and then weekly unless otherwise indicated. 29 patients completed median 40 days of RT, resulting in 478 CBCT and 1,099 US bladder volumes. 21 patients were treated to intact glands and 8 to the post

prostatectomy bed. Median patient age was 70 years. Bladder volume on CBCT and US positively correlated (r = 0.85), with average bladder volume for all patients of 162 mL versus 149 mL, respectively. Bladder volume during treatment was consistently lower than the volume at CT simulation (153 mL vs 194 mL, $p < 0.01$) and progressively declined during treatment. Patients older than 70 years presented with lower average bladder volumes than those < 70 years (122 mL vs 208 mL, respectively, $p<0.01$) and progressively declined during treatment. Patients older than 70 years presented with lower average bladder volumes than those < 70 years (122 mL vs 208 mL, respectively, $p < 0.01$). Patients with the highest agreement between CBCT and US (<10% variability) had higher average bladder volumes (192 mL vs 120 mL, $p=0.01$). US was found to be an accurate measure of bladder volume and may be used to monitor daily bladder volumes in patients being treated with radiation for prostate cancer.

The paper of **Deng-Yu Kuo, MD, MPH, Chen-Yang Hsu, MD, PhD et all [17]** bladder scan feedback method in predicting bladder filling for prostate radiotherapy: a prospective study showing an ultrasound based bladder volume scanner, has the potential to evaluate bladder volume during radiotherapy; thus, a prospective pilot study was initiated. There were 314 sets of data from 11 patients. The correlation coefficient between V^{BS} and V^{CT} was 0.87, where V^{BS} is the mean volume of 3 measurements by Bladder Scan and V^{CT} is the bladder volume derived from MVCT. The Bladder Scan feedback method resulted in a significant larger bladder volume than the conventional method, with a mean difference of 36.9 mL, when the failure was defined as V^{CT} <80% of planned volume, the Bladder Scan feedback method brought about a relative reduction in the failure rate with an odds ratio of 0.44 and an absolute reduction of 9.1%.

The accuracy of Bladder Scan was validated by MVCT in this study. The Bladder Scan feedback method can help patients fill the bladder adequately, with a larger bladder volume and a lower failure rate

Yat Man Tsang, Peter Hoskin [18] impact of bladder preparation protocols on post treatment toxicity in radiotherapy for localized prostate cancer patients This study compares the post radiotherapy related toxicity between the use of an empty and a full bladder preparation protocol in patients receiving radical radiotherapy for localized prostate cancer. 20 patients were included. There were significant differences in IGRT setup between the two groups. Although treatment times of the two groups were not significantly different, 5/200 (2.5%) ses sions were longer than 20 min in the full bladder group while this was not found in the other group. Associations between bladder preparation protocols and GI ($p = 1.0$) and GU ($p = 0.6$) toxicities were not statistically significant. The bladder size on planning CT was not significantly correlated to the GI ($R = 0.06$, $p = 0.8$) or GU ($R =0.27$, $p=0.3$) toxicity scores. No significant differences were found in OAR DVHs between patients with and without GI and GU toxicities. No grade 3/4 toxicities were reported. The empty bladder preparation approach has non-inferior acute and intermediate post RT GI and GUtoxicities in patients treated for localised prostate cancer with advanced radiotherapy techniques compared to the full bladder preparation.

Karin Braide, Jon Kindblom, Ulrika Lindencrona et all [19] value of a bladder-filling protocol for patients with prostate cancer who receive post-operative radiation: results from a prospective

clinical tria. This study compares two different strategies for maintaining a constant bladder volume during a course of postoperative radiotherapy in prostate cancer. In addition, we studied how changes in bladder filling affect the clinical target volume (CTV) and the coverage hereof. The estimated median bladder volumes were 120ml (95% CI: (93, 154)) and 123ml (95% CI: (98, 155)) in groups 1 and 2, respectively. The intra-individual variation in bladder volume, assessed as the standard deviation, was 64ml (95% CI: (46, 105)) in Group 1 and 61 (95% CI: (45, 94)) ml in Group 2. Increasing the bladder volume extended the bladder cranially while the caudal extension was almost constant. The correlation between bladder volume and V95%, bladder was 3.5ml per 100ml in group 1 and 1.2ml per 100ml in group 2 with no significant difference.

The intention to maintain a constant volume for the bladder is not fulfilled with either of the protocols in this study, and changes in bladder volumes does not seem to affect the position, or the coverage of the CTV.

S Sithamparam, R Ahmad, A Sabarudin, Z Othman, M Ismail [20] bladder filling variation during conformal radiotherapy for rectal cancer. This study also measured the association between bladder volume changes and diarrhea. Twenty two rectal cancer patients were recruited prospectively. Patients were planned for treatment with full bladder following departmental bladder filling protocol and the planning bladder volume was measured during CT-simulation. During radiotherapy, the bladder volume was measured weekly using cone-beam computed tomography (CBCT) and compared to planning bladder volume. Incidence and severity of diarrhea were recorded during the weekly patient review. There was a negative time trend for bladder volume throughout five weeks treatment. The mean bladder volume decreased 18 % from 123 mL (SD 54 mL) during CT- simulation to 101 mL (SD 71 mL) on the 5th week of radiotherapy, but the decrease is not statistically significant. However, there was a large variation of bladder volume within each patient during treatment. This study showed an association between changes of bladder volume and diarrhea (P = 0.045). In conclusion bladder volume reduced throughout radiotherapy treatment for conformal radiotherapy for rectal cancer and there was a large variation of bladder volume within patients.

Lori Holden, BSc, MRT(T), Julie Stanford, BKin, MRT(T), Laura D'Alimonte, BSc, MRT(T) et all [21] timing variability of bladder volumes in men receiving radiotherapy to the prostate. The primary objective of this study was to determine the time required for 95% of patients on a dose-escalated external-beam radiotherapy protocol to comfortably achieve a 180-cc bladder volume. In addition, measurement of patients' subjective assessment of urgency related to bladder filling was obtained to determine the feasibility of bladder-filling instructions.

The time required for 95% of patients to achieve a bladder volume of 180 cc was 75 and 57 minutes for groups 1 and 2, respectively (p=.03). Serum creatinine and use of bladder medications did not influence time to optimal bladder filling. Participants in group 2 reported moderate to severe urinary severity more frequently than participants in group 1.

Time to optimal bladder volume was highly varied and was faster with a 500-cc fluid preload. Customizing the wait times based on calculated ultrasound-based filling rates appears feasible in a busy radiotherapy department.

Heng, SP; Low, SH; Sivamany, K [22] influence of the bowel and bladder preparation protocol for radiotherapy of prostate cancer using kilo voltage cone beam CT. This study is to determine the influence of bladder and bowel preparation protocols on the dose-volume histograms (DVHs) of these organs using the cone beam computed tomography (CBCT)-based intensity modulated radiotherapy (IMRT) treatment planning for prostate cancer patients. The pelvic DVHs of 12 prostate cancer patients were studied using CBCT images obtained immediately before each treatment. Six patients had bladder and bowel preparation protocol whilst the other six patients were the control group. Contoured bladder and rectal volumes on CBCT images were compared with planning computed tomography. All patients were treated with IMRT with 7800 cGy in 39 fractions over 8 weeks. Compared with the patient with bladder preparation protocol, patients without bladder preparation instruction had higher bladder volume and dose variation. The maximum variation in bladder volume was as high as 98% in the control group. Without bowel preparation protocol, the rectal volumes were more variability. Owing to changes in rectal filling on the day of treatment, the maximum variation in rectal volume was as high as + 96%. With bowel preparation protocol, the maximum rectum volume variations were less than 25%. The changes in prostate target dose compared with planning dose were minimal as would be expected from positioning with daily image guidance and gold seed implanted.

Ali Hosni, Tara Rosewall, Timothy Craig, Vickie Kong, Andrew Bayley et all [23] effect of bowel preparation regime on inter-fraction rectal filling variation during image guided radiotherapy for prostate cancer. In the present study we investigated the impact of MoM on the interfraction dierences in rectal filling and assessed its tolerability. In the MoM group, the median volume of MoM taken by patients was 30 cm (range, 15–45 cm^3) in the first week and 15 cm (range, 0–30 cm) in the last week. The proportion of patients who took MoM decreased from 100% in the rst week to 60% in the last week (Fig. 2). Acute RTOG lower GI toxicity in MoM vs. non-MoM groups consisted of G2 diarrhea in 3 patients (7.5%) vs. 2 patients (5%) and G1 diarrhea in 21 patients (52.5%) vs. 7 patients (17.5%). In both groups, the onset of diarrhea was reported in the second week of RT, however with higher probability among patients who took MoM (the number of patients who had G1 diarrhea in the second week of RT in MoM vs. non MoM group was 9 [22.5%] vs. 5 [12.5%]). The addition of MoM to an anti atulence diet did not reduce interfraction variation in rectal lling and may cause diarrhea resulting in a substantial proportion of patients discontinuing its use. Simple dietary instructions appeared to be just as e ective at reducing interfraction rectal variability, and MoM should be omitted from routine use during prostate radiotherapy.

David J. Byun, Sarah Burleson, Daniel J. Gorovets, Lauren M. Jacobs, Zhigang Zhang, Margie Hunt, et all [24] strict bladder filling and rectal emptying during prostate SBRT: Does it make a dosimetric or clinical difference? To evaluate inter-fractional variations in bladder and rectum during prostate stereotactic body ra diation therapy (SBRT) and determine dosimetric and clinical consequences. Bladder and rectum volume reductions were observed throughout the

course of SBRT, with largest mean reductions of 86.9 mL (19.0%) for bladder and 6.4 mL (8.7%) for rectum noted at fraction #5 compared to Sim-CT ($p < 0.01$). Higher initial Sim-CT bladder volumes were predictive for greater reduction in absolute bladder volume during treatment ($\rho = -0.69$; $P < 0.01$). Over the course of SBRT, there was a small but significant increase in bladder mean dose ($+4.5 \pm 12.8\%$; $P < 0.01$) but no signi icant change in the D2cc ($+0.8 \pm 4.0\%$; $P = 0.28$). The mean bladder trigone displacement was in the anterior direction ($+4.02 \pm 6.59$ mm) with a corresponding decrease in mean trigone dose ($-3.6 \pm 9.6\%$; $P < 0.01$) and D2cc (-6.2 ± 15.6%, $P < 0.01$). There was a small but signi icant increase in mean rectal dose ($+7.0 \pm 12.9\%$, $P < 0.01$) but a decrease in rectal D2cc ($-2.2 \pm 10.1\%$; $P = 0.04$). No signi icant correlations were found between relative bladder volume changes, bladder trigone displacements, or rectum volume changes with rates of genitourinary or rectal toxicities

Alison Stillie, Tomas Kron, Chris Fox, Alan Herschtal [25] rectal filling at planning does not predict stability of the prostate gland during a course of radical radiotherapy if patients with large rectal filling are re-imaged. It has been suggested that large rectal filling is associated with an increased risk of prostate motion in radiotherapy. The aim of the present study was to determine if there is a correlation between rectal distension on planning computed tomography and the intrafraction and interfraction stability of the prostate gland during a course of radical radiotherapy for prostate cancer if a protocol was used to rescan patients with excessive rectal diameter during planning. The computed tomography planning scans of 89 patients with adenocarcinoma of the prostate treated with conformal radiotherapy were reviewed. All patients had three gold seed fiducial markers implanted into the prostate before planning computed tomography. About one in five patients had repeat computed tomography because their rectum was judged to be too large at the time of the first planning computed tomography. Rectal distension was assessed on planning computed tomography using outlines following European Organization for Research and Treatment of Cancer guidelines by measuring the rectal volume, the average cross-sectional area and the mean anterior-posterior diameter of the rectum. Daily kV images were obtained before and after treatment delivery to determine positional matching of the fiducial markers in the superior-inferior, anterior-posterior and right-left dimensions. In total, 2860 pre- and post treatment daily kV image pairs were obtained of 89 patients (average 32.1 image pairs per patient). The median rectal cross-sectional area was 7.3cm (2) (range 2.8-17.1), the median rectal volume was 54.8cm(3) (range 20.9-128.2), and the median anterior-posterior rectal diameter was 3.03cm (range 1.58-8.30). Unifactor linear regression models showed no statistically significant relationship between intra- and interfraction prostate stability and rectal volume on planning computed tomography. No statistically significant relationship between rectal distension on planning computed tomography and the intra- and interfraction stability of the prostate gland was identified if patients with a large rectal volume were rescanned for planning.

Lindsay S. Rowe, MD, Jeremy J. Mandia, MD, Kilian E. Salerno, MD, UmaT.Shankavaram, PhD, et all [26] guided radiation therapy for prostate cancer: results of a patterns of practice survey. Optimal management of patients with prostate cancer (PCa) to achieve bowel and bladder reproducibility for radiation therapy (RT) and the appropriate planning target volume (PTV) expansions for use with modern image guidance is uncertain. They surveyed American Society

of Radiation Oncology radiation oncologists to ascertain practice patterns for definitive PCa RT with respect to patient instructions and set up, daily image guidance, and subsequent PTV expansions. Of 190 respondents, 158 reported using conventional fractionation (CFx), 49 moderate hypofractionation (MHFx), and 61 stereotactic body radiation therapy (SBRT). Diet modifications during RT were advised by 84% of respondents, treatment with full bladder by 96%, and bowel instructions by 78%. Prescription of bowel medication was higher for respondents using SBRT (95.1%) versus those using CFx/MHFx (55.1%; 34.7%). The most common implantable device reported was fiducial markers, with increased use in SBRT (86.0%; 68.9%) versus CFx/MHFx. Cone beam computed tomography was the most common daily imaging technique across fractionation regimens. SBRT showed correlation between PTV margin expansions, fiducial marker use, and image guidance. Survey results indicate heterogeneity in treatment modality, dose, patient instructions, and PTV expansions used by radiation oncologists in the treatment of patients with PCa.

Mike Darud, BSc, RTT, ACT, Alison Giddings, BSc, RTT, ACT, Mira Keyes, MD, FRCP(C), Colleen McGahan, MSc, and Scott Tyldesely, MD, FRCP(C) [27] evaluation of a protocol to reduce rectal volume and prostate motion for external beam radiation therapy of the prostate. To assess the impact of an empty rectum protocol on pros tate movement at the time of radiation treatment delivery and the ef fect of such a protocol on bowel habits. Differences in isocenter position between planned field placement and actual field placement (before corrective action) were compared. Analysis of directional displacement showed a statis tically significant trend for the prostate to be more likely to be shifted inferior and left ($P < .001$ and $P= .0004$), regardless of use of ERP. Patients not on the ERP tended to have less displacement in the right direction ($P = .054$). Patients on the ERP had larger displacements in the inferior direction ($P = .02$) and tended to have larger displace ments over the course of treatment ($P = .04$). There was no statisti cally significant trend in the anteroposterior direction. Time lapse between last bowel movement and treatment was not significantly different between the two groups. The ERP group mean was 3.9 hours (95% CI 3.4, 4.4) compared with the non-ERP group mean of 3.8 hours (95% CI 3.4, 4.3). Time since last bowel movement did not have an impact on prostate position.Prostate displacement evaluated in this study is similar to that found by others. The empty rectum protocol used in this study has some impact on prostate displacement in some dimensions, but overall does not reduce the variability of prostate position or have an influence on patients' bowel habits.

Most of the literature has focused on either the bladder, rectum, or bowel. Additionally, many studies have divided patients into two groups, where the first group strictly follows the filling protocol, while the second group does not adhere to it. Some studies have reported that for bladder filling, a catheter is used to instill normal saline into the bladder before scanning. In most cases, research has been conducted on patients receiving treatment with either IMRT or VMAT techniques.

However, in our study, we have examined both the bladder and rectum simultaneously. Moreover, we have ensured that the same filling protocol is followed for both groups. Additionally, our study includes patients who have undergone treatment using three different techniques: 3DCRT, IMRT, and VMAT.



CHAPTER-2: THEORITICAL BACKGROUND

2.1 Overview of Prostate Cancer

Prostate cancer is the second most commonly diagnosed cancer in men worldwide, following lung cancer. It is characterized by the uncontrolled growth of cells in the prostate gland, which is responsible for producing seminal fluid. The disease often progresses slowly and may not present symptoms in its early stages. However, when symptoms do occur, they can include difficulty urinating, blood in urine or semen, and pelvic discomfort.

In 2022, there were approximately 1,467,854 new cases of prostate cancer globally, making it the fourth most common cancer worldwide [1]. The age-standardized incidence rate (ASR) varies significantly across different regions, influenced by factors such as genetics, lifestyle, and access to healthcare services.

In Bangladesh, prostate cancer is less prevalent compared to other regions. The age-standardized incidence rate is approximately 1.7 per 100,000 men, placing it among the lowest in Asia [2]. This lower incidence rate may be attributed to factors such as limited screening programs, underreporting, and differences in healthcare infrastructure. However, as awareness and healthcare access improve, the detection rates of prostate cancer are expected to rise.

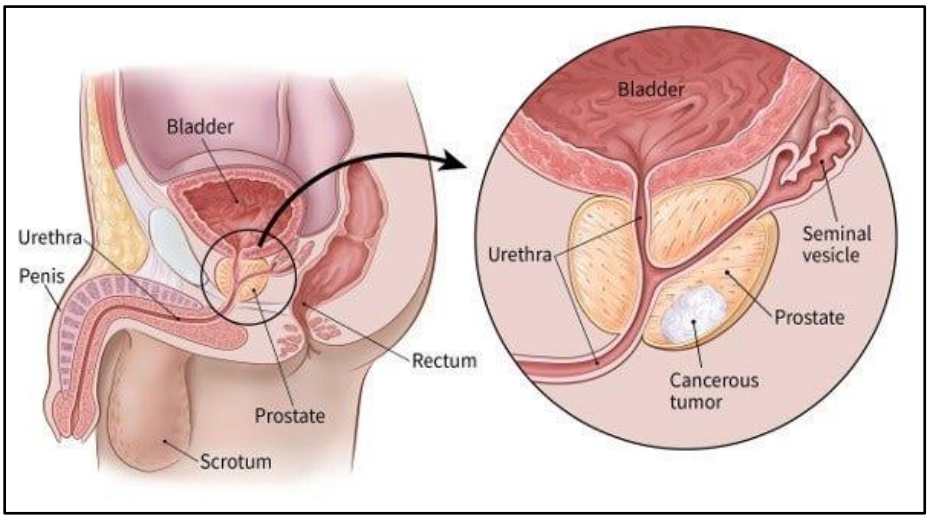


Figure 1: Prostate Cancer

2.1.1 Cause of Prostate Cancer

The exact cause of prostate cancer remains unknown, but several factors are believed to contribute to its development. These factors include genetic, environmental, and lifestyle influences, as well as certain biological processes.

1. **Age**: The risk of prostate cancer increases with age. It is rare in men under 40, but the majority of cases occur in men aged 50 and older. The median age of diagnosis is 66 years [1].

2. **Genetics and Family History**: Men with a family history of prostate cancer are at a higher risk. Certain inherited genetic mutations, such as BRCA1 and BRCA2 (which are also

linked to breast and ovarian cancers), can increase the risk of prostate cancer. First-degree relatives (father, brother) with prostate cancer significantly increase a man's risk [2].

3. **Ethnicity**: Prostate cancer is more common in African-American men compared to men of other ethnicities. This group also tends to be diagnosed at a younger age and may experience more aggressive forms of the disease [3].

4. **Hormones**: Androgens (male hormones like testosterone) play a role in the growth of prostate cancer cells. High levels of these hormones may promote the development and progression of prostate cancer [4].

5. **Diet and Lifestyle**: Diets high in red meat, high-fat dairy, and processed foods, as well as low intake of fruits and vegetables, may increase the risk of prostate cancer. A lack of physical activity and obesity have also been associated with a higher risk of developing the disease. On the other hand, a healthy diet with high fiber and low-fat intake may help reduce risk [5].

6. **Environmental Factors**: Exposure to certain chemicals, such as those in pesticides or other industrial chemicals, has been suggested as a possible risk factor for prostate cancer. Research is still ongoing to determine the extent of the environmental impact [6].

7. **Inflammation and Infection**: Chronic inflammation of the prostate (prostatitis) has been linked to an increased risk of developing prostate cancer. Some studies suggest that infections may play a role in triggering inflammation that may lead to cancer, although the exact mechanisms remain unclear [7].

8. **Gene Mutations**: Mutations in certain genes can predispose individuals to prostate cancer. Genetic mutations inherited from parents, or changes that happen over time, may lead to the uncontrolled growth of prostate cells, resulting in cancer [8].

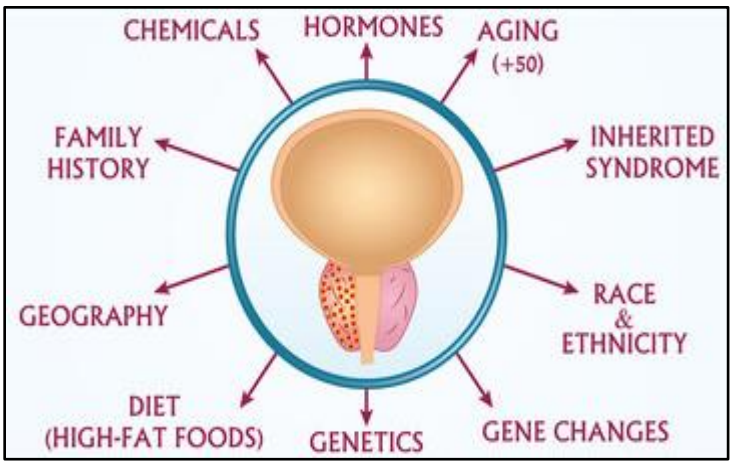


Figure 2: Causes of Prostate Cancer

2.1.2 Stages and Grades of Prostate Cancer

Prostate cancer is classified into stages and grades to help determine the extent of the disease and its aggressiveness, which is critical for treatment planning and prognosis.

Stages of Prostate Cancer

The stage of prostate cancer indicates how far the cancer has spread within the prostate and to other parts of the body. Staging is typically done using the TNM (Tumor, Node, Metastasis) system and may also involve imaging tests.

- **Stage I**:
  - Cancer is confined to a small area within the prostate. It is usually diagnosed through a biopsy and is not detectable via physical examination or imaging.
  - Tumor is classified as T1 (non-palpable) and the PSA levels are generally low.
- **Stage II**:
  - Cancer remains confined to the prostate but may be larger or involve more areas within the gland.
  - It may be palpable during a digital rectal exam (DRE). This stage is also divided into two sub-stages, IIA (where cancer is confined to one side of the prostate) and IIB (when it affects both sides of the prostate).
- **Stage III**:
  - Cancer has spread beyond the prostate to the seminal vesicles or nearby tissues but has not yet reached distant parts of the body.
  - Tumors in this stage are classified as T3 (cancer has spread through the outer layer of the prostate) and T4 (tumor has spread to nearby structures such as the bladder or rectum).
- **Stage IV**:
  - Cancer has metastasized to distant parts of the body such as the bones or lymph nodes. This is the most advanced stage of prostate cancer.
  - Stage IV is further categorized into IVA (spread to nearby lymph nodes or structures) and IVB (metastasis to distant organs such as bones or lungs).

Grades of Prostate Cancer (Gleason Score)

The grade of prostate cancer refers to how abnormal the cancer cells appear under a microscope. The Gleason grading system is commonly used to assess the aggressiveness of prostate cancer.

- **Gleason Score 6 or below**:
  - Considered low-grade prostate cancer. Cancer cells resemble normal prostate tissue and tend to grow slowly.
  - Usually associated with favorable outcomes and may be managed with active surveillance.
- **Gleason Score 7**:

Moderate-grade prostate cancer. Cancer cells have moderate abnormalities and grow faster than low-grade cancer.

This grade is divided into two categories:

  - Gleason 3+4 = Tumor has a primary grade of 3 and a secondary grade of 4.
  - Gleason 4+3 = Tumor has a primary grade of 4 and a secondary grade of 3, indicating a more aggressive form.
- **Gleason Score 8-10**:
  - High-grade prostate cancer. The cells appear very abnormal and tend to grow and spread more aggressively.
  - Prostate cancers with this score may require more aggressive treatments such as surgery, radiation, and/or chemotherapy.

Grade Group	Gleason Score	Combined Gleason Score	Aggressiveness degree
Grade Group 1	3 + 3	6	Low risk
Grade Group 2	3 + 4	7	Intermediate risk but closer to low risk
Grade Group 3	4 + 3	7	Intermediate risk but closer to high risk
Grade Group 4	4 + 4, 3 + 5, 5 + 3	8	High risk
Grade Group 5	4 + 5, 5 + 4, 5 + 5	9 and 10	High risk

These two different systems are mapped together using the table that was provided and simplified based on the NCCN guidelines version 4.2018 prostate cancer.[27]

Figure 3: Grades of Prostate Cancer

2.2 Overview of Cervical Cancer

Cervical cancer is a major global health concern, ranking as the fourth most common cancer in women worldwide. It originates in the cervix, the lower part of the uterus, and is primarily caused by persistent infection with high-risk types of the human papillomavirus (HPV). HPV infection leads to cellular changes that can eventually progress to cancer. Early stages of cervical cancer may be asymptomatic, but as the disease advances, symptoms such as abnormal vaginal bleeding, pelvic pain, and pain during intercourse may occur.

In 2020, there were an estimated 604,000 new cases of cervical cancer worldwide, with the highest incidence rates seen in low- and middle-income countries due to limited access to screening and vaccination programs [28]. The age-standardized incidence rate (ASR) of cervical cancer varies significantly across different regions, with the highest rates found in sub-Saharan Africa and South Asia [29]. In Bangladesh, cervical cancer is a leading cause of cancer-related deaths among women, with an age-standardized incidence rate of approximately 12.9 per 100,000 women [30]. This elevated rate may be due to low levels of HPV vaccination coverage and inadequate screening programs. However, with improvements in screening and vaccination, the burden of cervical cancer is expected to decrease in the coming years.

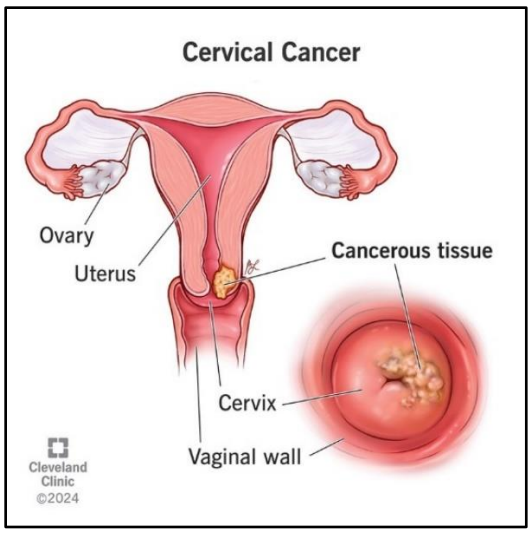


Figure 4: Cervical Cancer

2.2.1 Cause of Cervical Cancer

Cervical cancer is primarily caused by persistent infection with high-risk types of the human papillomavirus (HPV), a group of viruses commonly transmitted through sexual contact. HPV infection leads to changes in cervical cells, which can evolve into pre-cancerous lesions and, over time, potentially progress to invasive cancer. The most common high-risk HPV types linked to cervical cancer are HPV types 16 and 18, which together account for about 70% of all cervical cancer cases worldwide [31].

However, not all HPV infections result in cervical cancer. In most cases, the immune system clears the virus within a couple of years. Only persistent infections with high-risk HPV strains are likely to progress to cancer.

In addition to HPV infection, several other factors can increase the risk of developing cervical cancer:

1. **Smoking**: Smoking weakens the immune system, making it less effective in clearing HPV infections and increasing the likelihood of cervical cancer [32].
2. **Weakened immune system**: Conditions such as HIV/AIDS or immunosuppressive treatments (e.g., organ transplants) impair the immune response, increasing the risk of persistent HPV infection and cancer progression [33].
3. **Multiple sexual partners**: Having multiple sexual partners increases the likelihood of exposure to HPV [34].
4. **Early sexual activity**: Early initiation of sexual activity increases the risk of HPV exposure and infection [35].
5. **Oral contraceptive use**: Long-term use of oral contraceptives has been associated with an increased risk of cervical cancer, though the risk decreases after discontinuation of contraceptive use [36].
6. **History of sexually transmitted infections (STIs)**: Other STIs, such as chlamydia, can increase the risk of HPV infection and cervical cancer [37].

7. **Lack of screening**: Insufficient cervical cancer screening, such as Pap tests or HPV testing, may lead to late-stage detection of pre-cancerous lesions, allowing them to progress to cancer [38].

HPV vaccination is a preventive measure that significantly reduces the risk of HPV infection and cervical cancer. Regular screening through Pap smears and HPV testing can help detect precancerous changes early, facilitating early intervention and reducing the risk of cancer development.

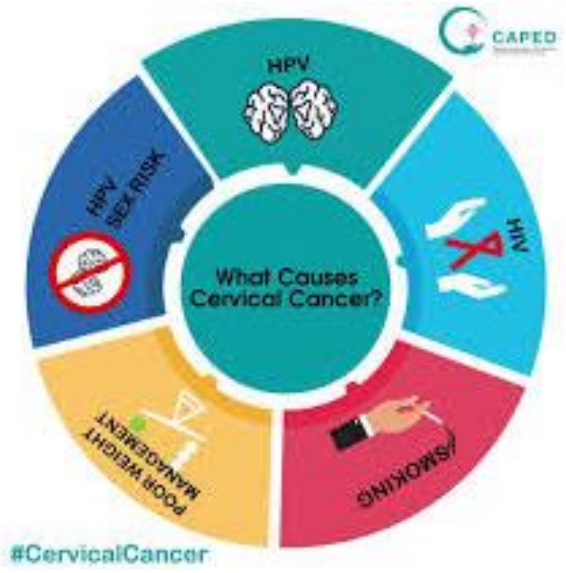


Figure 5: Causes of Cervical Cancer

2.2.2 Stages and Grades of Cervical Cancer

Cervical cancer is categorized into stages and grades to help determine the extent of the disease and guide treatment decisions. The staging system most commonly used is the FIGO (International Federation of Gynecology and Obstetrics) staging system, while the grading system refers to the appearance of the cancer cells under a microscope and helps predict the behavior of the tumor.

2.2.2.1 Stages of Cervical Cancer

1. **Stage 0 (Carcinoma in situ)**:
    - The cancer is confined to the surface layer of the cervix and has not spread into deeper tissues.
    - This stage is also known as pre-cancerous or localized abnormal cell growth.
2. **Stage I**:
    - **Stage IA**: The cancer is microscopic and can only be seen under a microscope. It is limited to the cervix.

- **Stage IB**: The cancer is larger than in stage IA and can be seen without a microscope, but it is still confined to the cervix.

3. **Stage II**:
   - The cancer has spread beyond the cervix but has not yet reached the pelvic wall or the lower part of the vagina.
   - **Stage IIA**: The cancer has spread to the upper two-thirds of the vagina or the tissues around the cervix but has not affected the pelvic wall.
   - **Stage IIB**: The cancer has spread to the tissues around the cervix but has not reached the pelvic wall.
4. **Stage III**:
   - The cancer has spread to the lower third of the vagina or to the pelvic wall. It may also cause kidney problems due to blockage of the ureters.
   - **Stage IIIA**: The cancer has spread to the lower third of the vagina but not to the pelvic wall.
   - **Stage IIIB**: The cancer has spread to the pelvic wall and/or caused the ureters to become blocked.
5. **Stage IV**:
   - The cancer has spread to distant organs.
   - **Stage IVA**: The cancer has spread to the bladder or rectum.
   - **Stage IVB**: The cancer has spread to distant organs such as the lungs, liver, or bones.

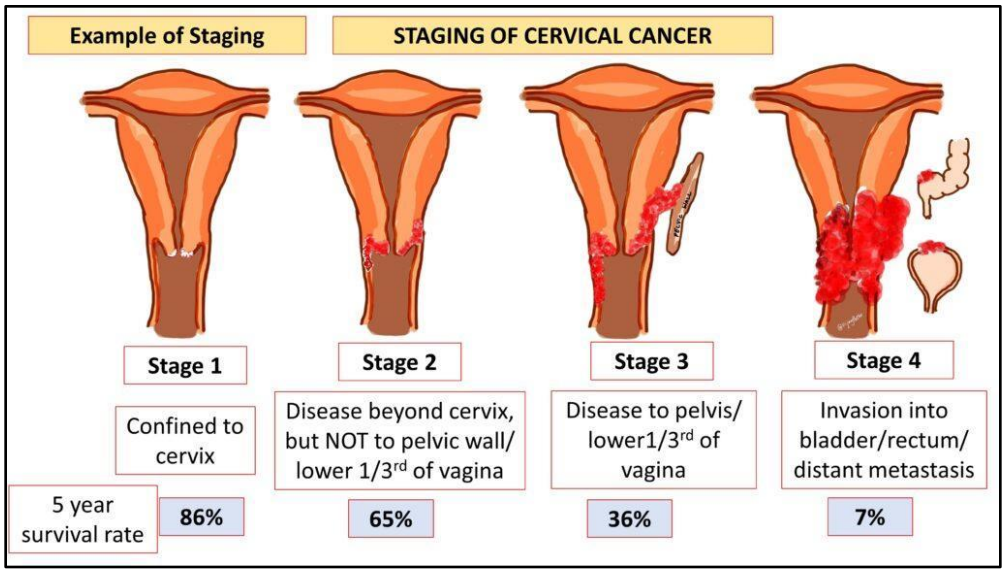


Figure 6: Stages of Cervical Cancer

2.2.2.2 Grades of Cervical Cancer

The grade of a cervical cancer refers to how abnormal the cancer cells look under a microscope, which can help predict how quickly the tumor is likely to grow and spread. The grades are:

1. **Grade 1 (Poorly differentiated)**:
    - The cancer cells look very abnormal and tend to grow and spread more aggressively.
    - This type of cancer is more likely to spread and cause recurrence.
2. **Grade 2 (Moderately differentiated)**:
    - The cancer cells look somewhat abnormal and are growing more quickly than grade 1.
    - This grade has an intermediate likelihood of spreading.
3. **Grade 3 (Well-differentiated)**:
    - The cancer cells look somewhat like normal cells and tend to grow slowly.
    - This type of cancer is less likely to spread.

2.3 OARs for Pelvic region

Organs at Risk (OARs) in the pelvic region are critical structures that could be damaged by radiation or other therapies during treatment for various pelvic cancers (e.g., prostate, cervical, rectal cancers). Protecting these organs is vital to minimize side effects and preserve function. Below are the OARs for the pelvic region, with distinctions for males and females.

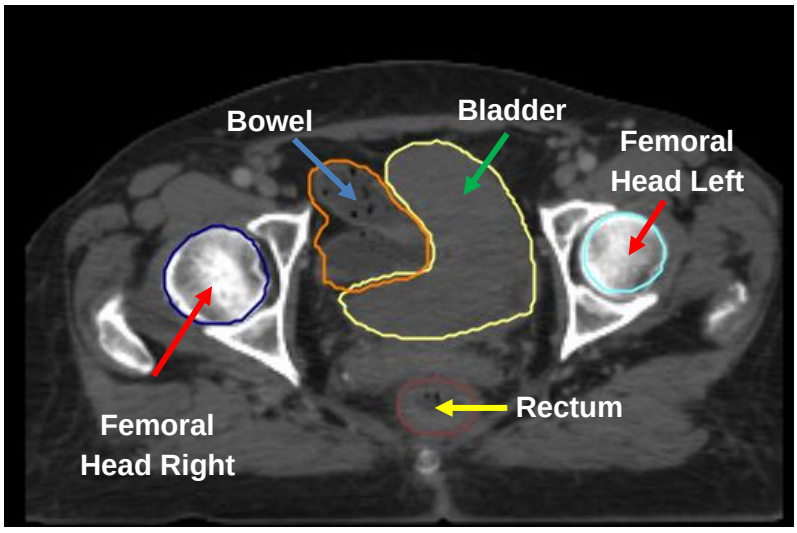


Figure 7: Organs at Risk (OARs)

2.3.1 Common OARs for the Pelvic Region (Both Male and Female)

- **Bladder**: The bladder is situated in the anterior part of the pelvic cavity. It is at risk of radiation damage, leading to symptoms like cystitis, urinary frequency, urgency, or incontinence.
- **Rectum**: The rectum is located posteriorly in the pelvic cavity and is a significant OAR during treatments aimed at pelvic tumors (e.g., prostate or gynecological cancers).
- **Small Intestine**: The small intestine can extend into the pelvic cavity. Radiation may cause complications such as nausea, vomiting, and chronic gastrointestinal symptoms.
- **Femoral Heads (Both Left and Right)**: The femoral heads are the ball-and-socket joints in the pelvis. They are critical for mobility and can be affected by radiation, leading to conditions like osteonecrosis (bone death).

2.3.2 OARs for the Pelvic Region in Females

- **Uterus:** The uterus is a central organ for fertility and reproduction. Radiation can damage the uterine lining and lead to infertility or other reproductive issues.
- **Ovaries:**The ovaries are located laterally to the uterus and produce eggs and hormones (estrogen, progesterone). They are highly sensitive to radiation.
- **Bladder Neck and Urethra:** Located at the junction of the bladder and urethra, the bladder neck is essential for urinary function. Radiation can lead to symptoms such as urinary incontinence or retention.
- **Fallopian Tubes:** The fallopian tubes are critical for female fertility. Radiation may damage these structures, leading to infertility.

2.3.3 OARs for the Pelvic Region in Males

- **Prostate:** The prostate gland is a key organ in male pelvic anatomy, involved in sperm production and hormone regulation. Radiotherapy targeting the prostate may inadvertently expose nearby organs.
- **Seminal Vesicles:** The seminal vesicles are responsible for producing semen. Radiation can affect their function, leading to infertility.
- **Testes:** The testes, located in the scrotum, are responsible for producing sperm and hormones. They are sensitive to radiation.
- **Penis:** The penis is an external organ that can be affected by radiation, particularly when treating prostate or pelvic cancers. Radiation may cause erectile dysfunction.

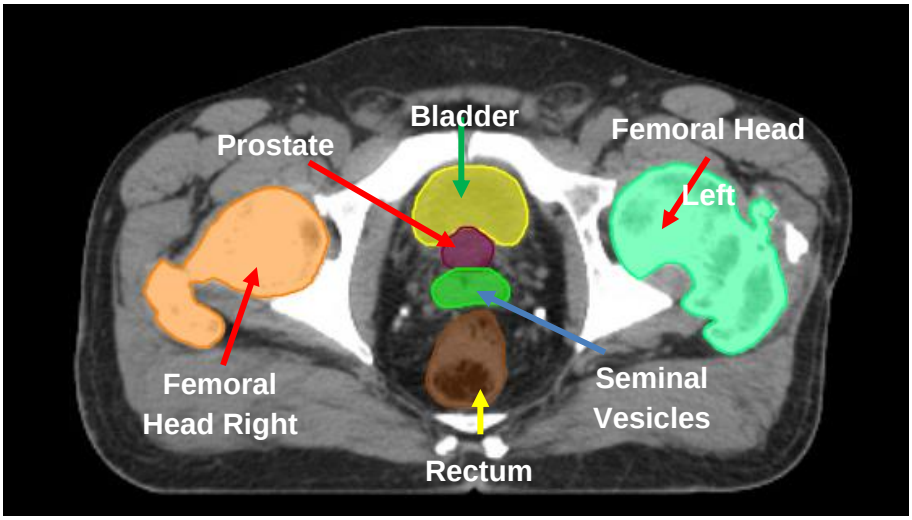


Figure 8: OARs for Pelvic region in Male

2.4 Geometric and Dosimetric Concepts in Radiotherapy

Geometric and Dosimetric concepts in radiotherapy are essential for achieving the goal of delivering the prescribed dose to the tumor while minimizing radiation exposure to surrounding healthy tissues and organs at risk (OARs). These concepts ensure precision and effectiveness in radiation therapy, especially in the treatment of complex and critical regions like the pelvic area.

2.4.1 Geometric Concepts in Radiotherapy

1. **Target Volume Delineation**:
    - **Gross Tumor Volume (GTV)**: This is the visible or palpable extent of the tumor, as seen on imaging such as CT, MRI, or PET scans.
    - **Clinical Target Volume (CTV)**: This includes the GTV and any surrounding areas that are at high risk of containing undetected cancer cells. It accounts for microscopic disease spread beyond the GTV.
    - **Planning Target Volume (PTV)**: The PTV accounts for the CTV along with any uncertainties in patient positioning, organ motion, and setup errors during radiation treatment. This ensures that the tumor receives an adequate dose, even if there is some variation in patient positioning.
2. **Patient Positioning and Immobilization**: Accurate **patient positioning** is crucial in ensuring that radiation is delivered to the target volume and not to healthy tissues. Small movements can lead to significant changes in dose distribution.
    - **Immobilization devices** (e.g., molds, masks, and headrests) are used to keep the patient in a reproducible position throughout the treatment, ensuring consistent alignment across treatment sessions.
    - **Image-guided radiotherapy (IGRT)** uses real-time imaging (e.g., X-ray, CT, MRI) to verify the position of the patient and the tumor before each treatment session, making adjustments as needed.

3. **Motion Management**: Many tumors, especially in the pelvic region, may move due to organ motion (e.g., bladder and rectal filling, respiratory motion).
    - **Motion management techniques** include strategies like **breath-hold techniques** (used for lung or upper abdominal tumors), **gating systems** (used for tumors affected by breathing), and **internal fiducial markers** to track the tumor's motion.
    - For pelvic cancers, **daily imaging** and techniques like **cone-beam CT (CBCT)** are used to assess and correct any motion before treatment.
4. **Treatment Planning and Optimization**:
    - **Treatment planning** involves defining the treatment fields (beam geometry) and the optimal dose distribution for delivering the prescribed dose to the target volume while minimizing radiation to normal tissues.
    - **Optimization** is the process of adjusting various treatment parameters (beam angles, intensities, and number of beams) to achieve the best dose distribution.
    - **Inverse planning** (used in techniques like IMRT and VMAT) involves defining the dose distribution and allowing the planning system to determine the optimal beam arrangement to meet these goals.

2.4.2 Dosimetric Concepts in Radiotherapy

1. **Dose Distribution**: It refers to how radiation is distributed within the treatment area. It is a critical concept in ensuring that the prescribed dose is delivered to the tumor while sparing surrounding healthy tissues.
    - **Isodose lines** are used to visualize the dose distribution in the treated volume. These lines represent regions receiving the same dose and help in assessing the adequacy of the dose delivered to the tumor.
2. **Radiation Dose and Fractionation**:
    - **Dose**: The amount of radiation delivered to a specific area is measured in **Gray (Gy)**, which represents the absorption of 1 joule of radiation energy by 1 kilogram of matter.
    - **Fractionation**: Radiation is delivered in multiple smaller doses (fractions) over a period of time, typically 5 days a week, to allow normal tissues to recover while keeping the tumor under control. A typical treatment fraction for pelvic cancers is between **1.8 and 2 Gy** per day.
    - **Hypofractionation** involves giving larger doses over fewer sessions, which is commonly used in specific cases, such as for prostate cancer or palliative treatments.
3. **Linear Energy Transfer (LET)**:
    - **LET** refers to the amount of energy deposited by a radiation particle per unit distance traveled through tissue. Higher LET radiation (such as protons or carbon ions) deposits more energy in the tissue, causing more damage to the DNA.

- **High-LET radiation** (e.g., proton therapy) has an advantage in certain clinical situations where precise targeting is necessary, and reducing radiation to surrounding tissues is critical.

4. **Radiobiology and Tumor Control**:
    - **Tumor Control Probability (TCP)** is a mathematical model used to predict the likelihood of tumor control following radiation treatment. It takes into account the dose delivered, the fractionation scheme, and the radiosensitivity of the tumor.
    - **Normal Tissue Complication Probability (NTCP)** is a model used to estimate the risk of complications in surrounding healthy tissues based on the radiation dose received.
    - **Biological Effective Dose (BED)**: This concept combines both dose and fractionation effects. It is used to compare the biological effects of different treatment regimens, especially when using hypofractionation or other non-standard regimens.
5. **Radiation Quality and Dose Calculation**:
    - **Radiation quality** refers to the type of radiation used, such as photon, electron, or proton therapy. Different radiation modalities have different penetration depths and dose distributions.
    - **Dose calculation** is done using advanced treatment planning systems (TPS), which simulate how radiation interacts with tissues. These systems use models like **Monte Carlo simulations** to predict dose distributions with high accuracy.
6. **Plan Evaluation**:
    - Treatment plans are evaluated using various **dosimetric metrics** to ensure that the dose distribution meets the prescribed treatment goals:
        - **Conformity Index (CI)**: Measures how well the radiation volume conforms to the tumor shape. A higher CI indicates better targeting.
        - **Homogeneity Index (HI)**: Indicates the uniformity of the dose within the target volume. A lower HI suggests a more uniform dose distribution.
7. **Dose-Volume Histograms (DVH)**:
    - **DVH** is a graphical representation that shows the volume of tissue receiving a specific dose of radiation. It is used to assess the trade-offs between tumor coverage and sparing normal tissues. DVH analysis helps in comparing different treatment plans and ensuring the optimal balance between efficacy and safety.

2.5 Bladder and Rectum Filling

Bladder and rectum filling are critical factors in **radiotherapy for pelvic cancers**. The volume and positioning of these organs during radiation treatment can significantly influence the **dose distribution** to the tumor and surrounding healthy tissues. Proper management of bladder and

rectum filling helps in **minimizing radiation exposure** to these organs and ensures that the target tumor receives the prescribed dose effectively. Inaccurate or inconsistent filling can lead to **increased toxicity** or decreased tumor control.

2.5.1 Bladder Filling

1. **Importance of Bladder Filling**:
    - The bladder is located anteriorly in the pelvic cavity, and its volume and position can affect the dose delivered to surrounding organs, such as the **rectum**, **femoral heads**, and the **reproductive organs**.
    - When the bladder is **full**, it displaces surrounding tissues, including the rectum, and moves the target area (e.g., prostate, cervix) into a more favorable position for radiation. A full bladder also reduces the amount of radiation delivered to the rectum, as it creates a separation between the radiation beams and the rectum.
    - In **prostate cancer** radiotherapy, a full bladder is used to push the small intestine and rectum away from the radiation field, reducing the risk of radiation-induced damage to these tissues.
2. **Techniques for Managing Bladder Filling**:
    - **Standardization of bladder volume**: Patients are typically asked to **drink a specific amount of water** (usually around **500–700 mL**) about **30 minutes** before their treatment session to ensure consistent bladder filling. This helps to achieve reproducible positioning and consistent organ alignment.
    - **Image-guided radiotherapy (IGRT)**: Imaging techniques such as **cone-beam CT (CBCT)** or **X-ray** are often used before treatment to verify bladder volume and positioning. If needed, adjustments are made to optimize bladder filling.
    - **Bladder-sparing techniques**: In some cases, the bladder is partially emptied or controlled to minimize radiation exposure to organs at risk, such as the bladder itself, rectum, and femoral heads.
3. **Effects of Bladder Filling on Dose Distribution**:
    - **Too full bladder**: A bladder that is excessively full can lead to displacement of surrounding organs and cause discomfort or other complications.
    - **Underfilled bladder**: Insufficient bladder filling can cause the bladder to be positioned closer to the target or critical structures like the rectum, which could lead to a less effective treatment plan and increased risk of side effects.

2.5.2 Rectum Filling

1. **Importance of Rectum Filling**:
    - The rectum is located posteriorly and can be affected by radiation when treating cancers in the pelvic region, such as **prostate**, **cervical**, or **rectal cancers**. Proper

rectum filling helps prevent excessive radiation exposure to the organ and surrounding healthy tissues, including the **bladder** and **small intestine**.

- During treatment, the rectum can change its shape and position depending on the amount of **gas and stool** it contains. A full rectum can push critical structures further from the radiation field and reduce the risk of damage to healthy tissues.

2. **Techniques for Managing Rectum Filling**:

   - **Emptying the rectum before treatment**: Patients are often asked to **empty their bowels** before radiotherapy sessions to ensure that the rectum is as empty as possible. This reduces the chances of variability in rectal positioning.
   - **Rectal balloon devices**: In some cases, a **rectal balloon** may be inserted to standardize the rectal volume and positioning. The balloon creates space between the rectum and nearby tissues (e.g., prostate, cervix) to help maintain the correct dose distribution and minimize toxicity to adjacent structures.
   - **Dietary management**: Patients may be advised to avoid foods that can cause bloating or excessive gas production before radiotherapy sessions to prevent variations in rectal filling.

3. **Effects of Rectum Filling on Dose Distribution**:

   - **Full rectum**: A full rectum can push the target tumor into a more favorable position, potentially increasing the dose delivered to the tumor. However, a very full rectum can also bring the rectum itself closer to the radiation field, increasing the risk of radiation-induced damage.
   - **Underfilled rectum**: An empty or partially empty rectum can lead to variations in its position and shape, which could affect the treatment plan's accuracy. It may bring the rectum and other organs at risk closer to the radiation field.

2.5.3 Managing Both Bladder and Rectum Filling Together

1. **Coordination of Bladder and Rectum Filling**:

   - In pelvic radiotherapy, managing the **bladder and rectum filling simultaneously** is essential for achieving the optimal dose distribution. Patients are instructed to achieve a balance between filling the bladder and emptying the rectum to minimize the overlap of radiation fields and critical structures.
   - **Simultaneous optimization** of bladder and rectum filling ensures that both the tumor receives adequate dose coverage and the surrounding healthy tissues, including the bladder and rectum, are spared as much as possible.

2. **Advanced Techniques**:

   - **Intensity-Modulated Radiation Therapy (IMRT)** and **Image-Guided Radiation Therapy (IGRT)** help in better controlling bladder and rectum positioning. These

techniques provide the flexibility to adjust treatment plans based on daily imaging and patient positioning.

3. **Patient Education**:
    - Clear instructions about bladder and rectum filling before each session are crucial. Patients are educated about the need for consistent hydration and bowel preparation to achieve reproducible treatment conditions and reduce uncertainties in the treatment process.

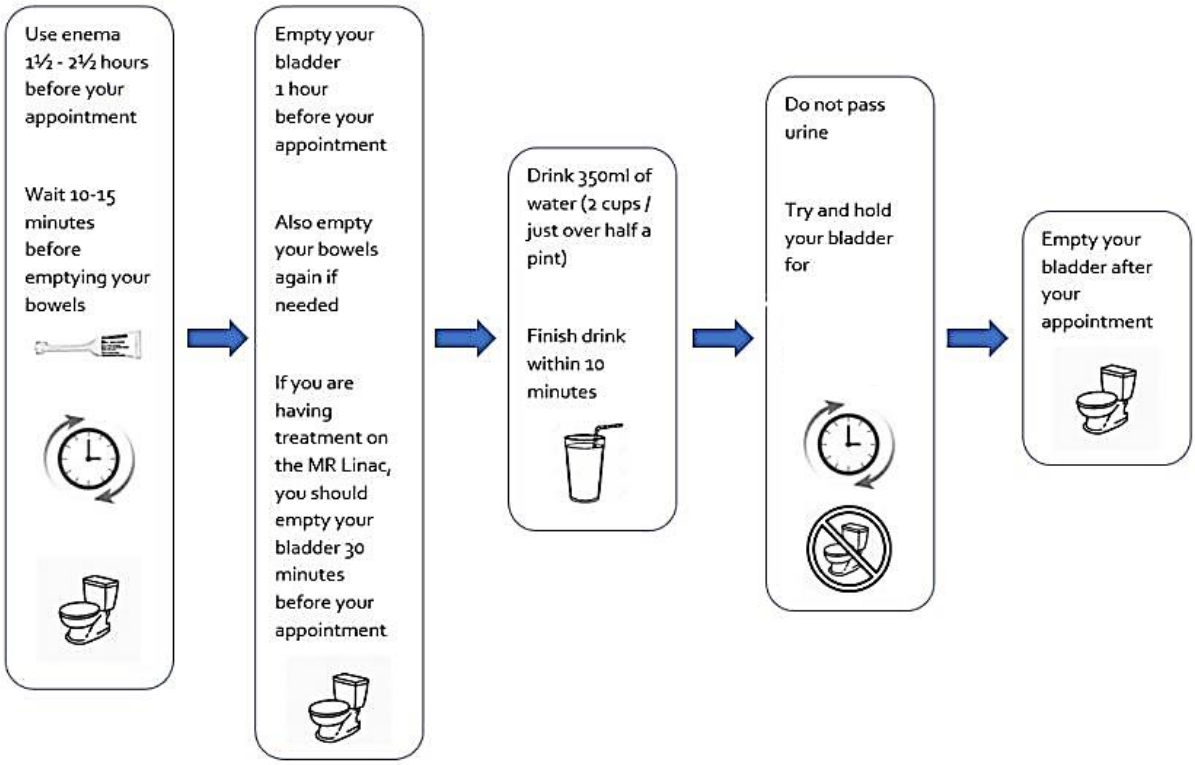


Figure 9: Bladder and Rectum Filling Protocol

2.6 Dice Similarity Coefficient (DSC)

The **Dice Similarity Coefficient (DSC)** is commonly used in **medical image analysis** to evaluate the accuracy of **image segmentation**. In medical imaging, DSC is particularly useful for assessing the similarity between **predicted segmentation** (e.g., from an automated algorithm) and **ground truth segmentation** (e.g., manually annotated by a radiologist). The DSC provides a numerical value between 0 and 1, where a value closer to 1 indicates a higher degree of similarity between the two segmentations.

2.6.1 Formula for the Dice Similarity Coefficient (DSC)

$$DSC = \frac{2|\mathbf{A} \cap \mathbf{B}|}{|\mathbf{A}| + |\mathbf{B}|}$$



Figure 10: Dice Similarity Coefficient Image

Where:

- ***A*** is the **set of pixels** corresponding to the predicted region (e.g., pixels identified as the tumor by an algorithm).
- ***B*** is the **set of pixels** corresponding to the ground truth (e.g., the pixels marked as the tumor by a radiologist).
- **|*A*|** and **|*B*|** are the total number of pixels in the predicted region and the ground truth region, respectively.
- **|*A*∩*B*|** is the number of overlapping pixels between the predicted region and the ground truth region.

2.6.2 Interpretation of the DSC Score

- **DSC = 1**: Perfect overlap, meaning the predicted segmentation matches the ground truth exactly. This indicates perfect accuracy.
- **DSC = 0**: No overlap, meaning the predicted segmentation and the ground truth do not intersect at all. This indicates complete failure in segmentation.
- **0 < DSC < 1**: Partial overlap, with higher values indicating better segmentation performance.

2.7 Housdoff Distance (HD)

The **Hausdorff Distance** (HD) is a commonly used metric in **image analysis** to compare two sets or shapes, particularly in tasks such as **image segmentation**, **shape matching**, and **boundary detection**. It is particularly useful for evaluating the similarity between **segmented regions** (e.g., tumor boundaries in medical imaging) and **ground truth regions** (manually annotated by experts). The Hausdorff Distance measures the **greatest discrepancy** between two sets of points, and thus it is sensitive to outliers or extreme deviations.

2.7.1 Formula for Hausdorff Distance (HD)

$$d_H(A,B) = \max\left(\sup_{a\in A}\inf_{b\in B}\|a-b\|, \sup_{b\in B}\inf_{a\in A}\|b-a\|\right)$$

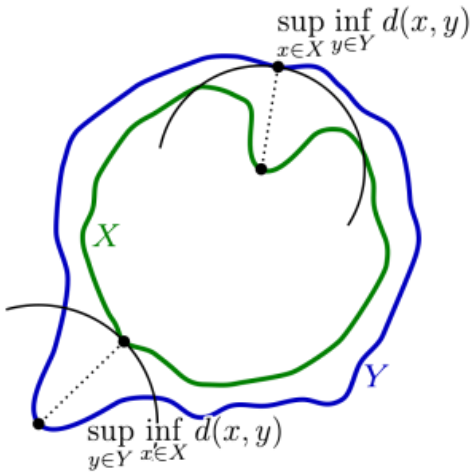


Figure 11: Hausdoff Distance Image

Where:

- A and B are the sets representing the points (pixels) of the predicted and ground truth boundaries, respectively.
- $\|a-b\|$ is the distance between two points aaa and bbb, typically the **Euclidean distance** in the context of image analysis.
- The first part calculates the greatest minimum distance from each point in set A to the nearest point in set B.
- The second part calculates the greatest minimum distance from each point in set B to the nearest point in set A.

2.7.2 Interpretation of Hausdorff Distance in Imaging

- **Hausdorff Distance = 0**: This indicates perfect overlap between the two sets, meaning the predicted segmentation exactly matches the ground truth.
- **Hausdorff Distance > 0**: The larger the Hausdorff distance, the greater the dissimilarity between the predicted and ground truth sets. The Hausdorff distance is sensitive to the worst-case discrepancy (i.e., the largest difference between points).
    - A small Hausdorff distance indicates that the boundaries are close, even if some minor discrepancies exist.
    - A large Hausdorff distance suggests significant mismatches, where parts of the predicted boundary are far from the ground truth.

2.8 Cone Beam Computed Tomography CBCT

Cone beam computed tomography (CBCT) is a radiographic imaging method that allows accurate, three-dimensional (3D) imaging of hard tissue structures. CBCT is the most significant

among the medical diagnostic imaging modalities that have emerged recently. This imaging modality is capable of providing sub-millimeter resolution (2-line pair/mm) images of higher diagnostic quality, with shorter scanning times (~60 s).

In **CBCT**, the X-ray tube rotates around the patient, capturing multiple 2D images from different angles. These images are then reconstructed into 3D images using computer algorithms. The **cone-shaped** X-ray beam allows for rapid acquisition of 3D data in a single rotation, as opposed to the fan-shaped beam used in traditional CT, which requires multiple rotations to collect enough data for 3D reconstruction.

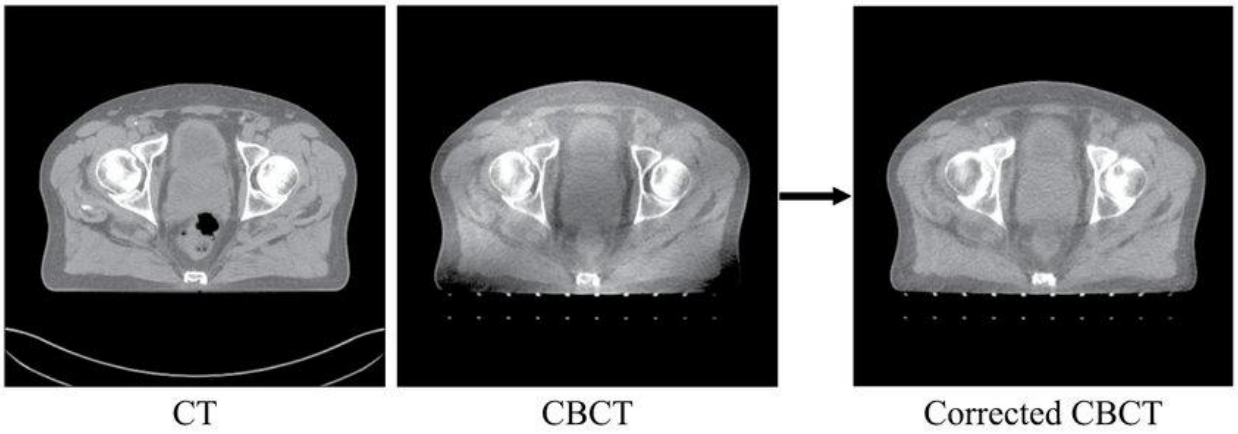


Figure 12: CT and CBCT Image

2.9 Radiation Therapy for Pelvic Cancer

In **radiation therapy** for pelvic cancers, the **dose** (amount of radiation) and **fractionation** (number of treatments) are carefully planned to maximize tumor control while minimizing harm to surrounding healthy tissues. The specific dose and fractionation schedule depend on the type of cancer, its stage, and the patient's overall health.

2.9.1 Types of Radiation Therapy for Pelvic Cancer

1. **External Beam Radiation Therapy (EBRT)**:
    - **Description**: EBRT is the most common form of radiation therapy used to treat pelvic cancers. A machine (linear accelerator) directs high-energy radiation beams from outside the body to the tumor site. It targets the cancerous cells while minimizing exposure to surrounding healthy tissues.
    - **Use**: EBRT is typically used for prostate cancer, cervical cancer, rectal cancer, and other pelvic malignancies. It can be used before surgery to shrink the tumor or after surgery to kill any remaining cancer cells.
    - **Technique**: Modern technologies like **Intensity-Modulated Radiation Therapy (IMRT)**, **Volumetric Modulated Arc Therapy (VMAT)** and **Image-Guided Radiation Therapy (IGRT)** enable precise targeting of the tumor and reduce radiation exposure to nearby organs at risk (OARs) such as the bladder, rectum, and femoral heads.

2. **Brachytherapy** (Internal Radiation Therapy):
    - Brachytherapy involves placing a radioactive source directly inside or very close to the tumor. In pelvic cancers, the radioactive material may be placed in the prostate (for prostate cancer) or in the vagina/uterus (for gynecological cancers like cervical or endometrial cancer).
    - **Use**: This type of radiation allows for higher doses to be delivered to the tumor while sparing surrounding healthy tissue.
    - **Technique**: Brachytherapy can be performed with temporary or permanent implants. For example, in prostate cancer, radioactive seeds are implanted directly into the prostate gland.

3. **Proton Therapy**:
    - **Description**: Proton therapy is a type of radiation treatment that uses protons (charged particles) instead of X-rays to treat cancer. Protons have mass and can deliver energy precisely to the tumor with minimal exit dose, which reduces the amount of radiation to surrounding healthy tissues.
    - **Use**: Proton therapy is particularly beneficial for tumors located near sensitive structures, such as the bladder, rectum, and femoral heads, making it an ideal treatment for pelvic cancers.
    - **Technique**: Proton therapy is less widely available than traditional radiation therapy, but its precision makes it an option for certain pelvic cancers, especially in pediatric cases or cases where normal tissue sparing is crucial.

4. **Stereotactic Body Radiation Therapy (SBRT)**:
    - **Description**: SBRT is a more advanced form of EBRT that delivers very high doses of radiation to a small, well-defined tumor in a limited number of treatments (usually 3–5 sessions). This method is particularly useful for tumors in areas that are difficult to treat with conventional radiation techniques.
    - **Use**: SBRT can be used for treating primary or metastatic tumors in the pelvic region, especially in cases where traditional radiotherapy may not be as effective.
    - **Technique**: SBRT requires precise imaging and positioning to ensure the radiation is accurately delivered to the tumor while sparing surrounding healthy tissues.

2.9.2 Dose and Fractionation in Radiation Therapy for Pelvic Cancer

- **External beam radiation:** this type of radiation delivers a beam of high-energy x-rays to a patient's tumor from outside the body. Radiation oncologists at Johns Hopkins may use intensity-modulated radiation therapy (IMRT), which delivers targeted doses of radiation in a pattern to match the shape of the tumor; or image-guided radiation therapy (IGRT), which uses images taken before and during the radiation delivery to guide the treatment. These techniques help the radiation oncologists focus radiation only on the tumor itself, and not healthy surrounding tissue.

- **Brachytherapy:** Used less often than external beam radiation to treat rectal cancer, this radiation treatment is a targeted, high-dose treatment that is delivered directly in or near the tumor. Johns Hopkins is one of the only hospitals in the country to explore endorectal high-dose rate brachytherapy for rectal cancer, and a team of Johns Hopkins scientists are leading a clinical trial to study this further.

- **Intraoperative radiation therapy (IORT):** Intraoperative radiation therapy is delivered directly to a tumor site after the tumor has been removed during surgery. The treatment comes from a radioactive source fed through wires that are placed on the tumor. The technique may be used for a rectal tumor that has infiltrated muscles or bones in the pelvis, or in cases where rectal cancer has returned after a tumor was previously treated by radiation or surgery. Johns Hopkins radiation oncologists specialize in high-dose intraoperative radiation therapy for recurrent rectal cancer patients.

2.10 External beam radiation therapy (EBRT)

External-beam radiation therapy (EBRT) delivers a highly focused beam of radiation directly to the tumor. EBRT is the most common type of radiation therapy for rectal cancer treatment. It is proven to reduce the chance that the tumor will regrow in the pelvis after surgery. Newer radiation techniques can reduce radiation exposure to the pelvic bones, decreasing the risk of osteoporosis [11].

EBRT using a special X-ray machine called a linear accelerator, it delivers high-energy rays to tumors. The machine delivers radiation from any angle and shapes radiation beams to the contour of the tumor. The machine moves around the body without touching the patient, aiming radiation at the cancer. Some types of focused EBRT target a tumor with higher, more precise doses of radiation, while reducing damage to healthy tissue and nearby organs. As a result, modern EBRT may help reduce the risk of side effects associated with traditional radiation treatment [12].



Figure 13: EBRT machine

2.10.1 Conventional Radiation Therapy

Refers to the old techniques of radiation therapy where treatments would be planned by defining a limited number of beams with the boundaries delineated on orthogonal x-rays of the patient. It has been largely replaced by other highly conformal external beam radiation therapies, which use CT images to plan the treatment.

Beam shaping was limited and typically simple square or rectangular beams were used. A typical beam arrangement is the four-field box. Due to the low conformity of these treatments, adjacent tissues/organs often fall into the high dose region resulting in treatment side effects. Also, the amount of radiation delivered to the targeted tumor is usually not adequate resulting in less effective treatment.

2.10. 2 Intensity-modulated Radiation Therapy (IMRT)

IMRT is an advanced mode of high-precision radiotherapy that uses computer-controlled linear accelerators to deliver precise radiation doses to a malignant tumor or specific areas within the tumor, IMRT allows for the radiation dose to conform more precisely to the three-dimensional (3-D) shape of the tumor by modulating or controlling-the intensity of the radiation beam in multiple small volumes. IMRT also allows higher radiation doses to be focused to regions within the tumor while minimizing the dose to surrounding normal critical structures. Treatment is carefully planned by using 3-D computed tomography (CT) or magnetic resonance (MRI) images of the patient in conjunction with computerized dose calculations to determine the dose intensity pattern that will best conform to the tumor shape. Typically, combinations of multiple intensity-modulated fields coming from different beam directions produce a custom-tailored radiation dose that maximizes tumor dose while also minimizing the dose to adjacent normal tissues. Because the ratio of normal tissue dose to tumor dose is reduced to a minimum with the IMRT approach, higher and more effective radiation doses can safely be delivered to tumors with fewer side effects compared with conventional radiotherapy techniques. IMRT also has the potential to reduce treatment toxicity, even when doses are not increased. Due to its complexity, IMRT does require slightly longer daily treatment times and additional planning and safety checks before the patient can start the treatment than conventional radiotherapy.

Currently, IMRT is being used most extensively to treat cancers of the prostate, head and neck, and central nervous system. IMRT may also be beneficial for treating pediatric malignancies. Radiation therapy, including IMRT, stops cancer cells from dividing and growing, thus slowing or stopping tumor growth. In many cases, radiation therapy is capable of killing all of the cancer cells, thus shrinking or eliminating tumors [16].

2.10. 3 Volumetric Modulated Arc Therapy (VMAT)

VMAT is a new type of IMRT technique. Volumetric modulated arc therapy (VMAT) is a novel radiation technique, which can achieve highly conformal dose distributions with improved target volume coverage and sparing of normal tissues compared with conventional radiotherapy techniques. It is a next generation arc therapy technique that has established new standards for Radiation Therapy by reducing the treatment delivery time to few minutes. This reduces the potential for patient motion and enables better patient comfort and tolerance. A further development of IMRT is volumetric modulated arc therapy (VMAT), with which radiation is delivered as the gantry rotates around the patient. There is a continuous modulation of the rotation speed of the gantry, the MLC field shape and the delivered dose rate. The treatment is given as one or more arcs, where each arc can cover up to 360 and go either clockwise or counterclockwise. The term dual arc denotes the delivery of two oppositely oriented arcs covering the same gantry angles, where the optimization seeks to minimize the leaf travel. VMAT also has the potential to offer additional advantages, such as reduced treatment delivery time compared with conventional static field intensity modulated radiotherapy (IMRT). The clinical worldwide use of VMAT is increasing significantly. Currently the majority of published data on VMAT are limited

to planning and feasibility studies, although there is emerging clinical outcome data in several tumor sites. [16].

2.11 Brachytherapy

Brachytherapy is a form of localized radiation therapy where radioactive sources are placed directly inside or very close to the tumor, providing targeted treatment to cancerous tissues while minimizing exposure to surrounding healthy tissues. This technique is commonly used for cancers of the prostate, cervix, breast, skin, and eye, among others [1].

In **prostate cancer**, brachytherapy is typically administered through the insertion of small radioactive seeds into the prostate gland, delivering precise radiation doses over a period of time [2]. For **cervical cancer**, high-dose-rate brachytherapy is often used after external beam radiation to target the tumor located within the cervix [3].

One of the main advantages of brachytherapy is its ability to deliver high radiation doses directly to the tumor, leading to effective cancer control with a relatively low risk of side effects. Additionally, it offers a shorter treatment duration compared to external radiation therapies, typically requiring fewer sessions [4].

2.11.1 Types of Brachytherapy:

- **Low-Dose-Rate (LDR)**: Seeds are implanted and deliver radiation slowly over several weeks or months.
- **High-Dose-Rate (HDR)**: A temporary catheter is inserted, and a high dose of radiation is delivered over a short period of time.
- **Pulsed Dose-Rate (PDR)**: A combination of LDR and HDR, where the radiation is delivered in pulses [6].



CHAPTER-3: MATERIALS AND METHODS

3.1 Materials

The following materials are used for this study

1) **Patient's Data**
2) **Treatment Planning**
3) **CT Image**
4) **CBCT Image**
5) **Bladder and rectum-filling protocol**
6) **RTOG Protocols for Contouring**
7) **Python (Programming Language)**

3.1.1 Patient's Data

This study enrolled 20 pelvic cancer patients, divided into two groups: the First Group and the Controlled Group, to evaluate the impact of bladder and rectal filling on the dosimetric quality of pelvic radiotherapy.

- **First Group:** Patients in this group were selected based on available clinical data from January 2018 to December 31, 2023.
- **Controlled Group:** Patients in this group were recently selected and strictly followed the hospital's bladder and rectal filling protocol.

The clinical profiles of these patients are presented in Table-01.

Table 1: Clinical Profiles of Patients

Criteria	No (%)
Age, average (range)	65 (62-79) First Group 71 (63-78) Controlled Group
Male	10 (50%)
Female	10 (50%)
3DCRT Patients	4 (20%)
IMRT Patients	6 (30%)
VMAT Patients	10 (50%)
Radiotherapy Dose (Gy)	60 Gy (For Prostate) 50 Gy (For Cervix)
Fractions	20 (for prostate) 25 (for cervix)

3.1.2 Treatment Planning:

3.1.2.1 Radiotherapy planning and dose constraints

All patients were immobilized and underwent CT simulation and treatment planning according to departmental policy. These planning CT images were sent to the Varian Eclipse contouring station

(version 16.3) where delineation and treatment planning were performed. Target volumes and the organs at risk (OARs) were contoured according to Radiation Therapy Oncology Group (RTOG) contouring guidelines.

3.1.2.2 Target Volume Delineation

- Gross Tumor Volume (GTV): Including grossly visible tumor and/or palpable inguinal lymph nodes identified by clinical examination or imaging.
- Clinical Target Volume (CTV): Areas at high risk of microscopic disease, contoured according to institutional guidelines.
- Planning Target Volume (PTV): This PTV margin variation was dependent upon accounting for differences in both set-up and organ motion, allowing for targeted delivery of radiation.

3.1.2.3 OARs and Dose Constraints

The only OARs delineated with dose–volume constraints applied as follows:

- Spinal Cord: Dmax < 45 Gy
- Bladder: V45 Gy ≤ 35%
- Bowel: V40 Gy ≤ 30%
- Rectum: V30 Gy ≤ 60%
- Both Femoral Heads: V30 Gy ≤ 15%

Allowable deviations from the prescribed dose constraints were made according to institutional practices.

3.1.2.4 Normalization of Plans and Coverage of Doses

In each plan, the treatment plans were normalized to ensure that 95% of the PTV received the prescribed dose while maintaining an acceptable balance between achieving good tumor control and sparing normal tissue.

3.1.3 CT Image

In radiotherapy, accurate tumor delineation is crucial for delivering the prescribed dose to the targeted tissue while minimizing damage to the surrounding healthy structures. Contouring, the process of outlining and segmenting specific regions of interest (ROIs) on medical images, is an essential step in this process. CT imaging plays a pivotal role in radiotherapy planning due to its high spatial resolution and ability to visualize anatomical structures with great clarity. The use of CT images for contouring allows radiation oncologists and medical physicists to define precise boundaries for tumor volumes, critical organs, and other important structures, ensuring that radiation treatment is both effective and safe.

In radiotherapy, accurate contouring on CT images is essential not only for defining the gross tumor volume (GTV), clinical target volume (CTV), and planning target volume (PTV) but also for outlining organs at risk (OARs). The GTV represents the visible tumor, while the CTV includes the area that may contain microscopic disease, and the PTV accounts for potential uncertainties such as patient movement and setup errors. Contouring OARs—such as the spinal cord, lungs,

heart, kidneys, and liver—is equally important, as these structures must be protected from excessive radiation to minimize toxicity and long-term complications [1].

CT imaging provides detailed, cross-sectional views of the body, allowing radiation oncologists to delineate these volumes with great precision. This enables a more accurate treatment plan that targets the tumor while sparing healthy tissue [2]. By using advanced software tools, clinicians can manually or semi-automatically contour both tumor volumes and OARs based on anatomical landmarks and tissue characteristics [3]. The accuracy of this process is critical, as errors in contouring could result in suboptimal treatment, with either insufficient tumor control or unnecessary radiation exposure to surrounding organs [4].

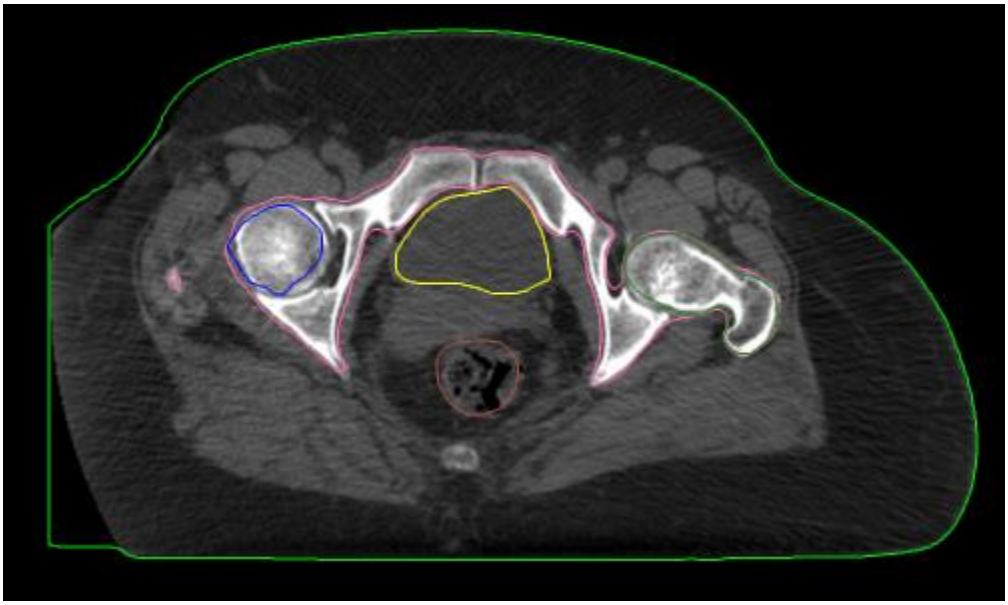


Figure 14: CT Image with OARs

3.1.4 CBCT Image

Cone-beam computed tomography (CBCT) was utilized during treatment as the **gold standard** for accurately defining bladder and rectum volumes. Patients underwent **daily CBCT scans** based on the discretion of the treating physician. If daily imaging was not required, CBCT was performed on **days 1 to 3**, followed by **weekly scan** throughout the treatment period. Bladder and rectum volumes were systematically contoured for analysis and treatment planning.

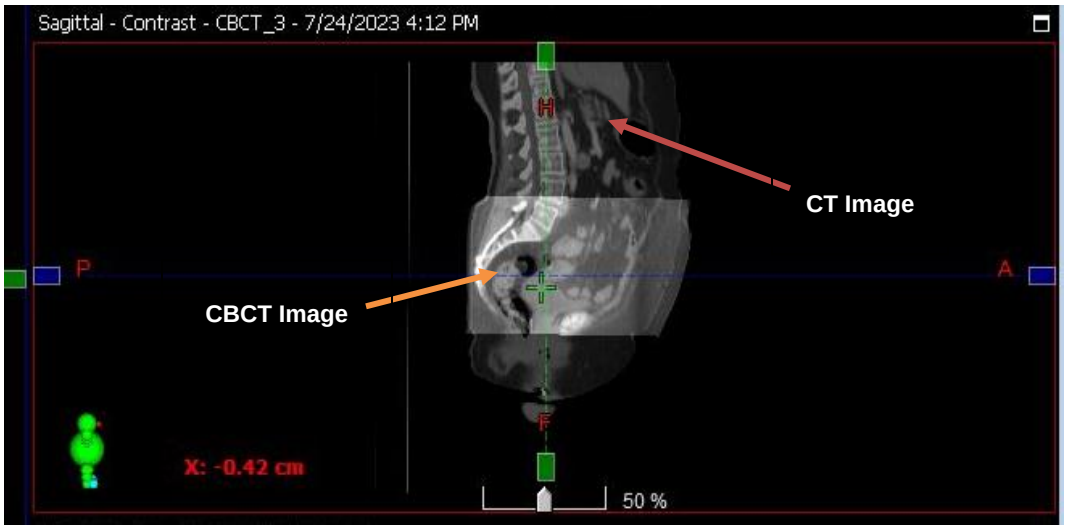


Figure 15: CT and CBCT image

Eclipse treatment planning system (Version 16.1, Varian Medical System, Inc, Palo Alto, California, USA) provides software tools for planning the treatment of malignant or benign diseases with radiation. Eclipse TPS is a computer-based software device used by trained medical professionals to design and simulate radiation therapy treatments. The Eclipse Treatment Planning System (Eclipse TPS) is used to plan radiotherapy treatments for patients with malignant or benign diseases. Eclipse TPS is used to plan external beam irradiation with photon, electron and proton beams, as well as for internal irradiation (brachytherapy) treatments. In this research VMAT plans were generated using Eclipse treatment planning system (TPS).

3.1.5 Bladder and Rectum volume measurement

During planning, bladder and rectum volume was measured from CT planning images. Planning bladder and rectum volume was used as the reference to measure the variation of bladder and rectum volume during treatment. Treatment bladder and rectum volume was derived every week from offline treatment Cone Beam CT (CBCT) images. In this study the both the CT-simulation and treatment bladder and rectum volume were measured using calculation based on ellipsoid formula. This is because bladder and rectum volume could not be measured directly from CBCT images. For this we contouring both structures on CBCT image. Each patient has 6 CBCT images.

3.1.6 Bladder and Rectum filling protocol

1. **For Simulation Day:**
   - **Step 1: Void** (empty the bladder and rectum).
   - **Step 2: Drink** a specified amount of water (300mL).
   - **Step 3:** Wait for 30-45 minutes before CT scan (varies for patient to patient)

This protocol is same for both first group (n=14) and second (n=6) group.

2. **For Treatment Day:**

- **First Group – Standard Approach:**
  - Upon arrival, patients were given **300 mL of water to drink**.
  - They waited **30 minutes** before being sent for treatment.



- Patients were also instructed to **empty their rectum** if necessary.
- Treatment commenced **30–45 minutes** after hydration.

- **Second Group – Strict Adherence to Protocol:**
  - Patients first **voided their bladder** upon arrival.
  - They were then given **300 mL of water** and instructed to wait **30-45 minutes**.
  - Patients were assessed for bowel status:
    - If they reported discomfort or irregularity, an **enema was administered** to empty the rectum.
    - After voiding, they **drank another 300 mL of water** and waited another **30-40 minutes** before proceeding to treatment.

3.1.7 RTOG

The **Radiation Therapy Oncology Group (RTOG)** has established specific protocols for **contouring pelvic organs**, including the **bladder** and **rectum**, during radiotherapy for various cancers. These guidelines are essential for ensuring accurate target delineation and minimizing radiation exposure to surrounding healthy tissues.

Regarding the RTOG (Radiation Therapy Oncology Group) protocols, the RTOG 0418 protocol is frequently used as a guideline for contouring the bladder and rectum for pelvic radiotherapy. The RTOG 0418 outlines specific contouring guidelines, such as:

- **Bladder:** The bladder is contoured from the top of the pubic symphysis to the apex of the bladder.
- **Rectum:** The rectum is contoured starting from the anus and continuing to the rectosigmoid junction, with a specific emphasis on maintaining accurate delineation around the pelvic bones and other surrounding structures.

These protocols ensure that the bladder and rectum are accurately contoured for optimal treatment planning and dose distribution, while minimizing potential damage to surrounding organs.

1. **Bladder Contouring:**

   - **Start point:** Begin at the S2-S3 vertebra level (just above the pubic symphysis).
   - This is where the bladder is typically visible on CT scans and can be contoured starting from the superior aspect of the bladder as it sits above the pubic symphysis.
   - **End point:** Continue contouring to the top of the bladder dome (which usually corresponds to the L5-S1 level, depending on bladder filling).
   - We can outline the bladder shape from its anterior to posterior wall, ensuring you include the upper portion of the bladder but not the surrounding pelvic structures.

2. **Rectum Contouring:**

   - **Start point:** Begin contouring at the bottom of the sacrum (around the S3-S4 level) or just above the puborectalis muscle, where the rectum becomes visible.
   - **End point:** Continue to the rectosigmoid junction (usually around the S2-S3 level), where the rectum transitions into the sigmoid colon.

- Focus on accurately delineating the rectum's shape and its boundaries with surrounding structures like the pelvic bones, bladder, and other tissues.

RTOG Protocols for Contouring Bladder and Rectum

1. **RTOG 0126:** This protocol provides detailed guidelines for the **contouring of the bladder and rectum** in patients undergoing **radiotherapy for prostate cancer**. The protocol outlines how the **bladder** should be contoured from the **outer wall** and the **rectum** from the **inner wall**, avoiding overlap with other OARs.
    - **Bladder Contouring**: Contour the bladder wall from the outermost boundary, including any air-filled regions. The bladder should be filled with a specific volume of fluid (typically 300–500 mL) during the CT simulation.
    - **Rectum Contouring**: Contour the rectum from the inner wall, ensuring that the delineation excludes the **pelvic walls** and surrounding tissues to avoid overlap with the bladder, small intestine, or femoral heads.
2. **RTOG 0815:**
    - **Bladder Contouring**: Contour the bladder from the outermost border, ensuring it is accurately represented in its filled state during treatment planning.
    - **Rectum Contouring**: The rectum is contoured from the **anus** up to the **rectosigmoid junction**, ensuring that the internal wall is accurately delineated.
    - The RTOG 0815 guidelines also emphasize using **cone-beam CT (CBCT)** or **X-ray imaging** for **image-guided radiotherapy (IGRT)** to further verify the bladder and rectum positions during treatment.
3. **RTOG 0938:** This protocol provides specific guidance for **rectal cancer** patients. The rectum should be contoured from the **anus** to the **rectosigmoid junction**, similar to the approach used in **prostate cancer** treatments.
    - **Bladder and Rectum Filling Protocol**: Patients are asked to empty their bladder and rectum before the CT scan to reduce variability during treatment.

3.1.8 Python

Python is a high-level, interpreted, object-oriented programming language with dynamic semantics. When you add built in data structures, dynamic typing and dynamic binding, you can see why it is so good for Rapid Application Development (RAD). Another reason of its popularity is Extensive Use of python as an embedded language to efficiently combine existing components.

- Simple and Readable Syntax → Python's syntax is simple. The readability means that the cost of maintenance on the program is minimized.
- Modules and Packages → Python provides support for code modules and packages, which helps in writing modular programs and promoting code reuse.
- Cross-Platform and Open Source → The Python interpreter and standard library are freely available in both source and binary forms across all major platforms.

3.2 Method

This project was conducted in **three phases**:

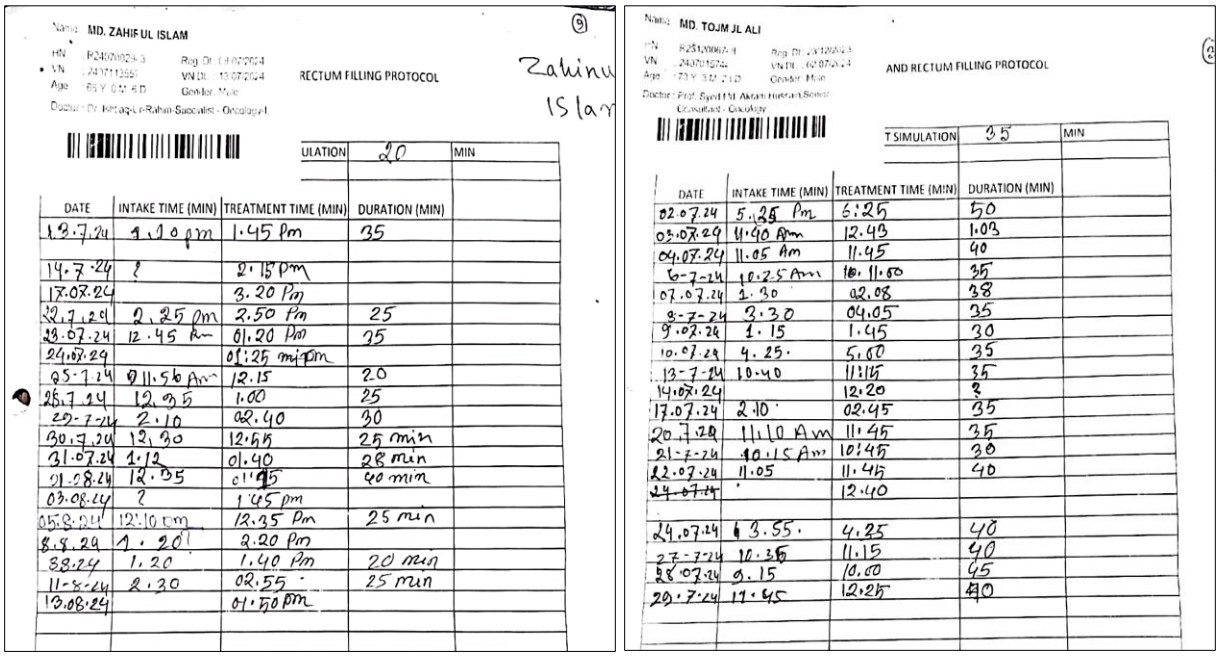
- **Phase 1: Patient Selection, Image Registration, and Contouring**
- **Phase 2: Dosimetric Analysis**
- **Phase 3: Geometric Changes Assessment**

3.2.1 Patients selection

First, a comprehensive list of cervical and prostate cancer patients treated from 2018 to 2023 was created. Therefore, we selected the patients with the largest amount of CBCT images and classified them. Among them, patients whose CBCT images fully visualized the bladder and rectum were screened for inclusion. From this selection, **14 patients** (First group) were chosen for the **project study**.

Second, a protocol form was generated for new patients treated during the period of July to September 2024. Six new patients (Second group) were selected for the project, following the same criteria as before.

Finally, for each of the selected patients, we chose six high-quality CBCT images for contouring.



Name: MD. ZAHIFUL ISLAM

RECTUM FILLING PROTOCOL

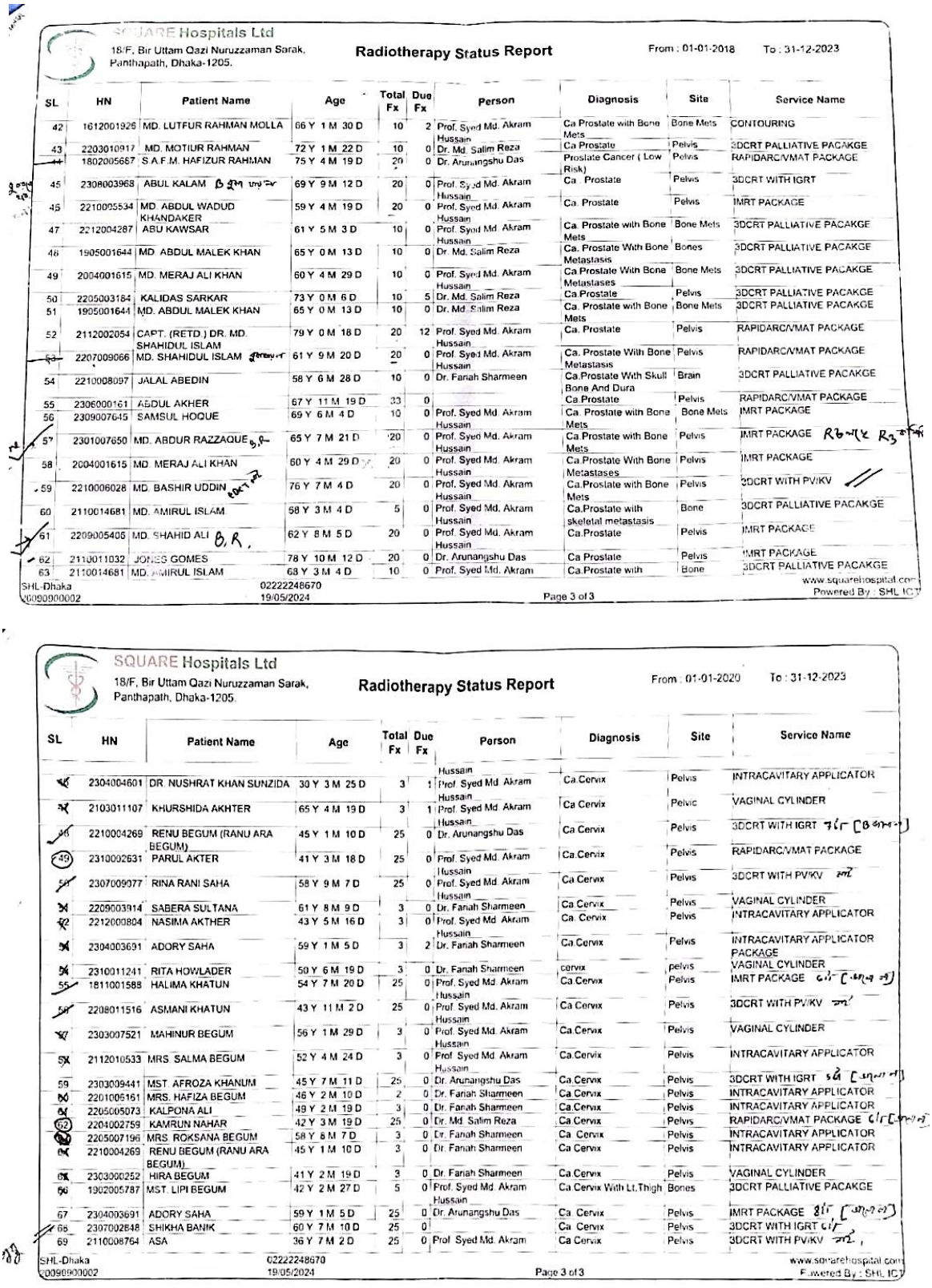
DATE	INTAKE TIME (MIN)	TREATMENT TIME (MIN)	DURATION (MIN)
19.7.24	1.10 pm	1.45 pm	35
14.7.24	?	2.15 pm	
17.07.24		3.20 Pm	
22.7.24	2.25 pm	2.50 Pm	25
23.07.24	12.45 Pm	01.20 Pm	35
24.07.24		01:25 mipm	
25.7.24	11.50 Am	12.15	20
26.7.24	12.05	1.00	25
29-7-24	2.10	02.40	30
30.7.24	12.30	12.55	25 min
31.07.24	1.12	01.40	28 min
21.08.24	12.05	01.45	40 min
03.08.24	?	1.45 pm	
05.8.24	12.10 pm	12.35 Pm	25 min
8.8.24	1.20	2.20 pm	
08.24	1.20	1.40 Pm	20 min
11-8-24	2.30	02.55	25 min
13.08.24		01.50 pm	

Name: MD. TOJM JL ALI

AND RECTUM FILLING PROTOCOL

DATE	INTAKE TIME (MIN)	TREATMENT TIME (MIN)	DURATION (MIN)
02.07.24	5.25 Pm	6:25	50
03.07.24	11.90 Am	12.43	1.03
04.07.24	11.05 Am	11.45	40
6-7-24	10.25 Am	10. 11.50	35
07.07.24	1.30	02.08	38
8-7-24	3.30	04.05	35
9.07.24	1.15	1.45	30
10.07.24	4.25.	5.50	35
13-7-24	10.40	11:15	35
14.07.24		12.20	?
17.07.24	2.10	02.45	35
20.7.24	11.10 Am	11.45	35
21-7-24	10.15 Am	10:45	30
22.07.24	11.05	11.45	40
24-07-24		12.40	
24.07.24	3.55.	4.25	40
27-7-24	10.35	11.15	40
28.07.24	9.15	10.00	45
29.7.24	12.45	12.25	40

Figure 16: Bladder and rectum filling protocol form for Second Group



SQUARE Hospitals Ltd
18/F, Bir Uttam Qazi Nuruzzaman Sarak, Panthapath, Dhaka-1205.

Radiotherapy Status Report From : 01-01-2018 To : 31-12-2023

SL	HN	Patient Name	Age	Total Fx	Due Fx	Person	Diagnosis	Site	Service Name
42	1612001926	MD. LUTFUR RAHMAN MOLLA	66 Y 1 M 30 D	10	2	Prof. Syed Md. Akram Hussain	Ca Prostate with Bone Mets	Bone Mets	CONTOURING
43	2203010917	MD. MOTIUR RAHMAN	72 Y 1 M 22 D	10	0	Dr. Md. Salim Reza	Ca Prostate	Pelvis	3DCRT PALLIATIVE PACAKGE
44	1802005687	S A F M. HAFIZUR RAHMAN	75 Y 4 M 19 D	20	0	Dr. Arunangshu Das	Prostate Cancer (Low Risk)	Pelvis	RAPIDARC/VMAT PACKAGE
45	2308003968	ABUL KALAM	69 Y 9 M 12 D	20	0	Prof. Syed Md. Akram Hussain	Ca. Prostate	Pelvis	3DCRT WITH IGRT
46	2210005534	MD. ABDUL WADUD KHANDAKER	59 Y 4 M 19 D	20	0	Prof. Syed Md. Akram Hussain	Ca. Prostate	Pelvis	IMRT PACKAGE
47	2212004287	ABU KAWSAR	61 Y 5 M 3 D	10	0	Prof. Syed Md. Akram Hussain	Ca. Prostate with Bone Mets	Bone Mets	3DCRT PALLIATIVE PACAKGE
48	1905001644	MD. ABDUL MALEK KHAN	65 Y 0 M 13 D	10	0	Dr. Md. Salim Reza	Ca. Prostate With Bone Metastasis	Bones	3DCRT PALLIATIVE PACAKGE
49	2004001615	MD. MERAJ ALI KHAN	60 Y 4 M 29 D	10	0	Prof. Syed Md. Akram Hussain	Ca.Prostate With Bone Metastases	Bone Mets	3DCRT PALLIATIVE PACAKGE
50	2205003184	KALIDAS SARKAR	73 Y 0 M 6 D	10	5	Dr. Md. Salim Reza	Ca.Prostate	Pelvis	3DCRT PALLIATIVE PACAKGE
51	1905001644	MD. ABDUL MALEK KHAN	65 Y 0 M 13 D	10	0	Dr. Md. Salim Reza	Ca. Prostate with Bone Mets	Bone Mets	3DCRT PALLIATIVE PACAKGE
52	2112002054	CAPT. (RETD.) DR. MD. SHAHIDUL ISLAM	79 Y 0 M 18 D	20	12	Prof. Syed Md. Akram Hussain	Ca. Prostate	Pelvis	RAPIDARC/VMAT PACKAGE
53	2207009066	MD. SHAHIDUL ISLAM	61 Y 9 M 20 D	20	0	Prof. Syed Md. Akram Hussain	Ca. Prostate With Bone Metastasis	Pelvis	RAPIDARC/VMAT PACKAGE
54	2210008097	JALAL ABEDIN	58 Y 6 M 28 D	10	0	Dr. Farhah Sharmeen	Ca.Prostate With Skull Bone And Dura	Brain	3DCRT PALLIATIVE PACAKGE
55	2306000161	ABDUL AKHER	67 Y 11 M 19 D	33	0		Ca.Prostate	Pelvis	RAPIDARC/VMAT PACKAGE
56	2309007645	SAMSUL HOQUE	69 Y 6 M 4 D	10	0	Prof. Syed Md. Akram Hussain	Ca. Prostate with Bone Mets	Bone Mets	IMRT PACKAGE
57	2301007650	MD. ABDUR RAZZAQUE	65 Y 7 M 21 D	20	0	Prof. Syed Md. Akram Hussain	Ca.Prostate with Bone Mets	Pelvis	IMRT PACKAGE
58	2004001615	MD. MERAJ ALI KHAN	60 Y 4 M 29 D	20	0	Prof. Syed Md. Akram Hussain	Ca.Prostate With Bone Metastases	Pelvis	IMRT PACKAGE
59	2210006028	MD. BASHIR UDDIN	76 Y 7 M 4 D	20	0	Prof. Syed Md. Akram Hussain	Ca.Prostate with Bone Mets	Pelvis	3DCRT WITH PVIKV
60	2110014681	MD. AMIRUL ISLAM	68 Y 3 M 4 D	5	0	Prof. Syed Md. Akram Hussain	Ca.Prostate with skeletal metastasis	Bone	3DCRT PALLIATIVE PACAKGE
61	2209005406	MD. SHAHID ALI	62 Y 8 M 5 D	20	0	Prof. Syed Md. Akram Hussain	Ca.Prostate	Pelvis	IMRT PACKAGE
62	2110011032	JONES GOMES	78 Y 10 M 12 D	20	0	Dr. Arunangshu Das	Ca Prostate	Pelvis	IMRT PACKAGE
63	2110014681	MD. AMIRUL ISLAM	68 Y 3 M 4 D	10	0	Prof. Syed Md. Akram	Ca.Prostate with	Bone	3DCRT PALLIATIVE PACAKGE

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SQUARE Hospitals Ltd
18/F, Bir Uttam Qazi Nuruzzaman Sarak, Panthapath, Dhaka-1205.

Radiotherapy Status Report From : 01-01-2020 To : 31-12-2023

SL	HN	Patient Name	Age	Total Fx	Due Fx	Person	Diagnosis	Site	Service Name
46	2304004601	DR. NUSHRAT KHAN SUNZIDA	30 Y 3 M 25 D	3	1	Prof. Syed Md. Akram Hussain	Ca.Cervix	Pelvis	INTRACAVITARY APPLICATOR
47	2103011107	KHURSHIDA AKHTER	65 Y 4 M 19 D	3	1	Prof. Syed Md. Akram Hussain	Ca Cervix	Pelvic	VAGINAL CYLINDER
48	2210004269	RENU BEGUM (RANU ARA BEGUM)	45 Y 1 M 10 D	25	0	Dr. Arunangshu Das	Ca Cervix	Pelvis	3DCRT WITH IGRT
49	2310002631	PARUL AKTER	41 Y 3 M 18 D	25	0	Prof. Syed Md. Akram Hussain	Ca.Cervix	Pelvis	RAPIDARC/VMAT PACKAGE
50	2307009077	RINA RANI SAHA	58 Y 9 M 7 D	25	0	Prof. Syed Md. Akram Hussain	Ca.Cervix	Pelvis	3DCRT WITH PVIKV
51	2209003914	SABERA SULTANA	61 Y 8 M 9 D	3	0	Dr. Farhah Sharmeen	Ca.Cervix	Pelvis	VAGINAL CYLINDER
52	2212000804	NASIMA AKTHER	43 Y 5 M 16 D	3	0	Prof. Syed Md. Akram Hussain	Ca. Cervix	Pelvis	INTRACAVITARY APPLICATOR
53	2304003691	ADORY SAHA	59 Y 1 M 5 D	3	2	Dr. Farhah Sharmeen	Ca.Cervix	Pelvis	INTRACAVITARY APPLICATOR PACKAGE
54	2310011241	RITA HOWLADER	50 Y 6 M 19 D	3	0	Dr. Farhah Sharmeen	cervix	pelvis	VAGINAL CYLINDER
55	1811001588	HALIMA KHATUN	54 Y 7 M 20 D	25	0	Prof. Syed Md. Akram Hussain	Ca.Cervix	Pelvis	IMRT PACKAGE
56	2208011516	ASMANI KHATUN	43 Y 11 M 2 D	25	0	Prof. Syed Md. Akram Hussain	Ca.Cervix	Pelvis	3DCRT WITH PVIKV
57	2303007521	MAHINUR BEGUM	56 Y 1 M 29 D	3	0	Prof. Syed Md. Akram Hussain	Ca.Cervix	Pelvis	VAGINAL CYLINDER
58	2112010533	MRS. SALMA BEGUM	52 Y 4 M 24 D	3	0	Prof. Syed Md. Akram Hussain	Ca.Cervix	Pelvis	INTRACAVITARY APPLICATOR
59	2303009441	MST. AFROZA KHANUM	45 Y 7 M 11 D	25	0	Dr. Arunangshu Das	Ca.Cervix	Pelvis	3DCRT WITH IGRT
60	2201005161	MRS. HAFIZA BEGUM	46 Y 2 M 10 D	2	0	Dr. Farhah Sharmeen	Ca.Cervix	Pelvis	INTRACAVITARY APPLICATOR
61	2205005073	KALPONA ALI	49 Y 2 M 19 D	3	0	Dr. Farhah Sharmeen	Ca.Cervix	Pelvis	INTRACAVITARY APPLICATOR
62	2204002759	KAMRUN NAHAR	42 Y 3 M 19 D	25	0	Dr. Md. Salim Reza	Ca.Cervix	Pelvis	RAPIDARC/VMAT PACKAGE
63	2205007196	MRS. ROKSANA BEGUM	58 Y 8 M 7 D	3	0	Dr. Farhah Sharmeen	Ca. Cervix	Pelvis	INTRACAVITARY APPLICATOR
64	2210004269	RENU BEGUM (RANU ARA BEGUM)	45 Y 1 M 10 D	3	0	Dr. Farhah Sharmeen	Ca Cervix	Pelvis	INTRACAVITARY APPLICATOR
65	2303000252	HIRA BEGUM	41 Y 2 M 19 D	3	0	Dr. Farhah Sharmeen	Ca.Cervix	Pelvis	VAGINAL CYLINDER
66	1902005787	MST. LIPI BEGUM	42 Y 2 M 27 D	5	0	Prof. Syed Md. Akram Hussain	Ca.Cervix With Lt.Thigh	Bones	3DCRT PALLIATIVE PACAKGE
67	2304003691	ADORY SAHA	59 Y 1 M 5 D	25	0	Dr. Arunangshu Das	Ca. Cervix	Pelvis	IMRT PACKAGE
68	2307002848	SHIKHA BANIK	60 Y 7 M 10 D	25	0		Ca. Cervix	Pelvis	3DCRT WITH IGRT
69	2110008764	ASA	36 Y 7 M 2 D	25	0	Prof. Syed Md. Akram	Ca Cervix	Pelvis	3DCRT WITH PVIKV

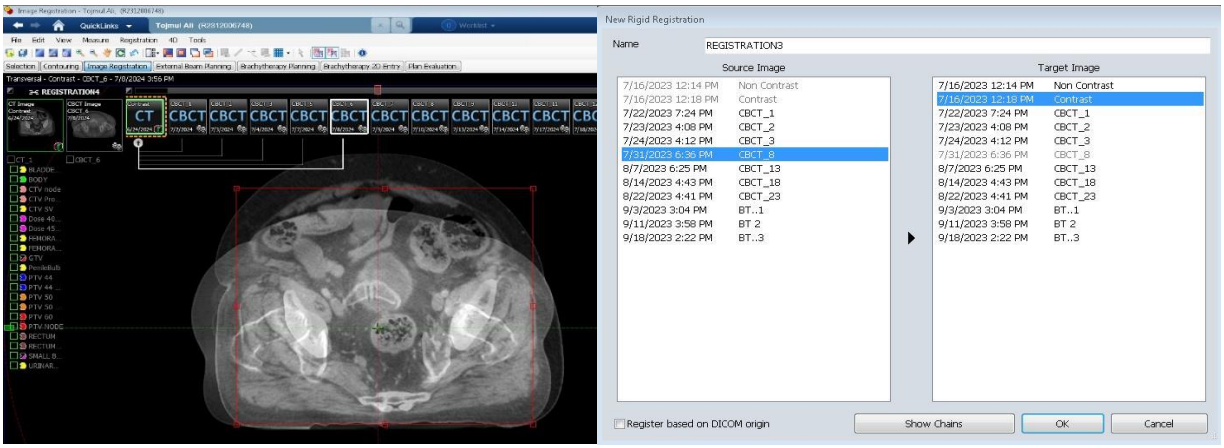
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Figure 17: Patients list for first group

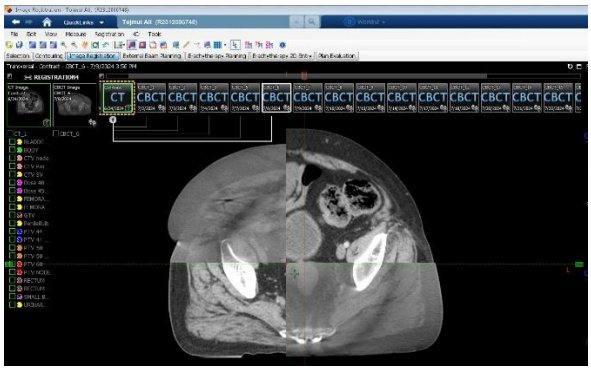
3.2.2 Image Registration

Image registration was performed to **accurately align CBCT images with CT simulation images**, ensuring proper **overlap and patient setup** during treatment. This process enabled the visualization of **CT and CBCT images in a single slice**, making **contouring more precise and efficient**. For each CBBCT image was need to registered with CT simulation image.

3.2.2.1 Image Registration process:



a) b)



c)

Figure 18: a) Before image registration, b) CT image to which CBCT image need to registered (registration in manual), c) After the registration

Image Setup Parameters:

- The **parameters** were set to **extended range** for better accuracy.
- The **intensity range** was set between **0-200 HU** to optimize image clarity.
- The **soft tissue from the CT image** was matched with the **soft tissue from the CBCT image** to achieve accurate body alignment between the two imaging modalities.

- In certain cases, **Structure VOI (Volume of Interest)** adjustments were required to achieve better alignment during image registration.

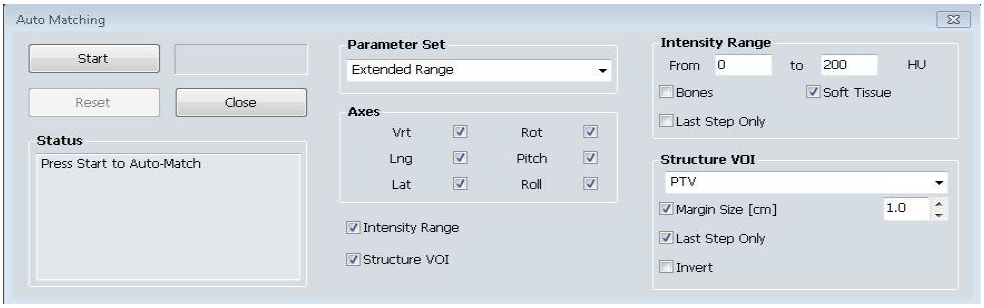


Figure 19: Image setup Parameters

This method ensured a **precise match between CT and CBCT images**, improving **treatment accuracy** and facilitating **contouring**.

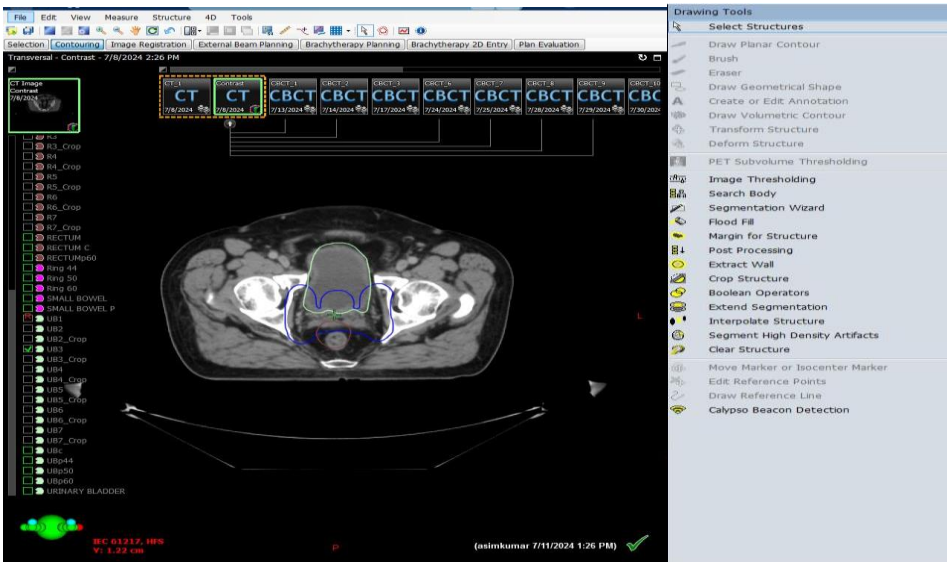
3.2.3 Contouring

Before starting the **contouring**, **six duplicate structures** for the **bladder** (given name UB1, UB2, ...UB7) **and rectum** (given name R1, R2……R7) were created for each of the **six CBCT images**. This ensured that separate structures could be identified and analyzed for each CBCT images.

During the **contouring process**, the following tools were used:

- **Draw Planar** – For outlining structures in a specific plane.
- **Brush Tool** – For refining and filling in the contours.
- **Eraser Tool** – To remove unwanted areas and correct contours.
- **Interpolate Structure** – To smooth and connect contour slices for consistency.

The **first phase** of the study followed the **same process** for two groups.



a) b)

Figure 20: a) Contouring part b) Contouring tools

3.2.4 Dosimetric Analysis

At the beginning of **Phase 2**, data was extracted from the **contoured images** of patients in both the **First Group** and **Second Group**. The key parameters analyzed included **structure volumes** of the **bladder and rectum**.

3.2.4.1 OARs Volume Changes and Dose coverage

- **Step-1:**
  - Bladder and rectum volumes were recorded on the **CT simulation day** to establish a baseline.
- **Step-2:**
  - The volumes from the **contoured structures** (generated during Phase 1) were systematically tabulated for comparative analysis.
- **Step-3:**
  - Dose coverage of the **bladder and rectum** was assessed using **DVH graphs** and the reference dose constraints based on protocol:
    - **Bladder:** V45Gy **≤ 35%**
    - **Rectum:** V30Gy **≤ 60%**

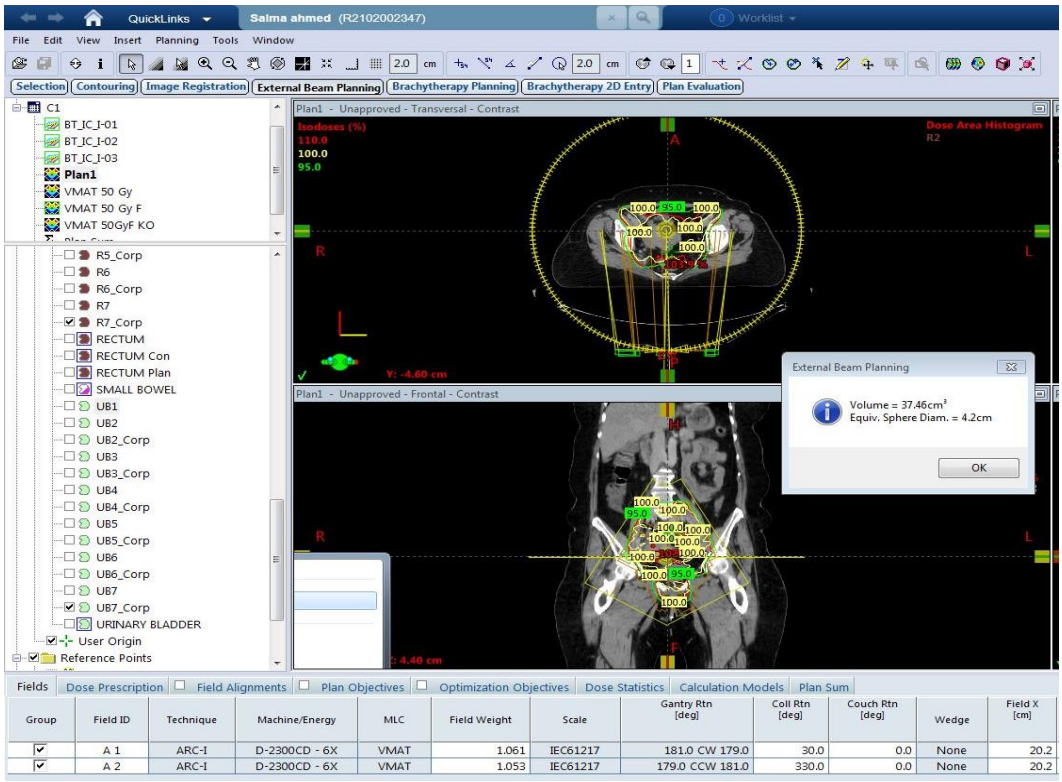


Figure 21: Measure new volume for bladder

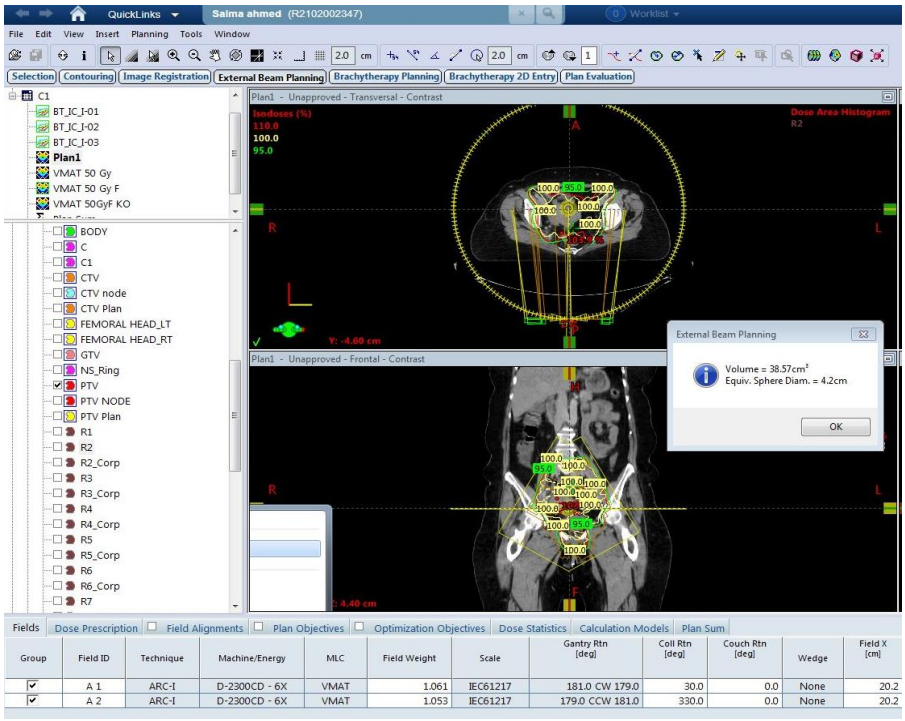


Figure 22: Measure new volume for rectum

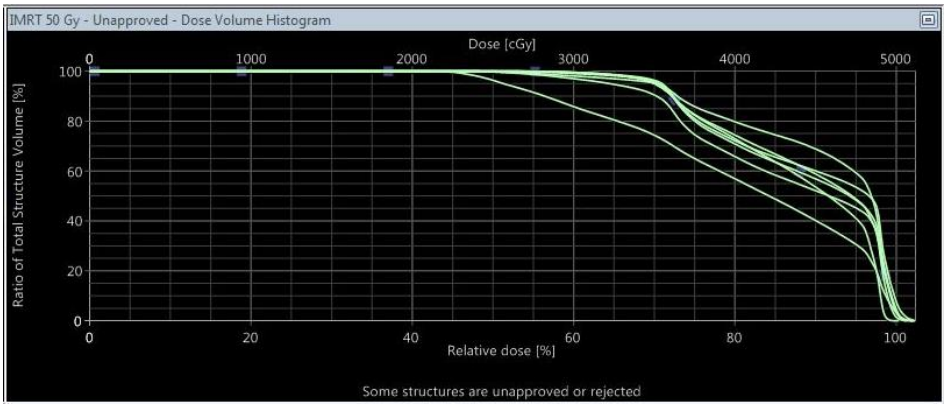


Figure 23: DVH graph show dose coverage for bladder

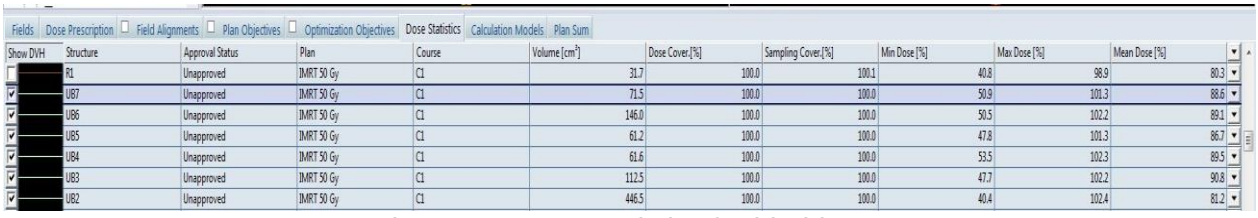


Figure 24: Dose statistics for bladder

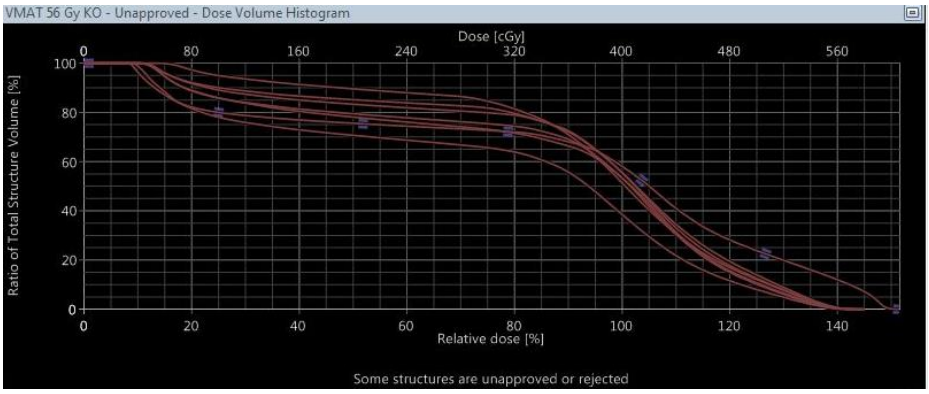
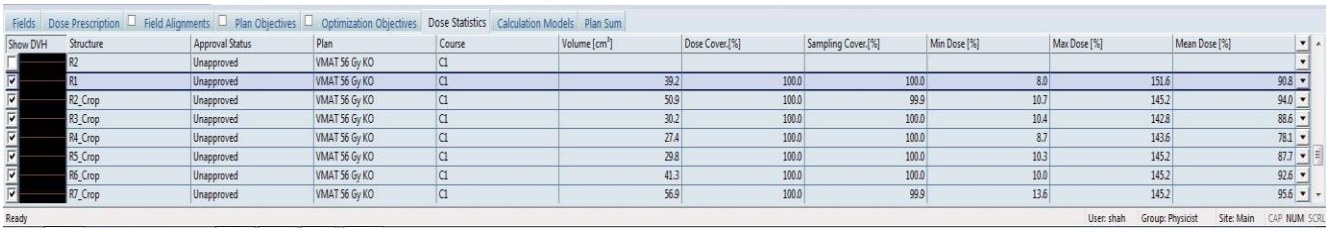


Figure 25: DVH graph show dose coverage for rectum



Fields | Dose Prescription | Field Alignments | Plan Objectives | Optimization Objectives | Dose Statistics | Calculation Models | Plan Sum

Show DVH	Structure	Approval Status	Plan	Course	Volume [cm³]	Dose Cover.[%]	Sampling Cover.[%]	Min Dose [%]	Max Dose [%]	Mean Dose [%]
	R2	Unapproved	VMAT 56 Gy KO	C1						
	R1	Unapproved	VMAT 56 Gy KO	C1	39.2	100.0	100.0	8.0	151.6	90.8
	R2_Crop	Unapproved	VMAT 56 Gy KO	C1	50.9	100.0	99.9	10.7	145.2	94.0
	R3_Crop	Unapproved	VMAT 56 Gy KO	C1	30.2	100.0	100.0	10.4	142.8	88.6
	R4_Crop	Unapproved	VMAT 56 Gy KO	C1	27.4	100.0	100.0	8.7	143.6	78.1
	R5_Crop	Unapproved	VMAT 56 Gy KO	C1	29.8	100.0	100.0	10.3	145.2	87.7
	R6_Crop	Unapproved	VMAT 56 Gy KO	C1	41.3	100.0	100.0	10.0	145.2	92.6
	R7_Crop	Unapproved	VMAT 56 Gy KO	C1	56.9	100.0	99.9	13.6	145.2	95.6

Ready | User: shah | Group: Physicist | Site: Main | CAP NUM SCRL

Figure 26: Dose statistics for rectum

After extracting coverage data from the DVH, six new treatment plans were generated for each patient to evaluate the impact of structural changes on MU variations, as six CBCT images were acquired per patient.

For patients treated with IMRT and VMAT, re-planning was feasible due to the availability of optimization options. However, for 3DCRT, where re-optimization was not possible, the changes in mean dose resulting from structural modifications were documented.

During the planning process, the bladder and rectum in the CT images were assigned the same priority and dose constraints as in the initial plan to ensure consistency in optimization. Additionally, if bladder and rectum cropping was performed in the original PTV, an equivalent adjustment was applied to the new structure before optimization.

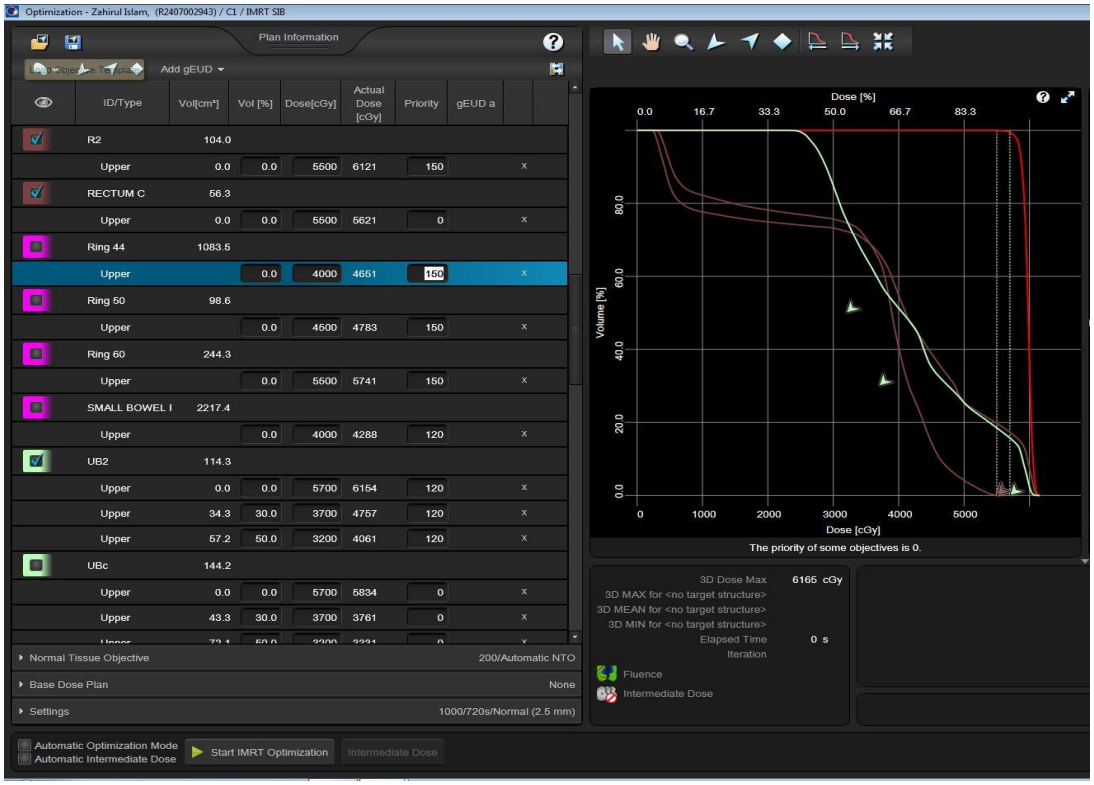


Figure 27: Re-optimization for IMRT patient's

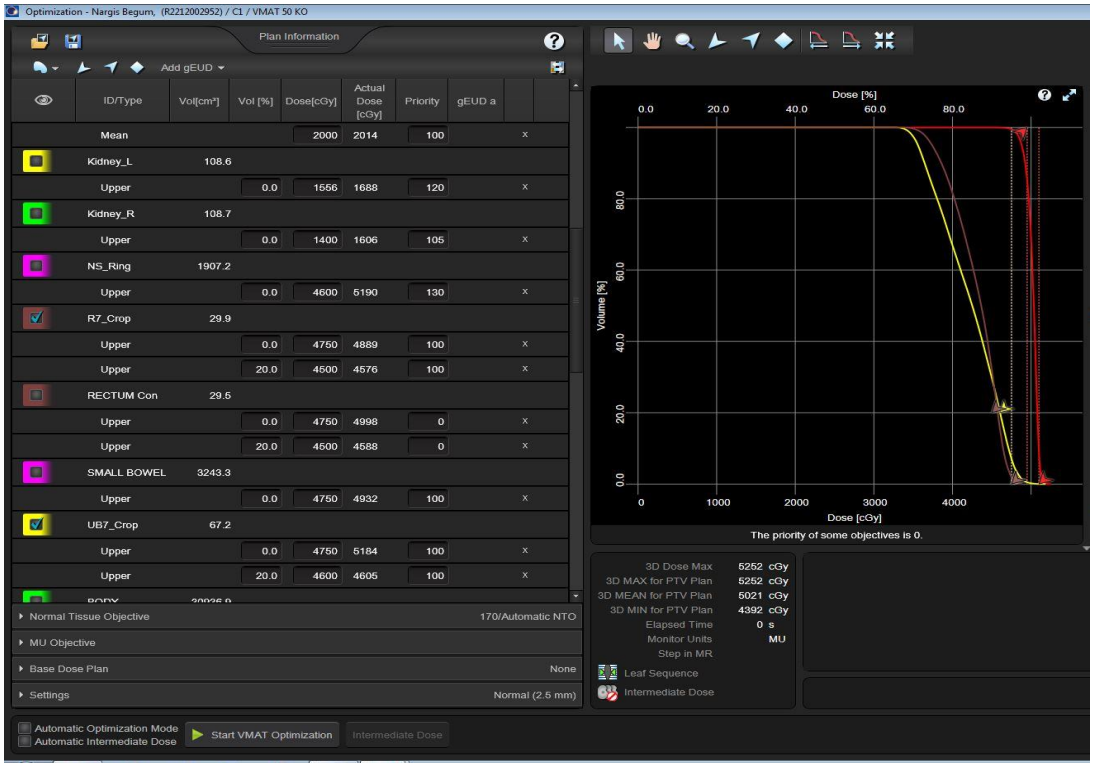


Figure 28: Re-optimization for VMAT patient's

3.2.5 Geometric Changes Assessment

In the 3rd part, we need to follow some steps to complete the process:

Step 1: Install Spyder for Python Development

Spyder is a powerful Python IDE, widely preferred in research due to its user-friendly interface and robust debugging tools. It is particularly useful for automating complex calculations such as Hausdorff Distance (HD) and Dice Similarity Coefficient (DSC), which are challenging to compute manually—especially when analyzing 25–35 image slices in medical imaging studies.

Step 2: Create a Control Folder

Establish a structured directory to store patient data for analysis. This ensures systematic data organization and facilitates easy access to CT and CBCT scans.

Step 3: Develop a Python Script for HD and DSC Calculation

Implement a validated Python script to compute HD and DSC, comparing CT and CBCT structures. Define:

- CT reference structures: UB1 (bladder) and R1 (rectum)
- CBCT structures: UB2–UB7 (bladder) and R2–R7 (rectum)
- UB1 and R1 serve as the reference structures for comparative analysis.

Step 4: Execute the Program

Run the Python script in Spyder to automate the calculation of HD and DSC. This ensures efficient, accurate, and repeatable analysis for medical imaging research.

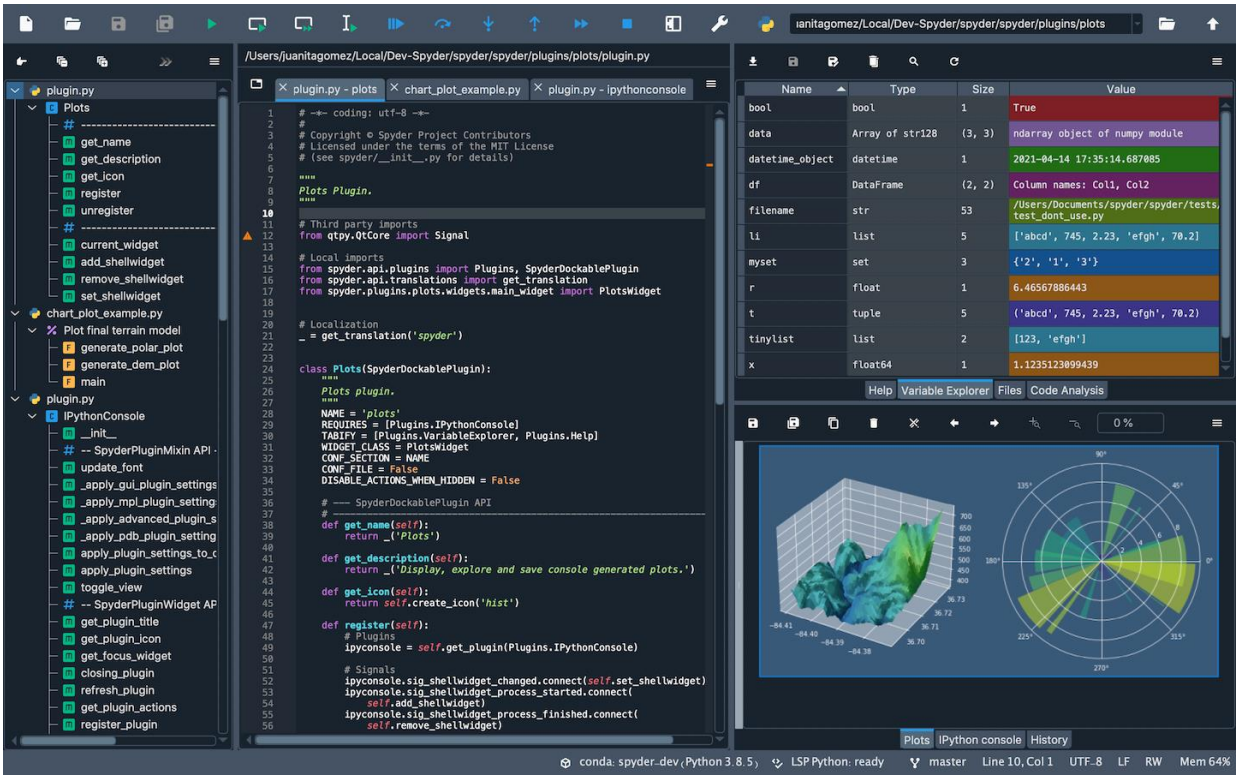


Figure 29: Spyder Phyton app interface

```
# Set your control folder path
control_folder = 'Control'  # Modify this to your actual control folder path

# Reference structure (manually changeable)
reference_structure_name = 'UB1'

# List of structures to compare (manually changeable)
structure_names = ['UB2', 'UB3', 'UB4', 'UB5', 'UB6', 'UB7']  # Add all your structure names here

# Process all subfolders in the control folder
process_control_folder(control_folder, reference_structure_name, structure_names)
```

Figure 30: Some part of the Python code for bladder

```
# Set your control folder path
control_folder = 'Control'  # Modify this to your actual control folder path

# Reference structure (manually changeable)
reference_structure_name = 'R1'

# List of structures to compare (manually changeable)
structure_names = ['R2', 'R3', 'R4', 'R5', 'R6', 'R7']  # Add all your structure names here

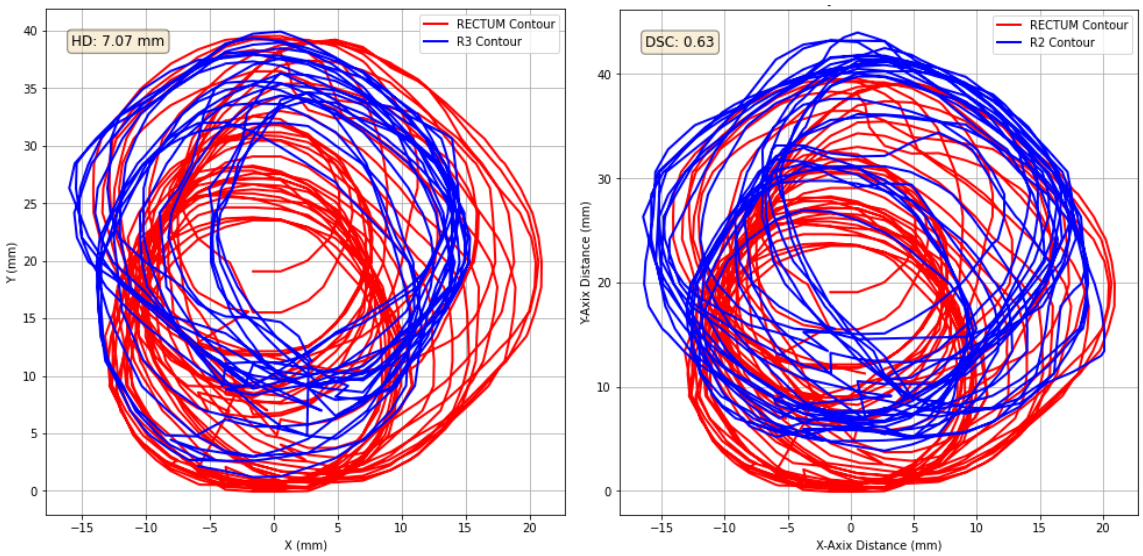
# Process all subfolders in the control folder
process_control_folder(control_folder, reference_structure_name, structure_names)
```

Figure 31: Some part of the Python code for rectum

Patient ID: 2308003968	
Slice Number	HD for UB
Processing	Number of slices for UB1 and UB2 differ.
Slice 0	HD for UB2 = 10.03 mm
Slice 1	HD for UB2 = 13.99 mm
Slice 2	HD for UB2 = 4.56 mm
Slice 3	HD for UB2 = 8.00 mm
Slice 4	HD for UB2 = 15.84 mm
Slice 5	HD for UB2 = 11.92 mm
Slice 6	HD for UB2 = 5.88 mm
Slice 7	HD for UB2 = 7.28 mm
Slice 8	HD for UB2 = 7.85 mm
Slice 9	HD for UB2 = 7.75 mm
Slice 10	HD for UB2 = 85.44 mm
Slice 11	HD for UB2 = 9.35 mm
Slice 12	HD for UB2 = 10.23 mm
Slice 13	HD for UB2 = 12.27 mm
Slice 14	HD for UB2 = 14.00 mm
Slice 15	HD for UB2 = 15.51 mm
Slice 16	HD for UB2 = 19.55 mm
Slice 17	HD for UB2 = 20.39 mm
Slice 18	HD for UB2 = 25.97 mm
Slice 19	HD for UB2 = 43.00 mm
Slice 20	HD for UB2 = 49.45 mm

Patient ID: R120506539	
Slice Number	HD for Rectum
Processing	Number of slices for R1 and R2 differ.
Slice 0	DSC for R2 = 0.7975
Slice 1	DSC for R2 = 0.9209
Slice 2	DSC for R2 = 0.8285
Slice 3	DSC for R2 = 0.7476
Slice 4	DSC for R2 = 0.8513
Slice 5	DSC for R2 = 0.5457
Slice 6	DSC for R2 = 0.3796
Slice 7	DSC for R2 = 0.3559
Slice 8	DSC for R2 = 0.3526
Slice 9	DSC for R2 = 0.3145
Slice 10	DSC for R2 = 0.2739
Slice 11	DSC for R2 = 0.2236
Slice 12	DSC for R2 = 0.2410
Slice 13	DSC for R2 = 0.3695
Slice 14	DSC for R2 = 0.4330
Slice 15	DSC for R2 = 0.4717
Slice 16	DSC for R2 = 0.5658
Slice 17	DSC for R2 = 0.0000
Slice 18	DSC for R2 = 0.5738
Slice 19	DSC for R2 = 0.5297
Slice 20	DSC for R2 = 0.6022

Figure 32: HD and DSC difference between CT and CBCT bladder (par slice)



a) b)

Figure 33: a) HD b) DSC difference between CT and CBCT rectum (par slices)

Patient ID: 2308003968 (Final Maximum Hausdorff Distance)		
Max HD for UB	Size (mm)	Slice Number
UB2	85.44 mm	10
UB3	51.63 mm	20
UB4	42.47 mm	20
UB5	74.73 mm	15
UB6	47.38 mm	20
UB7	49.33 mm	20

Figure 34: Max HD for bladder

Patient ID: 2308003968 (Final Maximum Dice Similarity Coefficient)		
Max DSC for UB	Size (mm)	Slice Number
UB2	0.98	7
UB3	0.89	11
UB4	0.9	12
UB5	0.94	17
UB6	0.96	18
UB7	0.44	10

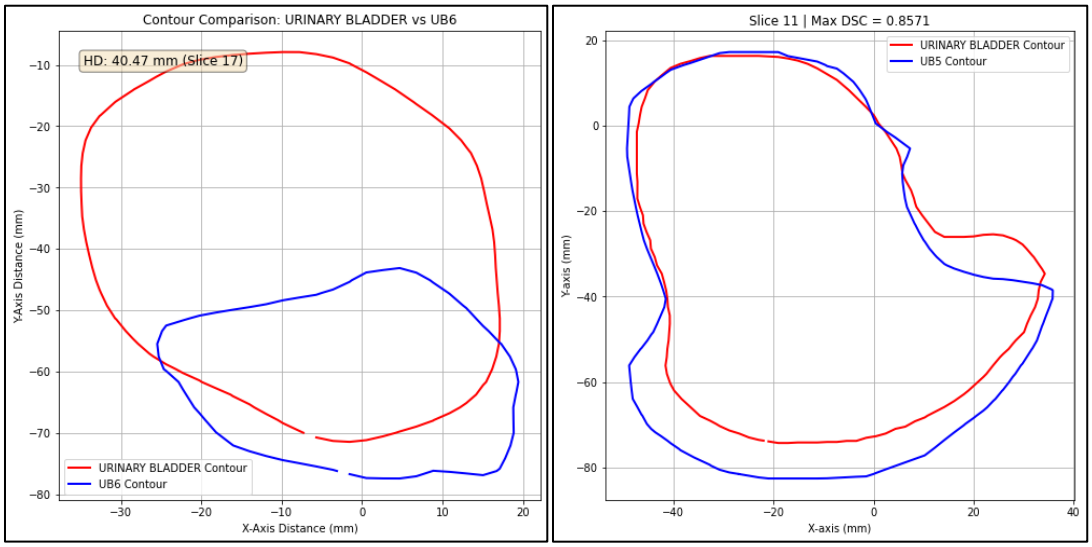
Figure 35: Max DSC for bladder

Patient ID: 1412005008 (Final Maximum Hausdorff Distance)		
Max HU for R	Size	Slice
R2	24.7 mm	27
R3	14.29 mm	26
R4	19.45 mm	24
R5	13.3 mm	8
R6	29.58 mm	23
R7	18.18 mm	25

Figure 36: Max HD for rectum

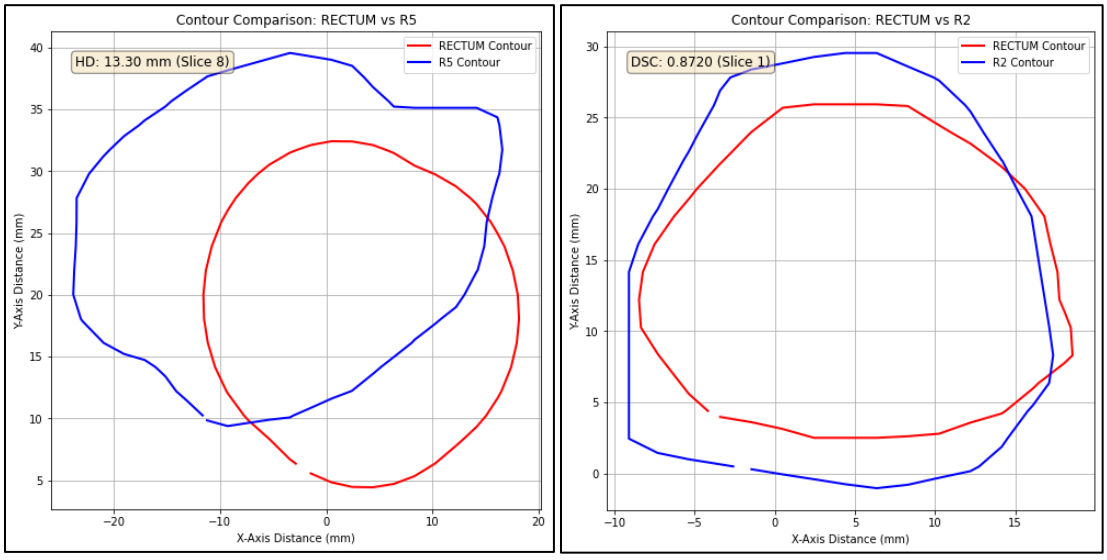
Patient ID: 1412005008 (Final Maximum Dice Similarity Coefficient)		
Max DSC for R	Size	Slice
R2	0.8720 mm	1
R3	0.9279 mm	17
R4	0.8927 mm	17
R5	0.9076 mm	24
R6	0.7146 mm	0
R7	0.8757 mm	18

Figure 37: Max DSC for rectum



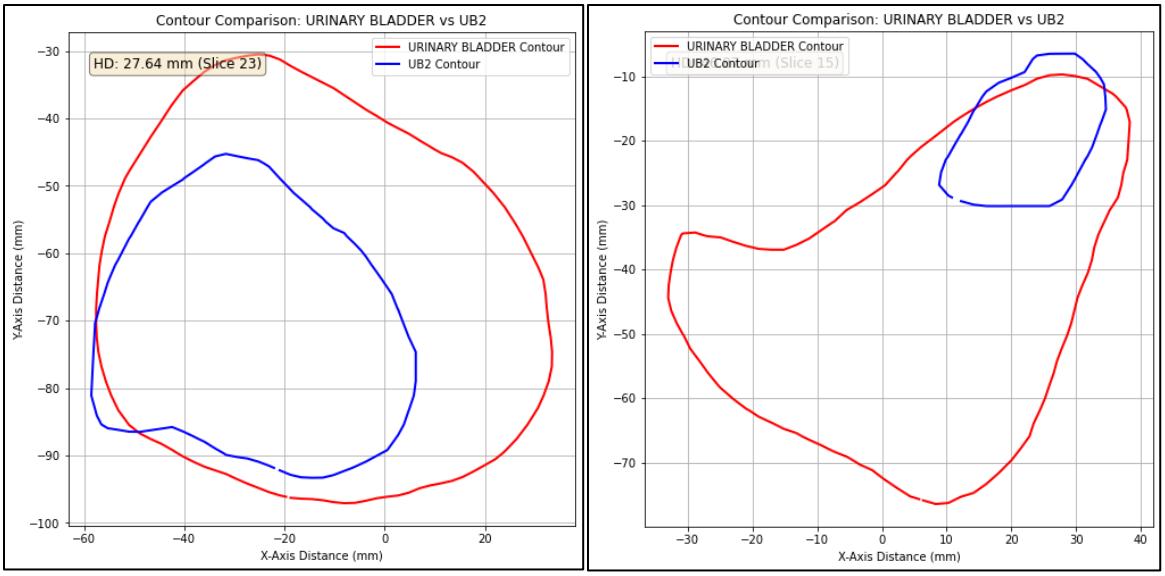
a) b)

Figure 38: a) HD difference, b) DSC difference between CT and CBCT bladder for 3DCRT cases



c) d)

Figure 39: c) HD difference, d) DSC difference between CT and CBCT rectum for IMRT cases



e) f)

Figure 40: c) HD difference, d) DSC difference between CT and CBCT rectum for VMAT cases

CHAPTER-4: RESULT

4.1 First Group Date

Since the first group consisted of 14 patients, with Seven males (prostate) and Seven female (cervix) patients, separate tables are provided for their data.

4.1.1 For Cervix Patients Date

Table 2: For Patient-1 (ID: R2102002347, Treatment Technique: VMAT)

Bladder					
Image	**Organ**	**Volume (cm^3)**	**Dose**	**HD**	**DSC**
CT	**UB1**	**37.46**	**98.95%**	**0**	**1**
CBCT	UB2	140.24	87.33%	37.16	0.0
	UB3	57.77	97.66%	25.5	0.0
	UB4	150.34	80.36%	32.26	0.0
	UB5	151.89	81.56%	32.01	0.0
	UB6	58.31	95.30%	24.8	0.0
	UB7	60.16	96.34%	23.78	0.0
Rectum					
CT	**R1**	**37.88**	**100%**	**0**	**1**
CBCT	R2	38.57	100%	17.13	0.9231
	R3	51.67	100%	29.88	0.8031
	R4	47.63	100%	22.99	0.8727
	R5	52.6	100%	25.61	0.8557
	R6	38.96	100%	22.9	0.9341
	R7	43.21	100%	19.88	0.918

Bladder				
	Volume	**Dose**	**HD**	**DSC**
Mean (CBCT)	103.1183333	0.897583333	29.25166667	0
S. D	44.52762417	0.070463097	4.882480301	0
abs VD	-175.2758498	9.289203301		

Rectum				
	Volume		**HD**	**DSC**
Mean (CBCT)	45.44	1	23.065	0.88445
S. D	5.613002761	0	4.04809729	0.045947425
abs VD	-19.95776135	0		

Table 3: Shows the amount of change in MU after re-planning with the new structure for P-1.

Fx	25	Dose/Fx	200	cGy					
Total	5000	cGy							
Crop UB and CropR								Avg	SD
Fields	**Normal MU**	2	3	4	5	6	7		
A1	**200**	198	198	197	198	196	198		
A2	**209**	195	202	205	195	200	203		
Sum	409	393	400	402	393	396	401		
Diff		3.91198	2.200489	1.711491	3.91198	3.178484	1.95599	2.811736	0.901114
Dose	200	192.176	195.599	196.577	192.176	193.643	196.088	194.3765	1.802229
Total	5000								
Actual Dose received	4859.413203								
abs DD	140.5867971								
% DD	2.811735941								

Table 4: For Patient-2 (ID: 1412005008, Treatment Technique: IMRT)

Bladder					
Image	**Organ**	**Volume (cm^3)**	**Dose**	**HD**	**DSC**
CT	**UB1**	**181.29**	**93.48%**	**0**	**1**
CBCT	UB2	360.21	88.28%	71.57	0.2623
	UB3	374.21	84.34%	75.74	0.7059
	UB4	319.28	85.51%	73.64	0
	UB5	256.17	88.43%	69.99	0.8571
	UB6	169.62	94.01%	61.4	0.7692
	UB7	99.59	97.36%	47.77	0.4
Rectum					
CT	**R1**	**32.83**	**100%**	**0**	**1**
CBCT	R2	41.18	100%	24.7	0.872
	R3	29.36	100%	14.29	0.9279
	R4	55.96	100%	19.45	0.8927
	R5	38.27	100%	13.3	0.9076
	R6	58.71	100%	29.58	0.7146
	R7	26.56	100%	1818	0.8757

Bladder				
	Volume		**HD**	**DSC**
Mean (CBCT)	263.18	0.89655	66.685	0.499083333
S.D	100.377615	0.046035231	9.581751319	0.305199418
abs VD	-45.17072094	4.091784339		



Rectum				
	Volume		**HD**	**DSC**
Mean (CBCT)	41.67333333	1	319.8866667	0.865083333
S.D	12.15328442	0	670.000446	0.069907163
abs VD	-26.93674485	0		

Table 5: Shows the amount of change in MU after re-planning with the new structure for P-2.

Fx	25	Dose/Fx	200	cGy					
Total	5000	cGy							
Crop UB and Crop R								Avg	SD
Fields	**Normal MU**	2	3	4	5	6	7		
A1	**204**	181	181	176	176	174	191		
A2	**164**	170	162	160	160	172	172		
A3	**189**	182	178	183	175	181	181		
A4	**273**	245	251	253	251	253	256		
A5	**193**	178	178	178	179	183	185		
A6	**148**	150	149	149	152	149	149		
A7	**218**	216	213	215	209	211	211		
Sum	1389	1322	1312	1314	1302	1323	1345		
Diff		4.823614	5.543557	5.399568	6.263499	4.75162	3.167747	4.991601	0.95752
Dose	200	190.3528	188.9129	189.2009	187.473	190.4968	193.6645	190.0168	1.915041
Total	5000								
Actual Dose received	4750.419966								
abs DD	249.5800336								
% DD	4.991600672								

Table 6: For Patient-3 (ID: 1409007843, Treatment Technique: 3DCRT with mean dose)

Bladder						
Image	**Organ**	**Volume (cm^3)**	**Dose**	**HD**	**DSC**	**Mean Dose**
CT	**UB1**	**129.16**	**95.45%**	**0**	**1**	**98.7**
CBCT	UB2	80.72	97.24%	46.87	No valid overlap found.	99
	UB3	148.52	95.04%	15.55	No valid overlap found.	98.6
	UB4	194.94	96.40%	19.64	No valid overlap found.	99.2
	UB5	101.72	97.27%	42.63	No valid overlap found.	99.2
	UB6	101.6	96.85%	40.77	No valid overlap found.	99.1
	UB7	177.92	94.51%	24.34	No valid overlap found.	98.5
Rectum						
CT	**R1**	**37.97**	**100%**	**0**	**1**	**100.3**
CBCT	R2	35.25	100%	24.35	0.76	98.1
	R3	41.04	100%	29.67	0.82	99.8
	R4	27.98	100%	16.19	0.8	98.8
	R5	30.07	100%	22.85	0.73	97.3
	R6	22.81	100%	15.2	0.84	99.3
	R7	25.46	100%	14.15	0.9	99.7



Bladder				
	Volume		**HD**	**DSC**
Mean (CBCT)	134.2366667	0.962183333	31.63333333	#DIV/0!
S.D	42.39430216	0.010715008	12.19472381	#DIV/0!
abs VD	-3.930525446	-0.804958966		

Rectum				
	Volume		**HD**	**DSC**
Mean (CBCT)	30.435	1	20.40166667	0.808333333
S.D	6.119664343	0	5.647484445	0.054898897
abs VD	19.84461417	0		

Table 7: Patient-4 (ID: 120506539, Treatment Technique: IMRT)

Bladder					
Image	**Organ**	**Volume (cm^3)**	**Dose**	**HD**	**DSC**
CT	**UB1**	**134.26**	**53.54%**	**0**	**1**
CBCT	UB2	446.52	40.13%	0	No valid overlap found.
	UB3	112.49	68.81%	24.96	No valid overlap found.
	UB4	61.37	60.01%	67.32	No valid overlap found.
	UB5	61.27	52.12%	50.8	No valid overlap found.
	UB6	146.03	58.83%	40.47	No valid overlap found.
	UB7	71.51	56.84%	50.76	No valid overlap found.
Rectum					
CT	**R1**	**44.95**	**95%**	**0**	**1**
CBCT	R2	37.57	100%	0	0.9209
	R3	38.97	100%	0	0.9307
	R4	45.99	100%	0	0.8585
	R5	36.91	100%	29.62	0.8224
	R6	42.06	100%	33.1	0.6403
	R7	37.89	100%	22.53	0.8611

Bladder				
	Volume		**HD**	**DSC**
Mean (CBCT)	149.865	0.561233333	39.05166667	#DIV/0!
S.D	136.1590565	0.087150324	21.60179963	#DIV/0!
abs VD	-11.62297036	-4.82505292		

Rectum				
	Volume		**HD**	**DSC**

Mean (CBCT)	39.89833333	100%	14.20833333	0.838983333
S.D	3.189244512	0.001166786	14.54472919	0.096386728
abs VD	11.23841305	-4.68329665		

Table 8: Shows the amount of change in MU after re-planning with the new structure for P-4.

Fx	25	Dose/Fx	200	cGy					
Total	5000	cGy							
Crop UB and Crop R								Avg	SD
Fields	**Normal MU**	2	3	4	5	6	7		
A1	**199**	190	194	194	194	196	194		
A2	**179**	179	179	175	182	180	184		
A3	**139**	140	144	141	138	138	141		
A4	**171**	171	167	166	168	170	166		
A5	**190**	181	182	181	180	183	182		
A6	**161**	154	152	152	153	155	153		
A7	**138**	142	142	140	139	139	141		
A8	**174**	175	175	177	176	175	177		
A9	**211**	197	203	206	204	207	202		
Sum	1562	1529	1538	1532	1534	1543	1540		
Diff		2.112676	1.536492	1.920615	1.792574	1.216389	1.408451	1.664533	0.307031
Dose	200	195.7746	196.927	196.1588	196.4149	197.5672	197.1831	196.6709	0.614063
Total	5000								
A.D.R	4916.773367								
abs DD	83.22663252								
% DD	1.66453265								

Table 9: Patient-5 (ID: 101006006, Treatment Technique: 3DCRT with mean dose)

Bladder						
Image	**Organ**	**Volume (cm^3)**	**Dose**	**HD**	**DSC**	**Mean Dose**
CT	**UB1**	**389.87**	**77.77%**	**0**	**1**	**4587.7**
CBCT	UB2	390.17	83.85%	27.64	No valid overlap found.	4708.6
	UB3	402.48	83.71%	25.67	No valid overlap found.	4696.5
	UB4	309.11	87.47%	33.08	No valid overlap found.	4770.6
	UB5	234.12	89.22%	41.19	No valid overlap found.	4799.3
	UB6	145.84	84.63%	80.94	No valid overlap found.	4734.9
	UB7	98.1	85.17%	75.22	No valid overlap found.	4757.2
Rectum						
CT	**R1**	**43.26**	**99%**	**0**	**1**	**4863.6**
CBCT	R2	38.72	100%	17.9	0.73	4926.4
	R3	25.26	100%	17.88	0.66	4956
	R4	30.32	100%	16.06	0.72	4962.8
	R5	29.34	100%	18.34	0.67	4970.9



	R6	33.58	100%	18.43	0.61	4979.9
	R7	36.03	100%	20.16	0.59	4975.5

Bladder				
	Volume		**HD**	**DSC**
Mean (CBCT)	263.3033333	85.68%	47.29	#DIV/0!
S.D	115.1122563	0.020136182	22.37963509	#DIV/0!
abs VD	32.46381272	-10.16458789		

Rectum				
	Volume		**HD**	**DSC**
Mean (CBCT)	32.20833333	1	18.12833333	0.663333333
S.D	4.45273106	0	1.201060874	0.051532083
abs VD	25.54707967	-0.654252642		

Table 10: Patient-6 (ID: 2307002848, Treatment Technique: 3DCRT with mean dose)

Bladder						
Image	**Organ**	**Volume (cm^3)**	**Dose%**	**HD**	**DSC**	**Mean Dose**
CT	**UB1**	**124.33**	**88.23%**	**0**	**1**	**96.4**
CBCT	UB2	189.25	80.86%	32.98	No valid overlap found.	94.1
	UB3	58.17	85.02%	34.47	No valid overlap found.	94.8
	UB4	113.36	75.89%	19.57	No valid overlap found.	91.9
	UB5	259.23	82.85%	44.42	No valid overlap found.	95
	UB6	131.66	77.16%	57.32	No valid overlap found.	92.3
	UB7	57.47	73.17%	29.71	No valid overlap found.	90.9
Rectum						
CT	**R1**	**63.17**	**100%**	**0**	**1**	**99.4**
CBCT	R2	71.02	100%	21.38	0.92	101.2
	R3	69.82	100%	17.34	0.93	100.4
	R4	96.75	100%	17.84	0.93	100.9
	R5	56.89	100%	22.52	0.91	100.3
	R6	64.71	100%	18.17	0.95	100.9
	R7	66.45	100%	19.23	0.94	100.1

Bladder				
	Volume		**HD**	**DSC**
Mean (CBCT)	134.8566667	79.16%	36.41166667	#DIV/0!
S.D	71.61148317	0.041113397	11.87007641	#DIV/0!
abs VD	-8.466714925	10.28183913		

Rectum				
	Volume		**HD**	**DSC**
Mean (CBCT)	70.94	1	19.41333333	0.93
S.D	12.40827949	0	1.909307612	0.012909944
abs VD	-12.30014247	0		

Table 11: Patient-7 (ID: 2212002952, Treatment Technique: VMAT)

Bladder					
Image	**Organ**	**Volume (cm^3)**	**Dose %**	**HD**	**DSC**
CT	**UB1**	**73.05**	**88.43%**	**0**	**1**
CBCT	UB2	503.12	68.61%	84.14	0.1433
	UB3	214.18	78.19%	49.25	0.491
	UB4	186.13	77.10%	46.78	0.8764
	UB5	246.95	82.61%	59.69	0.9444
	UB6	217.05	88.01%	46.09	0.7143
	UB7	296.07	83.82%	61.68	0.48
Rectum					
CT	**R1**	**50.56**	**92.50%**	**0**	**1**
CBCT	R2	61.29	100.00%	19.04	0.4337
	R3	64.2	99.70%	19.1	0.5128
	R4	98.37	98.67%	22.22	0.3725
	R5	50.72	99.99%	19.76	0.7952
	R6	77.3	98.91%	28.63	0.2762
	R7	61.48	100.00%	19.27	0.2981

Bladder				
	Volume		**HD**	**DSC**
Mean (CBCT)	277.25	0.797233333	57.93833333	0.608233333
S.D	106.6100829	0.06144422	13.19074349	0.271576255
abs VD	-279.5345654	9.845829093		

Rectum				
	Volume		**HD**	**DSC**
Mean (CBCT)	68.89333333	99.55%	21.33666667	0.448083333
S.D	15.30038961	0.005484752	3.440138886	0.174469113
abs VD	-36.26054852	-7.616216216		

Table 12: Shows the amount of change in MU after re-planning with the new structure for P-7.

Fx	25	Dose/Fx	200	cGy
Total	5000	cGy		

Crop UB and CropR							Avg	SD
Fields	**Normal MU**	2	3	4	5	6	7	
A1	**243**	239	244	251	242	244	240	

A2	**244**	240	244	245	247	234	239		
Sum	487	479	488	496	489	478	479		
Diff		1.64271	-0.20534	-1.84805	-0.41068	1.848049	1.64271	0.444901	1.369353
Dose	200	196.7146	200.4107	203.6961	200.8214	196.3039	196.7146	199.1102	2.738706
Total	5000								
A.D.R	4977.754962								
abs DD	22.24503765								
% DD	0.444901								

4.1.2 For Prostate Patients Date

Table 13: For Patient-8 (ID: 2209005406, Treatment Technique: IMRT)

Bladder					
Image	**Organ**	**Volume (cm^3)**	**Dose**	**HD**	**DSC**
CT	**UB1**	**187.49**	**13.11%**	**0**	**1**
CBCT	UB2	158.67	27.96%	86.5	'URINARY BLADDER' not found
	UB3	166.96	21.72%	46.86	'URINARY BLADDER' not found
	UB4	312.3	11.53%	25.97	'URINARY BLADDER' not found
	UB5	299.93	11.41%	22.95	'URINARY BLADDER' not found
	UB6	321.61	9.76%	27.27	'URINARY BLADDER' not found
	UB7	232.62	14.59%	20.33	'URINARY BLADDER' not found
Rectum					
CT	**R1**	**70.2**	**75.68%**	**0**	**1**
CBCT	R2	87.96	73.29%	14.56	0.7586
	R3	96.59	64.91%	21.43	0.8
	R4	91.82	73.21%	14.13	0.8824
	R5	119.4	75.79%	19.88	1
	R6	102.89	68.64%	18.13	0.8333
	R7	100.61	59.86%	18.02	0.9333

Bladder				
	Volume		**HD**	**DSC**
Mean (CBCT)	248.6816667	0.1616167	38.31333333	#DIV/0!
S.D	67.12020794	0.0655207	23.19750609	#DIV/0!
abs VD	0.326372962	-23.277396		

Rectum				
	Volume		**HD**	**DSC**
Mean (CBCT)	99.87833333	69.28%	17.69166667	0.867933
S.D	10.0705634	0.0550981	2.631751107	0.0712
abs VD	0.422768281	8.4522551		

Table 14: Shows the amount of change in MU after re-planning with the new structure for P-8.

Fx	20	Dose/Fx	300	cGy					
Total	6000	cGy							
Crop UB and Crop R								Avg	SD
Fields	**Normal MU**	2	3	4	5	6	7		
A1	**283**	280	278	278	281	278	283		
A2	**294**	281	281	294	281	293	294		
A3	**258**	257	251	258	257	258	258		
A4	**249**	249	249	249	249	249	249		
A5	**292**	292	292	292	292	292	292		
A6	**291**	293	291	292	292	293	293		
A7	**269**	269	269	269	270	270	269		
A8	**245**	245	245	245	245	245	245		
A9	**288**	288	288	288	288	288	288		
Sum	2469	2454	2444	2465	2455	2466	2471		
Diff		0.607533	1.012556	0.162009	0.567031	0.121507	-0.081	0.398272	0.367818
Dose	300	298.1774	296.9623	299.514	298.2989	299.6355	300.243	298.8052	1.103453
Total	6000								
A. D. R	5976.104								
abs DD	23.8963143								
% DD	0.398271905								

Table 15: Patient-9 (ID: 2301007650, Treatment Technique: IMRT)

Bladder					
Image	**Organ**	**Volume (cm^3)**	**Dose**	**HD**	**DSC**
CT	**UB1**	**243.93**	**23.46%**	**URINARY BLADDER' not found**	**URINARY BLADDER' not found**
CBCT	UB2	439.01	14.11%	'URINARY BLADDER' not found	URINARY BLADDER' not found
	UB3	254.42	20.50%	'URINARY BLADDER' not found	'URINARY BLADDER' not found
	UB4	276.41	21.24%	'URINARY BLADDER' not found	URINARY BLADDER' not found
	UB5	252.44	19.58%	'URINARY BLADDER' not found	'URINARY BLADDER' not found
	UB6	228.84	11.82%	'URINARY BLADDER' not found	'URINARY BLADDER' not found
	UB7	193.97	32.15%	'URINARY BLADDER' not found	'URINARY BLADDER' not found
Rectum					
CT	**R1**	**57.28**	**91.35%**	**'RECTUM' not found**	**'RECTUM' not found**
CBCT	R2	92.82	76.37%	'RECTUM' not found	'RECTUM' not found
	R3	87.7	79.48%	'RECTUM' not found	'RECTUM' not found
	R4	88.35	81.97%	'RECTUM' not found	'RECTUM' not found
	R5	107.95	83.46%	'RECTUM' not found	'RECTUM' not found
	R6	93.96	86.73%	'RECTUM' not found	'RECTUM' not found
	R7	103.98	80.88%	'RECTUM' not found	'RECTUM' not found

Bladder				
	Volume		**HD**	**DSC**
Mean (CBCT)	274.1816667	19.90%	#DIV/0!	#DIV/0!



S.D	78.02387101	0.0646956	#DIV/0!	#DIV/0!
abs VD	0.124017819	15.174766		

Rectum				
	Volume		**HD**	**DSC**
Mean (CBCT)	95.79333333	81.48%	#DIV/0!	#DIV/0!
S.D	7.614822534	0.032196	#DIV/0!	#DIV/0!
abs VD	0.672369646	10.802773		

Table 16: Shows the amount of change in MU after re-planning with the new structure for P-9.

Fx	20	Dose/Fx	300	cGy					
Total	6000	cGy							
Crop UB and R								Avg	SD
Fields	**Normal MU**	2	3	4	5	6	7		
A1	**289**	288	288	289	289	289	290		
A2	**262**	263	262	262	262	263	262		
A3	**242**	242	243	242	243	241	242		
A4	**278**	278	278	278	278	278	278		
A5	**333**	333	332	333	333	332	332		
A6	**281**	281	281	281	281	281	281		
A7	**288**	288	288	288	288	288	288		
A8	**274**	275	274	275	278	274	273		
A9	**298**	298	298	298	298	298	298		
Sum	2545	2546	2544	2546	2550	2544	2544		
Diff		-0.03929	0.039293	-0.03929	-0.19646	0.039293	0.039293	-0.0262	0.083865
Dose	300	300.1179	299.8821	300.1179	300.5894	299.8821	299.8821	300.0786	0.251596
Total	6000								
A. D. R.	6001.572								
abs DD	-1.571709234								
% DD	-0.026195154								

Table 17: Patient-10 (ID: 2106003366, Treatment Technique: IMRT)

Bladder					
Image	**Organ**	**Volume (cm^3)**	**Dose**	**HD**	**DSC**
CT	**UB1**	**260.78**	**29.29%**	**0**	**1**
CBCT	UB2	268.45	34.52%	33.52	0.8914
	UB3	292.09	30.80%	77.94	0.959
	UB4	285.1	31.50%	94.38	0.9557
	UB5	638.1	29.34%	48.25	0.941
	UB6	598.65	32.35%	49.32	0.8974
	UB7	652.2	31.39%	55.74	0.9363



Rectum					
CT	**R1**	**53.8**	**70.54%**	**0**	**1**
CBCT	R2	58.92	73.26%	10.49	0.9538
	R3	62.08	63.81%	9.82	0.9739
	R4	71.87	73.33%	13.56	0.9506
	R5	54.04	70.27%	11.32	0.9257
	R6	62.51	63.94%	11.18	0.9333
	R7	69.29	66.89%	10.74	0.9202

Bladder				
	Volume		**HD**	**DSC**
Mean (CBCT)	455.765	31.65%	59.85833333	0.930133
S.D	174.7626208	0.0157457	20.3107955	0.026504
abs VD	0.74769921	-8.0573575		

Rectum				
	Volume		**HD**	**DSC**
Mean (CBCT)	63.11833333	68.58%	11.185	0.942917
S.D	6.00238124	0.0396778	1.169269715	0.01845
abs VD	0.173203222	2.7738399		

Table 18: Shows the amount of change in MU after re-planning with the new structure for P-10.

Fx	20	Dose/Fx	300	cGy					
Total	6000	cGy							
Crop UB and R								Avg	SD
Fields	**Normal MU**	2	3	4	5	6	7		
A1	**237**	256	246	250	240	241	242		
A2	**219**	214	205	213	202	203	203		
A3	**228**	219	211	218	209	211	211		
A4	**212**	213	209	212	212	209	210		
A5	**240**	233	234	239	231	230	232		
A6	**206**	195	198	199	195	198	199		
A7	**204**	208	204	209	202	200	202		
A8	**218**	217	200	215	203	201	206		
A9	**221**	242	226	231	213	217	216		
Sum	1985	1997	1933	1986	1907	1910	1921		
Diff		-0.60453	2.619647	-0.05038	3.929471	3.778338	3.224181	2.149454	1.808386
Dose	300	301.8136	292.1411	300.1511	288.2116	288.665	290.3275	293.5516	5.425157
Total	6000								
ADR	5871.032746								
abs DD	128.9672544								
% DD	2.14945424								

Table 19: Patient-11 (ID: 1910000336, Treatment Technique: VMAT)

Bladder					
Image	**Organ**	**Volume (cm^3)**	**Dose**	**HD**	**DSC**
CT	**UB1**	**122.83**	**60.60%**	**'URINARY BLADDER' not found**	**'URINARY BLADDER' not found**
CBCT	UB2	201.58	45.15%	'URINARY BLADDER' not found	'URINARY BLADDER' not found
	UB3	241.53	39.14%	'URINARY BLADDER' not found	'URINARY BLADDER' not found
	UB4	292.85	41.67%	'URINARY BLADDER' not found	URINARY BLADDER' not found
	UB5	290.76	34.82%	'URINARY BLADDER' not found	'URINARY BLADDER' not found
	UB6	427.59	28.40%	'URINARY BLADDER' not found	'URINARY BLADDER' not found
	UB7	429.67	31.30%	'URINARY BLADDER' not found	'URINARY BLADDER' not found
Rectum					
CT	**R1**	**48.22**	**71%**	**'RECTUM' not found**	**'RECTUM' not found**
CBCT	R2	45.45	100%	RECTUM' not found	'RECTUM' not found
	R3	27.82	100%	'RECTUM' not found	'RECTUM' not found
	R4	25.17	100%	'RECTUM' not found	'RECTUM' not found
	R5	30.61	100%	'RECTUM' not found	'RECTUM' not found
	R6	46.69	100%	'RECTUM' not found	'RECTUM' not found
	R7	20.82	100%	'RECTUM' not found	'RECTUM' not found

Bladder				
	Volume		**HD**	**DSC**
Mean (CBCT)	313.9966667	36.75%	#DIV/0!	#DIV/0!
S.D	86.7561855	0.0582564	#DIV/0!	#DIV/0!
abs VD	1.556351597	39.361936		

Rectum				
	Volume		**HD**	**DSC**
Mean (CBCT)	32.76	100.00%	#DIV/0!	#DIV/0!
S.D	9.868120388	0	#DIV/0!	#DIV/0!
abs VD	0.320613853	-40.567894		

Table 20: Shows the amount of change in MU after re-planning with the new structure for P-11.

Fx	20	Dose/Fx	300	cGy					
Total	6000	cGy							
Crop UB and CropR								Avg	SD
Fields	**Normal MU**	2	3	4	5	6	7		
A1	**279**	302	312	294	304	316	300		
A2	**306**	282	296	301	314	306	313		
Sum	585	584	608	595	618	622	613		
Diff		0.17094	-3.93162	-1.7094	-5.64103	-6.32479	4.78632	-3.7037	2.267062
Dose	300	299.4872	311.7949	305.1282	316.9231	318.9744	314.359	311.1111	6.801185
Total	6000								
ADR	6222.222222								
abs DD	-222.2222222								
% DD	-3.703703704								

Table 21: Patient-12 (ID: 1909002860, Treatment Technique: VMAT)

Bladder					
Image	**Organ**	**Volume (cm^3)**	**Dose**	**HD**	**DSC**
CT	**UB1**	**229.92**	**27.78%**	**'URINARY BLADDER' not found**	**'URINARY BLADDER' not found**
CBCT	UB2	116.61	60.39%	URINARY BLADDER' not found	'URINARY BLADDER' not found
	UB3	114.18	40.05%	'URINARY BLADDER' not found	'URINARY BLADDER' not found
	UB4	185.48	30.87%	'URINARY BLADDER' not found	URINARY BLADDER' not found
	UB5	116.99	45.85%	'URINARY BLADDER' not found	'URINARY BLADDER' not found
	UB6	101.73	37.30%	'URINARY BLADDER' not found	'URINARY BLADDER' not found
	UB7	100.81	43.17%	'URINARY BLADDER' not found	'URINARY BLADDER' not found
Rectum					
CT	**R1**	**44.95**	**78.16%**	**'RECTUM' not found**	**'RECTUM' not found**
CBCT	R2	48.51	66.07%	RECTUM' not found	'RECTUM' not found
	R3	55.31	67.64%	'RECTUM' not found	'RECTUM' not found
	R4	53.99	65.53%	'RECTUM' not found	'RECTUM' not found
	R5	36.91	78.82%	'RECTUM' not found	'RECTUM' not found
	R6	55.94	64.31%	'RECTUM' not found	'RECTUM' not found
	R7	55.82	66.65%	'RECTUM' not found	'RECTUM' not found

Bladder				
	Volume		**HD**	**DSC**
Mean (CBCT)	122.6333333	42.94%	#DIV/0!	#DIV/0!
S.D	28.87472174	0.0911839	#DIV/0!	#DIV/0!
abs VD	0.466626073	-54.565635		

Rectum				
	Volume		**HD**	**DSC**
Mean (CBCT)	51.08	68.17%	#DIV/0!	#DIV/0!
S.D	6.829372836	0.0486986	#DIV/0!	#DIV/0!
abs VD	0.136373749	12.781474		

Table 22: Shows the amount of change in MU after re-planning with the new structure for P-12.

Fx	20	Dose/Fx	300	cGy					
Total	6000	cGy							
Crop UB and CropR								Avg	SD
Fields	**Normal MU**	2	3	4	5	6	7		
A1	**435**	644	662	630	664	637	651		
A2	**438**	636	619	663	634	665	608		
Sum	873	1280	1281	1293	1298	1302	1259		
Diff		-46.6208	-46.7354	-48.11	-48.6827	-49.1409	-44.2153	-47.2509	1.645064
Dose	300	439.8625	440.2062	444.3299	446.0481	447.4227	432.646	441.7526	4.935191
Total	6000								
ADR	8835.051546								
abs DD	-2835.051546								
% DD	-47.25085911								

Table 23: Patient-13 (ID: 2308003968, Treatment Technique: 3DCRT with mean dose)

Bladder						
Image	**Organ**	**Volume (cm^3)**	**Dose**	**HD**	**DSC**	**Mean Dose**
CT	**UB1**	**189**	**97.05%**	**0**	**1**	**27.78**
CBCT	UB2	245.75	87.91%	85.44	0.98	60.39
	UB3	295.27	76.51%	51.63	0.89	40.05
	UB4	221.4	92.38%	42.47	0.9	30.87
	UB5	204.31	90.17%	74.73	0.94	45.85
	UB6	268.47	77.66%	47.38	0.96	37.3
	UB7	215.19	83.88%	49.33	0.44	43.17
Rectum						
CT	**R1**	**69.73**	**77.46%**	**0**	**1**	**78.16**
CBCT	R2	104.11	81.61%	21.06	0.9	66.07
	R3	101.66	79.92%	21.35	0.94	67.64
	R4	111.22	85.30%	24.53	0.94	65.53
	R5	95.4	78.58%	23.62	0.93	78.82
	R6	109.99	81.56%	21.69	0.85	64.31
	R7	103.73	82.97%	21.89	0.88	66.65

Bladder				
	Volume		**HD**	**DSC**
Mean (CBCT)	241.7316667	84.75%	58.49666667	0.851667
S.D	31.92610344	0.0600696	15.81706638	0.186763
abs VD	0.279003527	12.672162		

Rectum				
	Volume		**HD**	**DSC**
Mean (CBCT)	104.3516667	81.66%	22.35666667	0.906667
S.D	5.272366378	0.0214057	1.269930007	0.0335
abs VD	0.496510349	-5.4178501		

Table 24: Patient-14 (ID: 2109015736, Treatment Technique: VMAT)

VMAT	**HN: 2109015736**		**Male**	
Bladder				
Image	**Organ**	**Volume (cm^3)**	**Dose**	**HD**	**DSC**
CT	**UB1**	**200**	**39.16%**	**'URINARY BLADDER' not found**	**'URINARY BLADDER' not found**
CBCT	UB2	95.55	49.77%	URINARY BLADDER' not found	'URINARY BLADDER' not found
	UB3	78.43	67.73%	'URINARY BLADDER' not found	'URINARY BLADDER' not found
	UB4	123.07	59.66%	'URINARY BLADDER' not found	URINARY BLADDER' not found
	UB5	114.64	61.34%	'URINARY BLADDER' not found	'URINARY BLADDER' not found
	UB6	65.9	68.87%	'URINARY BLADDER' not found	'URINARY BLADDER' not found
	UB7	97.13	54.39%	'URINARY BLADDER' not found	'URINARY BLADDER' not found



Rectum					
CT	**R1**	**39.2**	**70.39%**	**'RECTUM' not found**	**'RECTUM' not found**
CBCT	R2	81.12	86.98%	RECTUM' not found	'RECTUM' not found
	R3	34.23	75.42%	'RECTUM' not found	'RECTUM' not found
	R4	32.05	66.88%	'RECTUM' not found	'RECTUM' not found
	R5	33.42	72.66%	'RECTUM' not found	'RECTUM' not found
	R6	49.73	81.18%	'RECTUM' not found	'RECTUM' not found
	R7	78.6	87.73%	'RECTUM' not found	'RECTUM' not found

Bladder				
	Volume		**HD**	**DSC**
Mean (CBCT)	95.78666667	60.29%	#DIV/0!	#DIV/0!
S.D	19.56253108	0.0678171	#DIV/0!	#DIV/0!
abs VD	0.521066667	-53.966633		

Rectum				
	Volume		**HD**	**DSC**
Mean (CBCT)	51.525	78.48%	#DIV/0!	#DIV/0!
S.D	20.8898913	0.0755974	#DIV/0!	#DIV/0!
abs VD	0.314413265	-11.486007		

Table 25: Shows the amount of change in MU after re-planning with the new structure for P-14.

Fx	20	Dose/Fx	300	cGy					
Total	6000	cGy							
		Crop (UB+R)						Avg	SD
Fields	**Normal MU**	2	3	4	5	6	7		
A1	**296**	424	394	422	406	400	421		
A2	**259**	394	381	397	400	391	392		
Sum	555	818	775	819	806	791	813		
Diff		-47.3874	-39.6396	-47.5676	-45.2252	-42.5225	-46.4865	-44.8048	2.863423
Dose	300	442.1622	418.9189	442.7027	435.6757	427.5676	439.4595	434.4144	8.590268
Total	6000								
ADR	8688.288288								
abs DD	-2688.288288								
% DD	-44.8048048								

4.2 Second (Control) Group Data

Table 26: Patient-1 (ID: 2406005526, Gender: Female, Treatment Technique: VMAT)

Bladder					
Image	**Organ**	**Volume (cm^3)**	**Dose**	**HD**	**DSC**
CT	**UB1**	**39.41**	**77.15%**	**0**	**1**
CBCT	UB2	411.66	47.71%	75.59	0
	UB3	126.31	51.95%	64.33	0
	UB4	100.46	69.39%	48.77	0
	UB5	195.53	65.02%	64.94	0
	UB6	153.19	64.09%	63.98	0
	UB7	81.22	70.87%	40.36	0
Rectum					
CT	**R1**	**34.33**	**99.43%**	**0**	**1**
CBCT	R2	26.37	100.00%	24.38	0.9061
	R3	80.39	100.00%	27.06	0.6445
	R4	29.27	100.00%	26.43	0.9195
	R5	36.28	100.00%	22.94	0.7629
	R6	76.22	100.00%	24.04	0.7362
	R7	35.6	100.00%	20.63	0.9322

Bladder				
	Volume		**HD**	**DSC**
Mean (CBCT)	178.0616667	61.51%	59.66166667	0
S.D	110.7327758	0.086651981	11.64185469	0
abs VD	3.518184894	0.202786779		

Rectum				
	Volume		**HD**	**DSC**
Mean (CBCT)	47.355	100.00%	24.24666667	0.8169
S.D	22.01966521	0	2.142036933	0.110797
abs VD	0.379405768	-0.57326763		

Table 27: Shows the amount of change in MU after re-planning with the new structure for P-1.

Fx	20	Dose/Fx	300	cGy					
Total	6000	cGy							
No Crop								Avg	SD
Fields	**Normal MU**	2	3	4	5	6	7		
A1	**251**	245	249	240	228	236	222		
A2	**266**	237	236	240	250	242	253		
Sum	517	482	485	480	478	478	475		
Diff		6.769826	6.189555	7.156673	7.54352	7.54352	8.123791	7.221148	0.618418
Dose	300	279.6905	281.4313	278.53	277.3694	277.3694	275.6286	278.3366	1.855254
Total	6000								
ADR	5566.731141								
abs DD	433.2688588								
% DD	7.221147647								

Table 28: Patient-1 (ID: 2312006748, Gender: Male, Treatment Technique: VMAT)

Bladder					
Image	**Organ**	**Volume (cm^3)**	**Dose**	**HD**	**DSC**
CT	**UB1**	**243.57**	**7.92%**	**0**	**1**
CBCT	UB2	266.94	9.58%	22.35	0.8952
	UB3	112.75	13.57%	70.26	0.9061
	UB4	149.5	6.26%	71.36	0
	UB5	129.84	8.63%	95.3	0.4343
	UB6	192.59	11.07%	74.67	0.9446
	UB7	153.15	11.72%	71.24	0.9576
Rectum					
CT	**R1**	**106.95**	**80.68%**	**0**	**1**
CBCT	R2	121.89	77.32%	14.28	0.944
	R3	173.23	82.92%	18.3	0.9338
	R4	214.01	86.22%	27.72	0.889
	R5	201.46	82.95%	25.86	0.9625
	R6	105.67	69.63%	15.46	0.9421
	R7	129.51	75.75%	13.7	0.8976

Bladder				
	Volume		**HD**	**DSC**
Mean (CBCT)	167.4616667	10.14%	67.53	0.689633
S.D	50.76674352	0.023371736	21.98207906	0.357472
abs VD	0.312470063	-28.0092593		

Rectum				
	Volume		**HD**	**DSC**
Mean (CBCT)	157.6283333	79.13%	19.22	0.928167
S.D	41.42072602	0.055403833	5.602022492	0.027651
abs VD	0.473850709	1.91910428		

Table 29: Shows the amount of change in MU after re-planning with the new structure for P-2.

Fx	25	Dose/Fx	200	cGy					
Total	5000	cGy							
Crop UB and Crop R								Avg	SD
Fields	**Normal MU**	2	3	4	5	6	7		
A1	**245**	247	226	234	229	250	231		
A2	**237**	247	225	241	245	231	235		
A3	**235**	238	249	231	235	239	232		
Sum	717	732	700	706	709	720	698		
Diff		-2.09205	2.37099	1.53417	1.11576	-0.41841	2.64993	0.860065	1.65105
Dose	200	204.1841	195.258	196.9317	197.7685	200.8368	194.7001	198.2799	3.3021
Total	5000								
ADR	4956.996746								
abs DD	43.0032543								
% DD	0.860065086								

Table 30: Patient-3 (ID: 240702943, Gender: Male, Treatment Technique: IMRT)

Bladder					
Image	**Organ**	**Volume (cm^3)**	**Dose**	**HD**	**DSC**
CT	**UB1**	**176.68**	**21.38%**	**0**	**1**
CBCT	UB2	114.32	35.73%	40.3	0
	UB3	126.08	39.35%	58.77	0.6154
	UB4	133.26	34.53%	37.58	0.7692
	UB5	109.57	36.91%	61.84	0
	UB6	103.59	40.20%	63.98	0.1538
	UB7	128.4	40.23%	56.97	0.0485
Rectum					
CT	**R1**	**60.29**	**74.21%**	**0**	**1**
CBCT	R2	104.4	72.81%	19.19	0.93
	R3	115.61	73.68%	19.35	0.898
	R4	103.54	72.65%	14.11	0.8893
	R5	103.26	69.32%	15.84	0.9191
	R6	105.83	70.68%	11.27	0.8998
	R7	105.41	69.34%	13.88	0.8905

Bladder				
	Volume		**HD**	**DSC**
Mean (CBCT)	119.2033333	37.83%	53.24	0.264483
S.D	10.72288311	0.022298561	10.380114	0.310023
abs VD	0.325315071	-76.9176801		

Rectum				
	Volume		**HD**	**DSC**
Mean (CBCT)	106.3416667	71.41%	15.60666667	0.90445
S.D	4.245544396	0.017241584	2.913626987	0.015027
abs VD	0.763835904	3.768584647		

Table 31: Shows the amount of change in MU after re-planning with the new structure for P-3.

Fx	25	Dose/Fx	200	cGy					
Total	5000	cGy							
Crop UB and Crop R								Avg	SD
Fields	**Normal MU**	**2**	**3**	**4**	**5**	**6**	**7**		
A1	**380**	270	270	284	275	269	273		
A2	**328**	233	277	227	234	230	235		
A3	**264**	178	179	178	176	176	185		
A4	**401**	303	300	301	296	298	298		
A5	**420**	283	287	293	293	291	292		
A6	**400**	270	275	272	271	273	274		
A7	**277**	214	209	197	208	209	213		
A8	**357**	237	230	237	240	241	247		
A9	**433**	282	283	287	285	282	285		
Sum	3260	2270	2310	2276	2278	2269	2302		
Diff		30.3681	29.1411	30.18405	30.1227	30.39877	29.3865	29.93354	0.488368
Dose	200	139.2638	141.7178	139.6319	139.7546	139.2025	141.227	140.1329	0.976737
Total	5000								
ADR	3503.323108								
abs DD	1496.676892								
% DD	29.93353783								

Table 32: Patient-4 (ID: 2404000300, Gender: Male, Treatment Technique: VMAT)

Bladder					
Image	**Organ**	**Volume (cm^3)**	**Dose**	**HD**	**DSC**
CT	**UB1**	**136.11**	**43.60%**	**0**	**1**
CBCT	UB2	141.72	46.98%	36.74	0.9026
	UB3	129.02	71.82%	44.64	0.3709
	UB4	98.01	68.15%	32.12	0.3383
	UB5	141.66	46.67%	34.17	0.7778
	UB6	126.63	51.13%	37.04	0.7329
	UB7	73.61	50.59%	24.55	0.2406
Rectum					
CT	**R1**	**116.83**	**86.71%**	**0**	**1**
CBCT	R2	153.52	86.36%	37.64	0.9007
	R3	54.66	81.65%	33.74	0.7751
	R4	94.08	70.43%	30.23	0.8207
	R5	94.78	82.91%	37.05	0.8239
	R6	61.69	83.62%	39.21	0.8114
	R7	207.25	75.22%	21.53	0.9509

Bladder				
	Volume		**HD**	**DSC**
Mean (CBCT)	118.4416667	55.89%	34.87666667	0.560517
S.D	24.78996129	0.101588763	6.030662392	0.2522
abs VD	0.129809223	-28.1880734		

Rectum				
	Volume		**HD**	**DSC**
Mean (CBCT)	110.9966667	80.03%	33.23333333	0.847117
S.D	53.58336081	0.054646117	5.99931848	0.059639
abs VD	0.049930098	7.701918272		

Table 33: Shows the amount of change in MU after re-planning with the new structure for P-4.

Fx	25	Dose/Fx	200	cGy					
Total	5000	cGy							
Crop (UB+R)								Avg	SD
Fields	**Normal MU**	2	3	4	5	6	7		
A1	**365**	307	315	321	311	330	310		
A2	**346**	317	313	310	315	309	309		
Sum	711	624	628	631	626	639	619		
Diff		12.23629	11.6737	11.25176	11.95499	10.12658	12.93952	11.69714	0.872377
Dose	200	175.5274	176.6526	177.4965	176.09	179.7468	174.121	176.6057	1.744753
Total	5000								
ADR	4415.142991								
abs DD	584.8570089								
% DD	11.69714018								

Table 34: Patient-5 (ID: 1607003598, Gender: Male, Treatment Technique: VMAT)

Bladder					
Image	**Organ**	**Volume (cm^3)**	**Dose**	**HD**	**DSC**
CT	**UB1**	**265.54**	**8.32%**	**0**	**1**
CBCT	UB2	105.01	33.93%	70.94	0.25
	UB3	159.58	32.03%	78.65	0.1256
	UB4	142	26.98%	77.36	0.1186
	UB5	177.86	22.61%	77.19	0.1014
	UB6	150.55	24.41%	73.1	0
	UB7	120.95	23.50%	71.84	0
Rectum					
CT	**R1**	**93.72**	**83.53%**	**0**	**1**
CBCT	R2	140.86	80.36%	19.86	0.9702
	R3	95.26	64.43%	26.42	0.9373
	R4	102.88	72.73%	20.4	0.9655
	R5	99.96	68.08%	15.95	0.9062
	R6	104.52	68.09%	19.25	0.9818
	R7	105.07	73.22%	19.45	0.9263

Bladder				
	Volume		**HD**	**DSC**
Mean (CBCT)	142.6583333	27.24%	74.84666667	0.099267
S.D	24.07232395	0.043050577	2.989602352	0.085147
abs VD	0.462761417	-227.44391		

Rectum				
	Volume		**HD**	**DSC**
Mean (CBCT)	108.0916667	71.15%	20.22166667	0.947883
S.D	15.01992057	0.050916972	3.118992448	0.02669
abs VD	0.153346849	14.8190271		

Table 35: Shows the amount of change in MU after re-planning with the new structure for P-5.

Fx	20	Dose/Fx	300	cGy
Total	6000	cGy		

Crop (UB+R)								Avg	SD
Fields	**Normal MU**	2	3	4	5	6	7		
A1	**231**	219	231	226	226	245	237		
A2	**230**	219	228	247	232	234	238		
A3	224	219	227	220	233	219	228		
Sum	685	657	686	693	691	698	703		
Diff		4.087591	-0.14599	-1.16788	-0.87591	-1.89781	-2.62774	-0.43796	2.168592
Dose	200	191.8248	200.292	202.3358	201.7518	203.7956	205.2555	200.8759	4.337183
Total	5000								
ADR	4017.518248								
abs DD	982.4817518								
% DD	19.64963504								

Table 36: Patient-6 (ID: 2408004082, Gender: Female, Treatment Technique: VMAT)

Bladder					
Image	**Organ**	**Volume (cm^3)**	**Dose**	**HD**	**DSC**
CT	**UB1**	**156.49**	**69.20%**	**0**	**1**
CBCT	UB2	396.97	68.46%	89.81	0
	UB3	144.04	74.64%	63.5	0
	UB4	290.61	65.98%	83.67	0
	UB5	247.51	65.41%	84.97	0
	UB6	238.6	74.92%	74.79	0
	UB7	268.12	67.87%	82.78	0
Rectum					
CT	**R1**	**45.65**	**85.04%**	**0**	**1**
CBCT	R2	41.91	100.0%	22.86	0.6927
	R3	38.99	100.0%	21.5	0.6697
	R4	39.22	100.0%	22.1	0.7095
	R5	38.83	100.0%	16.6	0.7921
	R6	37.21	100.0%	27.25	0.6329
	R7	39.34	100.0%	20.04	0.6757

Bladder				
	Volume		**HD**	**DSC**
Mean (CBCT)	264.3083333	69.55%	79.92	0
S.D	74.95887615	0.03843608	8.577975674	0
abs VD	0.688979061	-0.50096339		

Rectum				
	Volume		**HD**	**DSC**
Mean (CBCT)	39.25	100.0%	21.725	0.695433
S.D	1.383847294	0	3.191393374	0.04918
abs VD	0.140197152	-17.5917215		

Table 37: Shows the amount of change in MU after re-planning with the new structure for P-6.

Fx	25	Dose/Fx	200	cGy					
Total	5000	cGy							
Crop (UB+R)								Avg	SD
Fields	**Normal MU**	2	3	4	5	6	7		
A1	**290**	295	300	292	268	311	272		
A2	**281**	260	282	271	287	273	269		
Sum	571	555	582	563	555	584	541		
Diff		2.802102	-1.92644	1.401051	2.802102	-2.27671	5.25394	1.342674	2.687884
Dose	200	194.3958	203.8529	197.1979	194.3958	204.5534	189.4921	197.3147	5.375769
Total	5000								
ADR	4932.866316								
abs DD	67.1336836								
% DD	1.342673672								

4.3 Comparison Between First and Control Group

Table 38: Dosimetry Data for first group

Dosimetry			
SL No.	**Avg MU**	**Overall abs DD**	**Overall % DD**
Case 1	2.811735941	140.5867971	2.811735941
Case 2	4.991600672	249.5800336	4.991600672
Case 4	1.66453265	83.22663252	1.66453265
Case 7	0.444900753	22.24503765	0.44490075
Case 1	0.398272	23.8963143	0.3982719
Case 2	0.0262	1.571709234	0.0261952
Case 3	2.149454	128.9672544	2.14945424
Case 4	3.7037	222.2222222	3.7037037
Case 5	47.2509	2835.051546	47.2509
Case 7	44.8048	2688.288288	44.8048
Avg	**10.82**	**639.56**	**10.82**
S.D	**17.67**	**1064.43**	**17.67**

Table 39: Dosimetry Data for control group

Dosimetry			
SL No.	**Avg MU**	**Overall abs DD**	**Overall % DD**
Case 1	7.221148	433.2688588	7.221147647
Case 2	0.860065	43.0032543	0.860065086
Case 3	29.93354	1496.676892	30
Case 4	11.69714	584.8570089	11.69714018
Case 5	19.64963504	982.4817518	-0.437956204
Case 6	1.342673672	67.1336836	1.342673672
AA MU	**11.78403362**	**601.2369082**	**8.436101368**
S.D.	**10.33162484**	**511.9461281**	**10.49374577**

Table 40: Geometric Data for first group

Geometry						
Number of Cases	Abs VD		HD Mean		DSC Mean	
	UB	R	UB	R	UB	R
Case 1	175.276	19.9578	29.25167	23.065	0	0.88445
Case 2	45.1707	26.9367	66.685	319.8867	0.499083	0.865083
Case 3	3.93053	19.84461	134.2367	30.435	0	0.808333
Case 4	11.623	11.23841	39.05167	14.20833	0	0.838983
Case 5	32.46381	25.54708	47.29	18.12833	0	0.663333
Case 6	8.46671	12.3001	36.41167	19.41333	0	0.93
Case 7	279.535	36.2605	57.93833	21.33667	0.608233	0.448083
Case 8	32.6373	42.2768	38.3133	17.6917	0	0.86793
Case 9	12.4018	67.237	0	0	0	0
Case 10	74.7699	17.3203	59.8583	11.185	0.93013	0.94292
Case 11	155.635	32.0614	0	0	0	0
Case 12	46.6626	13.6374	0	0	0	0
Case 13	27.9004	49.651	58.4967	22.3567	0.85167	0.90667
Case 14	52.1067	31.4413	0	0	0	0
Avg	**68.47**	**28.98**	**40.54**	**35.55**	**0.21**	**0.58**
S.D	**77.11**	**15.31**	**35.08**	**79.46**	**0.34**	**0.39**

Table 41: Geometric Data for control group

Geometry						
SL No.	Abs VD		HD Mean		DSC Mean	
	UB	R	UB	R	UB	R
Case 1	3.518184894	0.379405768	59.66167	24.24667	0	0.8169
Case 2	0.312470063	0.473850709	67.53	19.22	0.689633	0.928167
Case 3	0.325315071	0.763835904	53.24	15.60667	0.264483	0.90445
Case 4	0.129809223	0.049930098	34.87667	33.23333	0.560517	0.847117
Case 5	0.462761417	0.153346849	74.84667	20.22167	0.099267	0.947883
Case 6	0.688979061	0.140197152	79.92	21.725	0	0.695433
Avg	**0.906253288**	**0.32676108**	**61.67916833**	**22.37555667**	**0.26898**	**0.85666**
S.D.	**1.180325729**	**0.243910917**	**14.91088556**	**5.51105137**	**0.2694**	**0.08503**

Significant geometric and dosimetric differences were observed between the First and Control groups for bladder and rectum measurements. For the bladder, abs VD, HD, and DSC were 68.48 ± 77.11, 40.54 mm ± 35.08, 0.21 mm ± 0.34 and (First) versus 90.63 cc ± 118.03, 61.88 mm ± 14.91, and 0.27 ± 0.27 (Control). For the rectum, abs VD, HD, and DSC were 28.98 cc ± 15.31, 35.55 mm ± 79.46, and 0.58 ± 0.39 (First) versus 32.68 mm ± 24.39, 22.38 mm ± 5.51, and 0.86 ± 0.09 (Control). In dosimetry, the average MU difference was 10.82 ± 17.67 (First) and 7.96 ± 6.40 (Control). The abs DD was 639 cGy ± 1064 (First) and 601.23 cGy ± 511.94 (Control), while the % DD was 10.82% ± 17.67 (First) and 7.96% ± 6.40 (Control).

CHAPTER-5: DISCUSSIONS

5.1 Discussion

This study aimed to assess the geometric and dosimetric effects of bladder and rectal filling on the quality of pelvic radiotherapy treatments using three-dimensional conformal radiation therapy (3DCRT), intensity-modulated radiation therapy (IMRT), and volumetric modulated arc therapy (VMAT). The results provide valuable insights into the impact of bladder and rectal volume fluctuations on treatment accuracy, highlighting the critical need for precise organ filling protocols in optimizing treatment outcomes.

5.1.1 Geometric Effects

The geometric variations in bladder and rectum during treatment were evaluated using cone-beam computed tomography (CBCT), which allowed for the assessment of organ displacement over the course of treatment. The study found that bladder volume changes were more significant than rectal variations, leading to greater positional discrepancies. This was evident from the Hausdorff Distance (HD) and Dice Similarity Coefficient (DSC) metrics, which showed that patients who adhered to strict bladder and rectal filling protocols exhibited lower HD values and higher DSC values, indicating improved alignment between CT simulation and treatment images. These findings suggest that consistency in bladder and rectal filling can substantially enhance the accuracy of radiotherapy planning.

5.1.2 Dosimetric Impact

The dosimetric analysis focused on dose coverage to the planning target volume (PTV) and the organs at risk (OARs), including the bladder and rectum. Variations in bladder and rectal volume during treatment were shown to significantly alter dose distribution. In cases where IMRT and VMAT were employed, the ability to re-optimize treatment plans allowed for compensation of these variations, thereby minimizing dosimetric discrepancies. However, in 3DCRT treatments, where static field arrangements are used, the dose delivery became more susceptible to variations in organ positioning, leading to increased dose to OARs and potential under-dosage to the PTV.

The dose-volume histogram (DVH) analysis demonstrated that a highly distended bladder could result in displacement of the PTV, reducing the dose to the tumor and increasing the radiation dose delivered to surrounding healthy tissues. Conversely, insufficient bladder filling was associated with greater rectal exposure, potentially increasing the risk of toxicity. The findings underscore the importance of strict adherence to bladder and rectal filling protocols to mitigate these risks and achieve optimal treatment efficacy.

5.1.3 Clinical Implications

The results of this study emphasize the importance of patient compliance with bladder and rectal filling protocols in maintaining the accuracy and effectiveness of pelvic radiotherapy. Variations in bladder volume can lead to significant deviations in the dose received by the tumor and surrounding healthy tissues, underscoring the need for individualized patient management and real-time adaptive treatment strategies. The findings support the implementation of daily CBCT imaging, personalized bladder filling protocols, and adaptive re-planning to ensure the consistency of organ positioning throughout the treatment course.

Furthermore, the study highlights the superior capability of IMRT and VMAT over 3DCRT in adapting to anatomical changes. The ability to modulate the intensity and shape of the radiation beams in real-time offers a distinct advantage in minimizing dose discrepancies and improving the precision of radiotherapy.



CHAPTER-6: CONCLUSIONS

This study provides compelling evidence that variations in bladder and rectal filling significantly impact the geometric and dosimetric accuracy of pelvic radiotherapy treatment. The key findings are as follows:

Bladder and rectal filling significantly affect organ position and target volume coverage, with the greatest impact observed in patients who did not strictly adhere to filling protocols.

Geometric changes lead to dosimetric variations, as demonstrated by HD and DSC metrics, indicating that organ volume inconsistency can compromise treatment accuracy.

IMRT and VMAT techniques provide better adaptation to anatomical variations compared to 3DCRT, highlighting the importance of advanced treatment planning techniques in managing organ motion.

Strict adherence to bladder and rectal filling protocols leads to improved treatment outcomes, reducing dose deviations and minimizing toxicity to OARs.

Real-time adaptive strategies, including CBCT imaging and individualized hydration protocols, should be implemented to optimize dose conformity and mitigate the effects of anatomical variations.

CHAPTER-7: REFERENCES

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CHAPTER- 8: APPENDIX

Appendix-1: Python Machine Learning Coding for HD calculation for UB and R

```
import os
import pydicom
import numpy as np
from scipy.spatial.distance import directed_hausdorff
import matplotlib.pyplot as plt
from skimage.draw import polygon
# Function to load a DICOM file
def load_dicom_file(file_path):
    return pydicom.dcmread(file_path)
# Function to retrieve contour points for a given structure in the RTSS file, slice by slice
def get_contour_points_by_slice(rtss, structure_name):
    for structure in rtss.StructureSetROISequence:
        if structure.ROIName == structure_name:
            roi_number = structure.ROINumber
            break
    else:
        raise ValueError(f'Structure {structure_name} not found')
    contour_points_by_slice = []
    for contour in rtss.ROIContourSequence:
        if contour.ReferencedROINumber == roi_number:
            for contour_sequence in contour.ContourSequence:
                points = contour_sequence.ContourData
                if len(points) % 3 == 0:  # Ensure the points can be reshaped into (N, 3)
                    slice_points = np.array(points).reshape(-1, 3)
                    contour_points_by_slice.append(slice_points[:, :2])  # Keep only X and Y coordinates
if len(contour_points_by_slice) == 0:
        raise ValueError(f'No valid contours found for structure {structure_name}')
    return contour_points_by_slice
# Function to convert contour points into a binary mask
def contour_to_mask(contour_points, img_shape=(512, 512)):
    mask = np.zeros(img_shape, dtype=bool)
    for points in contour_points:
        rr, cc = polygon(points[:, 1], points[:, 0], shape=img_shape)  # Create polygon mask
        mask[rr, cc] = True
        return mask
# Function to calculate Hausdorff distance between two sets of contour points
def hausdorff_distance(points1, points2):
    if points1.shape[0] == 0 or points2.shape[0] == 0:
        # If one of the structures has no points, return NaN instead of inf
```



```
        return np.nan
    else:
        hd1 = directed_hausdorff(points1, points2)[0]
        hd2 = directed_hausdorff(points2, points1)[0]
        return max(hd1, hd2)
# Function to plot contour points of two structures
def plot_contour_points(reference_points, structure_points, reference_structure_name, structure_name, hd_value, slice_index):
    ref_x, ref_y = reference_points[:, 0], reference_points[:, 1]
    struct_x, struct_y = structure_points[:, 0], structure_points[:, 1]
    plt.figure(figsize=(8, 8))
    plt.plot(ref_x, ref_y, 'r-', label=f'{reference_structure_name} Contour', linewidth=2)
    plt.plot(struct_x, struct_y, 'b-', label=f'{structure_name} Contour', linewidth=2)
    plt.text(0.05, 0.95, f'HD: {hd_value:.2f} mm (Slice {slice_index})', transform=plt.gca().transAxes, fontsize=12,
             verticalalignment='top', bbox=dict(boxstyle="round", facecolor="wheat", alpha=0.5))
    plt.title(f'Contour Comparison: {reference_structure_name} vs {structure_name}')
    plt.xlabel('X-Axis Distance (mm)')
    plt.ylabel('Y-Axis Distance (mm)')
    plt.legend()
    plt.grid(True)
    plt.show()
# Function to find DICOM files in a folder
def find_dicom_files(folder_path):
    rtss_file = None
    for file in os.listdir(folder_path):
        if file.startswith('RS') and file.endswith('.dcm'):
            rtss_file = os.path.join(folder_path, file)
    return rtss_file
# Function to calculate contours for multiple structures, find the slice with max HD, and plot it
Def calculate_and_plot_max_hd_contour(folder_path, reference_structure_name, structure_name, hd_results):
    rtss_file = find_dicom_files(folder_path)
    if rtss_file:
        print(f'Processing folder: {folder_path}')
        rtss = load_dicom_file(rtss_file)
        try:
            all_structures = find_all_structures(rtss)
        except ValueError as e:
            print(f"Error in folder {folder_path}: {e}")
            return
        if reference_structure_name not in all_structures:
            print(f"Reference structure '{reference_structure_name}' not found in {folder_path}")
```



```
            return
        if structure_name not in all_structures:
            print(f"Structure '{structure_name}' not found in {folder_path}")
            return
        reference_points_slices = get_contour_points_by_slice(rtss, reference_structure_name)
        structure_points_slices = get_contour_points_by_slice(rtss, structure_name)
        if len(reference_points_slices) != len(structure_points_slices):
            print(f"Warning: Number of slices for {reference_structure_name} and {structure_name}
differ.")
        max_hd = 0
        max_slice_index = -1
        max_ref_points = None
        max_struct_points = None

        # Calculate HD for each slice and find the slice with the maximum HD
        for i, (ref_points, struct_points) in enumerate(zip(reference_points_slices,
structure_points_slices)):
            hd_value = hausdorff_distance(ref_points, struct_points)

            if np.isnan(hd_value):
                print(f"Slice {i}: Skipping HD calculation for {structure_name} (empty slice).")
                continue
                    print(f"Slice {i}: HD for {structure_name} = {hd_value:.2f} mm")

            if hd_value > max_hd:
                max_hd = hd_value
                max_slice_index = i
                max_ref_points = ref_points
                max_struct_points = struct_points
                # Store the max HD result with structure name for this folder
        hd_results.append((folder_path, structure_name, max_hd, max_slice_index))
        # Plot contour for the slice with the maximum HD
        if max_ref_points is not None and max_struct_points is not None:
            plot_contour_points(max_ref_points, max_struct_points, reference_structure_name,
structure_name, max_hd, max_slice_index)
    else:
        print(f'Missing DICOM files in folder: {folder_path}')
# Function to list all available structures in the RTSS file
def find_all_structures(rtss):
    structures = []
    for structure in rtss.StructureSetROISequence:
        structures.append(structure.ROIName)
       if not structures:
```



```
        raise ValueError('No structures found in the RTSS file')
        return structures
# Function to recursively process all subfolders in the 'control' folder
def process_control_folder(control_folder_path, reference_structure_name, structure_names):
    hd_results = []
    for root, dirs, files in os.walk(control_folder_path):
        for folder_name in dirs:
            folder_path = os.path.join(root, folder_name)
            if os.path.isdir(folder_path):
                for structure_name in structure_names:
                    calculate_and_plot_max_hd_contour(folder_path, reference_structure_name,
structure_name, hd_results)
    print("\nFinal Maximum Hausdorff Distance Results:")
    for folder, structure_name, hd_value, slice_index in hd_results:
        print(f"Folder: {folder} | Max HD for {structure_name}: {hd_value:.2f} mm | Slice:
{slice_index}")
# Set your control folder path
control_folder = 'Control'  # Modify with your actual control folder path
# Reference structure (manually changeable)
reference_structure_name = 'RECTUM'
# List of structures to compare (manually changeable)
structure_names = ['R2', 'R3','R4','R5','R6', 'R7']  # Add all your structure names here
# Process all subfolders in the control folder and compare with multiple structures
process_control_folder(control_folder, reference_structure_name, structure_names)
```



Appendix-2: Python Machine Learning Coding for DSC calculation for UB and R

```
import os
import pydicom
import numpy as np
from skimage.draw import polygon
import matplotlib.pyplot as plt
# Function to load a DICOM file
def load_dicom_file(file_path):
    return pydicom.dcmread(file_path)
# Function to retrieve contour points for a given structure in the RTSS file, slice by slice
def get_contour_points_by_slice(rtss, structure_name):
    for structure in rtss.StructureSetROISequence:
        if structure.ROIName == structure_name:
            roi_number = structure.ROINumber
            break
    else:
        raise ValueError(f"Structure '{structure_name}' not found in RTSS file.")
    contour_points_by_slice = []
    for contour in rtss.ROIContourSequence:
        if contour.ReferencedROINumber == roi_number:
            for contour_sequence in contour.ContourSequence:
                points = contour_sequence.ContourData
                if len(points) % 3 == 0:
                    slice_points = np.array(points).reshape(-1, 3)
                    contour_points_by_slice.append(slice_points[:, :2])
    if len(contour_points_by_slice) == 0:
        raise ValueError(f"No valid contours found for structure '{structure_name}'.")
    return contour_points_by_slice
# Function to convert contour points into a binary mask
def contour_to_mask(contour_points, img_shape=(512, 512)):
    mask = np.zeros(img_shape, dtype=bool)
    for points in contour_points:
        rr, cc = polygon(points[:, 1], points[:, 0], shape=img_shape)
        mask[rr, cc] = True
    return mask
# Function to calculate Dice Similarity Coefficient
def dice_coefficient(mask1, mask2):
    intersection = np.sum(mask1 & mask2)
    size1 = np.sum(mask1)
    size2 = np.sum(mask2)
    return (2.0 * intersection) / (size1 + size2) if (size1 + size2) > 0 else 0.0
# Function to plot contours for the slice with maximum DSC
def plot_contours(ref_points, struct_points, ref_name, struct_name, dsc_value, slice_idx):
```

```
    plt.figure(figsize=(8, 8))
    plt.plot(ref_points[:, 0], ref_points[:, 1], 'r-', label=f'{ref_name} Contour', linewidth=2)
    plt.plot(struct_points[:, 0], struct_points[:, 1], 'b-', label=f'{struct_name} Contour', linewidth=2)
    plt.title(f'Slice {slice_idx} | Max DSC = {dsc_value:.4f}')
    plt.xlabel('X-axis (mm)')
    plt.ylabel('Y-axis (mm)')
    plt.legend()
    plt.grid(True)
    plt.show()
# Function to process a folder for DSC calculation for UB-related structures
def process_ub_dsc(folder_path, reference_structure_name, structure_name):
    rtss_file = find_dicom_files(folder_path)
    if rtss_file:
        rtss = load_dicom_file(rtss_file)
        try:
            ref_slices = get_contour_points_by_slice(rtss, reference_structure_name)
            comp_slices = get_contour_points_by_slice(rtss, structure_name)
        except ValueError as e:
            print(f"Error in folder {folder_path}: {e}")
            return None
        max_dsc, max_slice_idx = 0, -1
        max_ref_points, max_struct_points = None, None
        # Loop through slices and calculate DSC
        for i, (ref_pts, comp_pts) in enumerate(zip(ref_slices, comp_slices)):
            mask_ref = contour_to_mask([ref_pts])
            mask_comp = contour_to_mask([comp_pts])
            dsc_value = dice_coefficient(mask_ref, mask_comp)
            if dsc_value > max_dsc:
                max_dsc, max_slice_idx = dsc_value, i
                max_ref_points, max_struct_points = ref_pts, comp_pts
        if max_dsc > 0:
            print(f"Folder: {folder_path} | Max DSC for {structure_name}: {max_dsc:.2f} | Slice:
{max_slice_idx}")
            plot_contours(max_ref_points, max_struct_points, reference_structure_name,
structure_name, max_dsc, max_slice_idx)
        else:
            print(f"Folder: {folder_path} | Max DSC for {structure_name}: No valid overlap found.")
    else:
        print(f"Missing RTSS file in folder: {folder_path}")
# Helper function to find DICOM RTSS file
def find_dicom_files(folder_path):
    for file in os.listdir(folder_path):
        if file.startswith('RS') and file.endswith('.dcm'):
```



```
            return os.path.join(folder_path, file)
    return None
# Main function to process UB-related structures
def process_ub_structures(control_folder, reference_structure_name, structure_names):
    for folder_name in os.listdir(control_folder):
        folder_path = os.path.join(control_folder, folder_name)
        if os.path.isdir(folder_path):
            print(f"\nProcessing folder: {folder_path}")
            for structure_name in structure_names:
                process_ub_dsc(folder_path, reference_structure_name, structure_name)
# Settings
control_folder = 'Control'  # Update with your folder path
reference_structure = 'URINARY BLADDER'
ub_structures = ['UB1','UB2', 'UB3', 'UB4','UB5','UB6','UB7']  # Add relevant UB-related structure
names
# Run the process
process_ub_structures(control_folder, reference_structure, ub_structures)
```

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